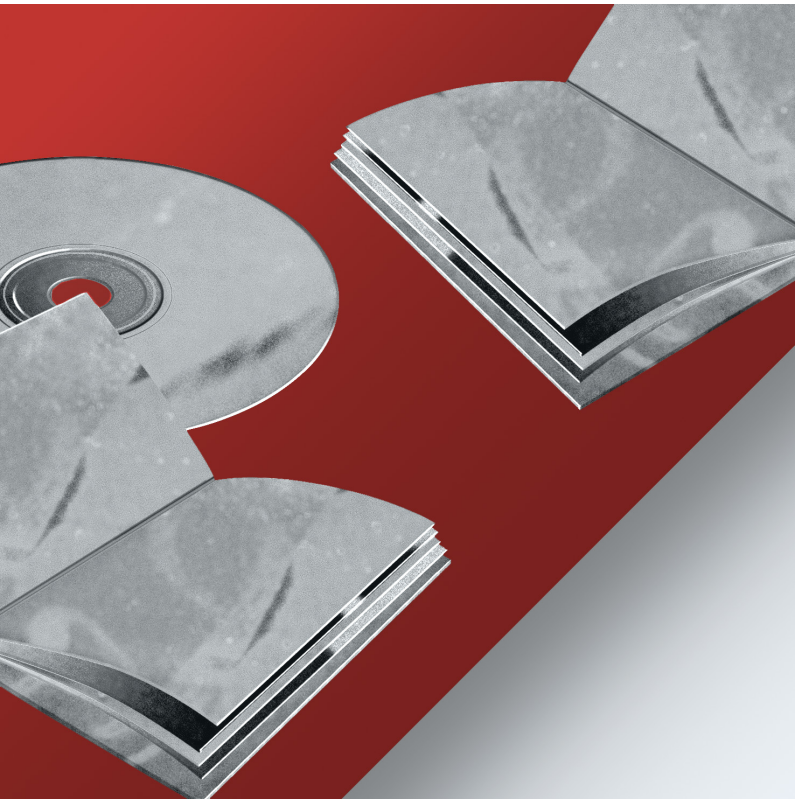


ASM Traceability 5.x (from R-20-2)

User Guide

www.asm-smt.com

11/2020 EN

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ASM Assembly Systems GmbH & Co. KG
Rupert-Mayer-Straße 44
81379 Munich
Germany

Item no.: 00900018-09

www.asm-smt.com

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1 Introduction

This document describes ASM Traceability. Before working with ASM Traceability, we recommend that you read this user guide. It contains all you need to know to learn about and use ASM Traceability.

The software interface uses the terms Traceability Service and Trace Service plus Traceability Monitor and Trace Monitor as synonyms.

This document uses the shorter terms Trace Service and Trace Monitor.

1.1 General

1.1.1 Symbols / Notes

Different symbols and notes have been used to make it easier to read and understand this document.

Words in bold/italic print	These refer to dialogs, dialog fields, menu items, and commands.
Step ...	Includes all actions to do with a particular dialog. A dialog may be composed of several actions ( ).
►	Requires an action (e.g. mouse click, open file)
	Indicates a warning. The information is essential for successful working.
	Explanatory information.

1.1.2 ASM on the World Wide Web

Log in to our ASM® homepage at **http://www.asm-smt.com**.

► You can choose between the German and English versions.

The different sections contain information about our products, services and contact persons.

In addition, registered customers can also access the *SIPLACE User Group*. Here you can call up special information about our placement systems e.g.

- Technical documentation,
- Technical information,
- Spare parts lists etc.

Registration for the user group is very simple:

► Click on the **Register** link.

► Fill in the registration form and send it off to us.

Soon afterwards, you will receive your access authorization with USER ID and password.

2 Overview of ASM Traceability

2.1 What is Traceability?

In the printed circuit board production industry, traceability means the ability to trace back production data over long periods. For example, the traceability data can be used even after years to determine which component was used in a particular batch/packaging unit, to place a certain board.

Traceability covers all the components needed to generate traceability data in a placement line. This includes the release of traceability data at the individual stations and the archiving/visualization of the same traceability data in Trace Monitor.

If required by the customer, splice sensors and board barcode readers can also be integrated into the traceability system.

NOTICE

Placement machines and printers

Printers are supported. This means that not only placement machine data but also printer data can be "traced". If this document mentions the terms **Station** and **Station software**, this refers to the placement machine. The document will otherwise use the terms **Printer** and **Printer software**.

NOTICE

Traceability data quality

The Traceability software is subjected to automated continuous tests before release. This involves virtual production of printed circuit boards in a continuous run of 48 hours. The Traceability data generated have a quality of 0 dpm.

In real production environments the quality of the Traceability data depend on the entire process chain.

By means of measures in the software (e.g. splice monitor), errors from the operational level and from the hardware process chain will be intercepted.

However, it is possible that the Traceability data may be faulty in rare cases.

The following error situations are known:

- ► Missing splice point (configurations with splice sensor)
- ► Scanning of wrong packaging unit information

2.1.1 Batch/Packaging Unit Change

A packaging unit is an individual reel or tray of components.

A component batch consists of one or more packaging units, with the same properties, which are either produced or purchased together.

A packaging unit change occurs at the moment when the first component of a new packaging unit (e.g. tape) is used.

A batch change occurs when a complete delivery of several packaging units i.e. a batch has been used.

NOTICE

Batch and packaging unit

In Traceability, the term "batch" is used for individual packaging units.

To ensure that you receive correct traceability data, the system must be informed about packaging unit changes. It needs to automatically recognize the end of a packaging unit.

There are two methods for achieving this:

- Tape end detection by splice detection
- Tape end detection by component level indicator

If a tape end is detected, and if barcode information has not been recorded for the successor packaging unit, the track concerned will be disabled. If there is no substitute track available, the machine will come to a standstill (error message "Barcode verification missing"). You then need to scan in the following packaging unit before you can restart the machine.

2.1.1.1 Detecting Packaging Unit Changes With Splice Detection

The packaging unit runs over a splice sensor, which detects the metal of the splice point and reports these detected splice points. For this to work properly, all splice points must always be connected with a metal strip.

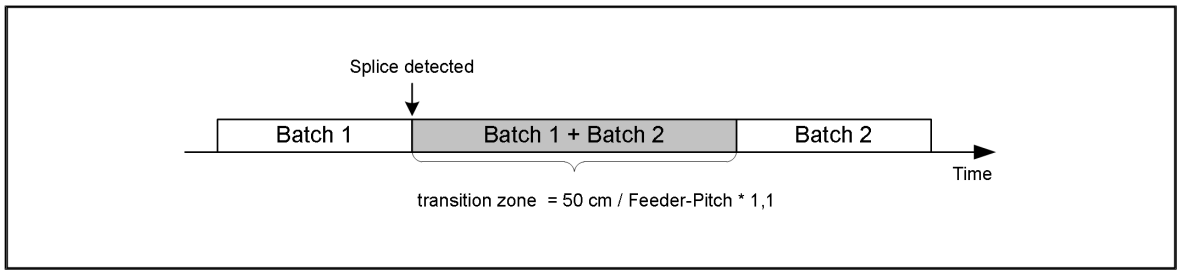
Until the splice point is detected, only packaging unit 1 will be used for the traceability data.

Once the splice point has been registered, the transition zone begins. During this transition zone, both packaging units are calculated (added) to the respective board. This means that it is possible for the board or panel traceability data to contain one packaging unit too much. The data will definitely contain the packaging unit actually used for the board.

When the transition zone comes to an end, only the new packaging unit will be used.

Packaging Unit Change with S Feeders

The following diagram illustrates the principle of a packaging unit change at a track with S feeders, monitored via a splice sensor.



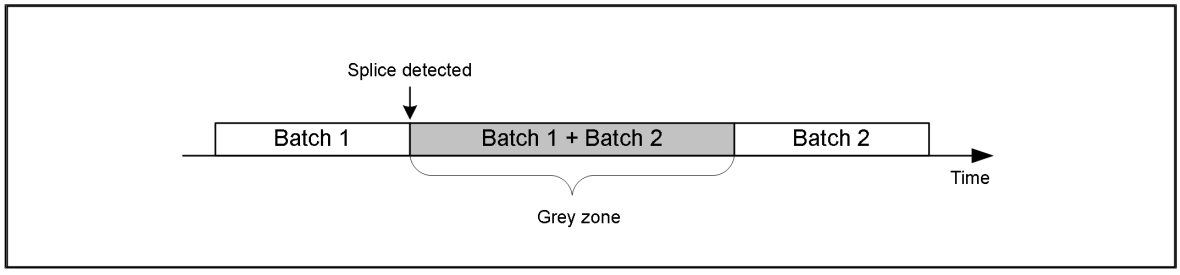
The "Feeder Pitch" or component pitch is defined for each component by the programming system. This value can be used to determine how many components of the old packaging unit are between the pickup position and the splice point. With the help of this number, the transition zone can be defined.

Firstly, packaging unit 1 is recorded in the traceability data. After the splice point has been detected, the transition zone begins, in which both packaging units (1 and 2) are recorded. The number of components in the transition zone can be calculated with the formula (50 cm/feeder pitch)x1.1.

After all components have been picked up from the transition zone, only packaging unit 2 will be recorded.

Packaging Unit Change with X Feeders

The following diagram illustrates the principle of a packaging unit change at a track with X feeders, monitored by a splice sensor.



Firstly, packaging unit 1 is recorded in the traceability data. After the splice point has been detected, the transition zone (gray zone) begins, in which both packaging units (1 and 2) are recorded. The length of the transition zone depends on the station software version. On 60x stations the length of the transition zone corresponds to 45 components, on 70x stations it corresponds to approx. the length of the splice plate.

After all components have been picked up from the transition zone, only packaging unit 2 will be recorded.

Regular checking of splice sensors

NOTICE

Regular checking of splice sensors

You should regularly check the function of your splice sensors. If a splice sensor does not function properly, this may mean that a packaging unit is not recognized correctly and that the traceability data contain incorrect packaging unit information for this track.

- **For S feeders**

To check the splice sensors, take a regular look at the LED displays in the sensor. Each track has an LED display which shines when the sensor is connected and currently recognizes no splice point. When the sensor detects a splice point, the LED goes out.
You can also configure a test mode (see Chapter "Splice test output active"), which will then display a message on your screen for each splice point detected.

- **For X feeders**

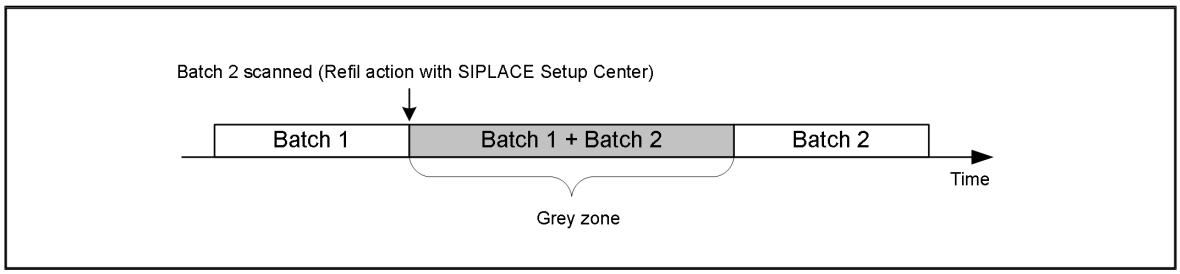
When X feeders are used, the display shows each splice point which is detected. For detailed information about testing splice sensors for X feeders, refer to the SIPLACE Service Manual for Tape Feeders 8 - 88 mm X (Item No.: 00194603-xx)

2.1.1.2 Detecting Packaging Unit Changes With Component Level Indicator

If there is no splice detection available for a particular track, the packaging unit change can be monitored with the component level indicator function.

The component level indicator knows the component levels of each track and updates these at the end of each board. If, after completion of a board, it calculates that a track must be empty and if you have not recorded a new packaging unit, the machine will be stopped. For this to work, you must have configured the component level indicator thresholds accordingly (see Component Level Indicator menu). If the new packaging unit was recorded in time, the system will switch over to this one, while taking a gray zone into account.

The following diagram illustrates the principle of a packaging unit change at a track which is only monitored via the component levels.



The transition zone begins at refill by the Setup Center. The length of the transition zone is at least 120% of the remaining component level of packaging unit 1 (depending on the station software version) at refill time.

This means that you can reduce the transition zone by refilling as late as possible.

In principle, it is also possible for bulkcase feeders and linear feeders to generate traceability data, with the help of the component level indicator. In this case, however, the operator must ensure that the feeder is completely empty before the refill.

NOTICE

Component Level Indicator

The component level indicator function is not as reliable as the splice detection function. Errors can occur when the component levels are entered or the component levels may only be roughly known.
The care you take when managing the component levels for the individual feeders will have a significant influence on the quality of the data.

2.1.2 Recording Packaging Unit Information with ASM Setup Center

The packaging unit information is recorded with ASM Setup Center.

With ASM Setup Center you can record different pieces of barcode information for each lane. Details of this can be found in the product information for ASM Setup Center.

To give you an overview of the traceability function, this section summarizes the situations in which packaging unit information can be recorded with Setup Center.

2.1.2.1 Recording After a Setup Change

In Setup Center you can set the way in which packaging unit information is adopted from a predecessor setup. We recommend using the delta setup function. The system also checks which tracks have changed since the predecessor setup. The track data which has remained the same will be accepted and does not need to be reentered.

For detailed information, refer to the ASM Setup Center User Guide.

Setup change with delta setup function

The tracks which have changed are re-recorded with the "Table check" function. All other data is taken from the predecessor setup.

Setup change without delta setup function

If you do not use this function, you will need to reenter all barcodes and component levels after each setup change. You will need to use the "Table check" function to scan the complete table.

CAUTION

Component levels

Please note that you will be redefining the component levels and may overwrite any existing component levels!

2.1.2.2 Refilling a Track

A track which is to be refilled is re-recorded with the "Refill" function. Make sure that you only use the "Refill" function for this! The system adds the new component level to the remaining component level of the old packaging unit.

2.1.2.3 Activating a Track

If the machine is unable to pick up from a particular track, after several attempts, it will assume that a feeder is empty and could stop with a "Empty track error". However, this may be due to a pickup error and there may still be enough components available.

In this case, you need to first solve the source of the error and then enable the track again. If you have selected the option **Activate pickup position with verification** in Setup Center, you will only be able to reactivate the track by scanning in the material barcodes. The component level remains the same until new material is configured as part of the troubleshooting process and verified as part of the activation process.

2.1.2.4 Setting the Component Level

With the "Correct component level" function you can correct the component level of a track, using Setup Center. Alternatively, the component level can be corrected on the SIPLACE user interface.

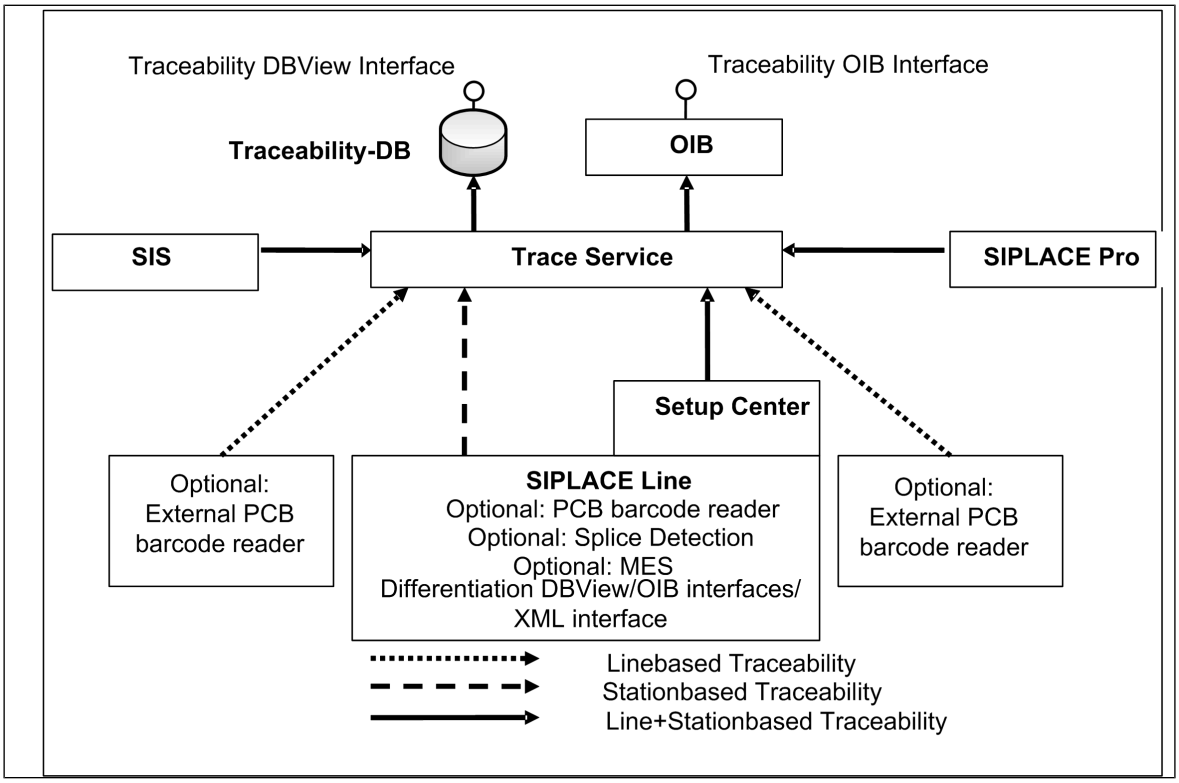
2.1.3 Recording MTC or WPC Traceability Data

Traceability data can be recorded for:

- **MTC (Matrix Tray Changer)**
- **WPC (Waffle Pack Changer)**
  In these cases, there are several rules which you must observe. If you do not observe these rules, correct traceability data cannot be guaranteed.
- Refilling of a tray always causes setting up.

- After a reboot or setup change, traceability checks whether tray barcodes are present for all MTC/WPCs. If barcodes are missing, the tray concerned will be disabled.
- The smallest unit for a packaging unit is the tray. All the components on a tray belong to the same packaging unit.
- If the traceability option is active, MTC areas can no longer be manually set to full. Instead, you need to record the tray with ASM Setup Center and release it for production this way.
- If a new packaging unit is reported for the MTC, this will be used immediately for the traceability data. The transition zone is = 0.
- Trays may only be changed when the machine is switched on.

2.2 The Traceability Principle



Traceability allows you to generate traceability data in various different ways. In principle we distinguish between station-based traceability and line-based traceability.

In station-based traceability, the data are produced in the stations, while line-based traceability generates the data without any direct communication to the stations.

Station-based traceability:

The stations are informed by Setup Center about currently used packaging units. During the production, the currently used packaging units are assigned to the respective board. This assignment can also be detailed on panel level or on individual placement position level.

After all data for a specific board have been collected, they are transferred to Trace Service.

In all operation modes, the packaging data used for production are recorded by the setup verification system ASM Setup Center, by scanning in the packaging unit barcode.

The machine recognizes packaging batch changes via the splice detection function (optional) or by counting the component levels.

The boards are identified by an external barcode reader or by an integrated SIPLACE barcode reader. The packaging data is assigned to a specific board via the individual board within the machine (in station-based traceability) or via the production period (in line-based traceability).

If the system is unable to detect whether the old package is still in use or whether the new one is already in use (packaging unit change), all possible ones are recorded in the traceability data. This is called a “gray zone”. If we increase the effort for collecting information (e.g. board barcode reader, splice detection) the gray zone gets smaller.

Line-based traceability:

The line configuration is imported from SIPLACE Pro. Trace Service is notified about incoming and outgoing boards by external board barcode readers (option). The system receives information from SIS about jobs and setups that were active at the line in the notified time range. With this information, Trace Service can search the Setup Center database to find all packaging data set up within this time range. The data is then correlated and assigned to a specific board or customer order in Trace Service. If no external board barcode readers are available, the data is correlated using timestamp information and the respective production order.

NOTICE

Restriction

In the "line-based traceability" mode, printers are treated in the same way as in the "station-based traceability" mode. That is, there is one data package for each printer in the line (e.g. an XML file) and one data package for the remainder of the line.

2.3 Operation Modes

Traceability supports various modes of operation, which are described below with details of their configurations.

2.3.1 Station-based traceability

In this operation mode, each machine assigns the data for the packaging unit of its current setup to the boards being produced. This gives one data package per machine and board.

Trace Service provides the correlated traceability data from external systems (OIB, file system) or via the traceability database. Data from the traceability database can be visualized and evaluated in Trace Monitor.

In station-based traceability, the packaging data can be assigned either to the whole board, or to single panels or placement positions on the board.

The recording accuracy of station-based traceability data can be improved by using integrated SIPLACE board barcode readers or splice detection sensors.

Station-based traceability can also be executed without board barcode readers. However, in this case, the traceability data package can not be assigned to a specific individual board.

2.3.2 Line Based Traceability - Board Correlated

In this operation mode, the traceability data are generated not at the placement machine but at a line level, through correlation of the information available there.

The packaging data is registered via ASM Setup Center. The information about the time when specific jobs were active is imported from SIS database.

Board barcode readers read the board barcodes and send these on to the Traceability system.

In consideration of the sequence of boards or the time stamps of the board production the packaging data can be assigned to the boards.

The following configurations are possible:

- External board barcode reader in front of AND behind the SIPLACE line
  - The packaging data is assigned to the boards regarding the timeframe between the PCB barcode scans in front of and behind the line.
- External board barcode reader in front of the SIPLACE line
  - The sequence of the boards and the maximum number of boards in the line (configurable) are used to calculate the production time of a board. The packaging data configured in this period can be assigned to the respective boards using the production start (scan time) and the production duration.

- PCB barcode reader behind the SIPLACE line

  The production start is determined according to two principles:

  – At production start (after download of the current job in LCGUI), the starting time for the job is set for the board which is being moved out at the back.

  – As soon as the number of boards being moved out (barcode read at back) exceeds a threshold, the starting time for boards scanned at the back can be determined more precisely. For example, if we have 10 possible boards in the SIPLACE line, we would use the scan time (at back) for the 10th board as the starting time for the current board. This process ensures that there is no negative influence on the traceability data quality, even during longer production stops.

Line-based traceability supports:

- Only external board barcode readers which are connected a standard interface connected to the SIPLACE system.
- SIPLACE board barcode readers, which are operated as external barcode readers (from Traceability 3.2 and station software 70x).
- Barcode readers, which can be operated via the CogiScan product flow controller.

2.3.3 Line Based Traceability - Order Correlated

This is the simplest operation mode but also the mode with the lowest accuracy.

The generation of Traceability data is performed without any communication to the placement machines and without scanning board barcodes.

The packaging data are exported from ASM Setup Center. The information about active jobs on the line is imported from the SIS database.

The timeframe is determined in which a specific customer job was active. The data for the packaging unit used during this timeframe can then be assigned to this customer order.

In this mode there is no data assignment to individual boards.

2.3.4 Process Data Traceability

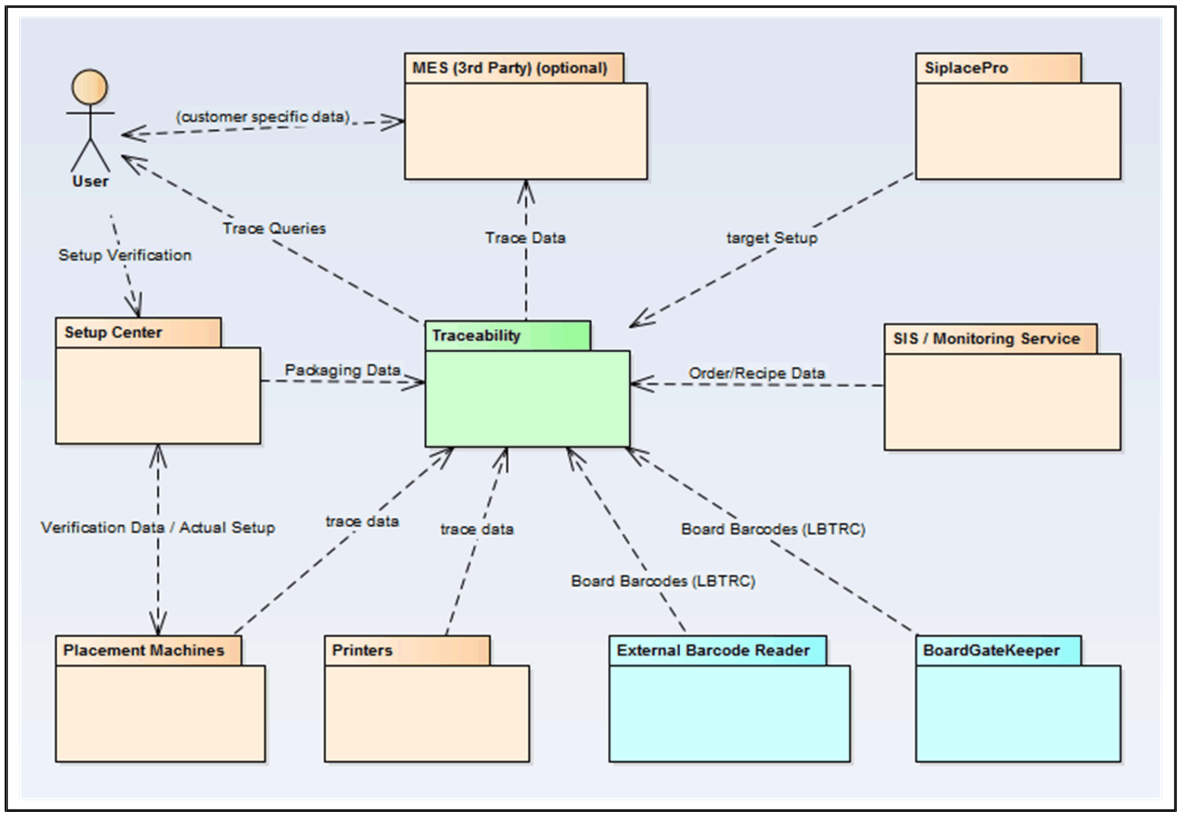
In this operating mode, the process data of each placement machine are collected. There are different types of data packages per processing area. The first package is sent as soon as processing the board has started in this area. During placement, a data package is sent after a maximum of 1000 components were placed. Another data package is sent as soon as processing the board is finished or aborted.

The data are not being correlated.

Process data do not depend on the material traceability data and can therefore be made available independently of the selected mode (station-based or line-based) and even without material data.

2.4 System Environment for Traceability

2.4.1 System Components



The traceability system environment consists of the following components:

SIPLACE Pro (engineering system)

- Defines boards and setups
- Defines which placement positions are to be traced for (station-based) traceability

SIS / monitoring service

- Provides the history of jobs at the line

Station software (machine control)

- Reads board barcodes
- Manages component levels
- Detects splice points
- Collects traceability data in one data package. One data package is generated for each board and station. This contains various data, including the relevant placement information (i.e. which batch or packaging unit is used at the individual placement positions).

ASM Setup Center (material control)

- Setup verification
- Allocation of the packaging unit to the individual divisions/feeders, the feeders to the tables and the tables to the machines
- Records all packaging unit information relevant for traceability

External board barcode reader

- Reads the board barcodes and sends them via a standard interface to the SIPLACE system
- External board barcode readers are devices from other manufacturers
- SIPLACE board barcode readers can be connected in the same way as external readers.

User

- Scans in the packaging unit data relevant for traceability as parts of setup verification.
- Can run Trace Monitor queries for traceability data if DB Export has been enabled.

BoardGateKeeper

- Connects external barcode readers and the CogiScan product flow controller.
- Freezes boards until they are released by an MES in the conveyor.

MES system

The MES system can import and process traceability data via the standard interface (OIB interface).

The Traceability OIB data interface is provided in two variants:

- OIB Synchronous
  - Direct communication takes place between Trace Service and the MES System. The MES system uses a callback interface to lock (interlocking option) boards on the conveyor. This interface requires additional licenses. Information about the required licenses can be found in the ASM Traceability version description.
- OIB Eventing
  - This is an asynchronous communication procedure, which supports consistent data transfer to multiple recipients and which also saves the traceability messages in an intermediate memory ("OIB Eventing Queue") if the target system (such as MES) is not available. This prevents loss of data due to infrastructure or operational failures and ensures that spooled data are delivered as soon as the target system is available again. This interface requires additional licenses.
  - Information about the required licenses can be found in the ASM Traceability version description.

Information about system configuration can be found in the ASM Traceability installation manual.

2.5 Traceability Levels

Traceability levels

Traceability data contains information about the batches/packaging units used for placing components on a board. There are various ways of recording traceability data.

- Level 1: Location/date-related data entry
- Level 2: Production job/recipe-related data entry
- Level 3: Board-related data entry
- Level 4: Placement position-related data entry

The traceability levels and the corresponding system configurations are defined below:

- **Level 1: Location/date-related data entry**
  - Splice sensors, optional
  - Implemented by line-based traceability (order correlated)
- **Level 2: Production job/recipe-related data entry**
  - Splice sensors, optional
  - Implemented by line-based traceability (order correlated)
- **Level 3: Board-related data entry**
  - Implemented by line-based traceability (board correlated) or station-based traceability
  - Station-based traceability: barcode reader optional
  - Line-based traceability: barcode reader required
  - Splice sensors, optional
- **Level 4: Placement position-related data entry**
  - Can only be realized with station-based traceability

- Splice sensors (optional)
- Forced setup verification and component level indicator

2.6 External PCB Barcode Readers

External board barcode readers can be used for operation of line-based traceability.

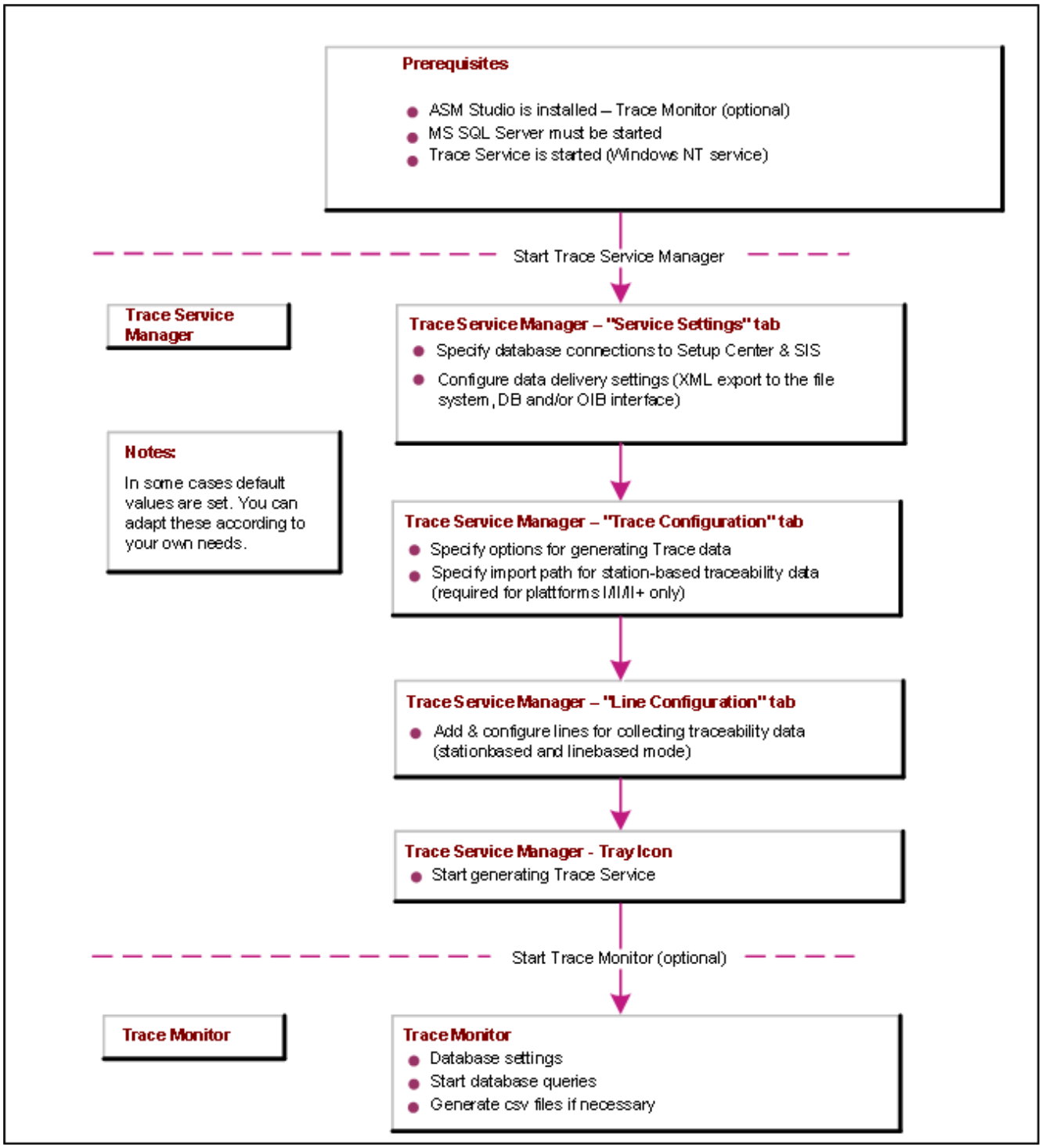
SIPLACE board barcode readers can also be used as external board barcode readers.

This enables SIPLACE board barcode readers to be

- integrated for operation of station-based traceability (1 reader per conveyor lane in each machine). Configuration of this option is performed in SIPLACE Pro (not in traceability).
- operated as external readers, so that line-based traceability (board-related) can be used (1 reader per conveyor lane before and optionally behind the line). Configuration of this option is performed in SIPLACE Pro AND in traceability.

In addition, you can connect more scanners, by integrating them via the ASM BoardGateKeeper product e.g. the CogiScan product flow controller.

2.7 Initial Configuration of ASM Traceability



3 Trace Service

The Trace Service Server runs as a Windows service and is automatically started each time your reboot your PC. You will see the relevant icon on your Windows console.



In order to process traceability data, you need to activate the Trace Service Server. Once the Trace Service Server has been enabled, traceability data processing will run until this function is manually disabled.

While the Trace Service Server is active, other parameters can be configured. Confirm to apply the changes to the lines concerned (and only to these).

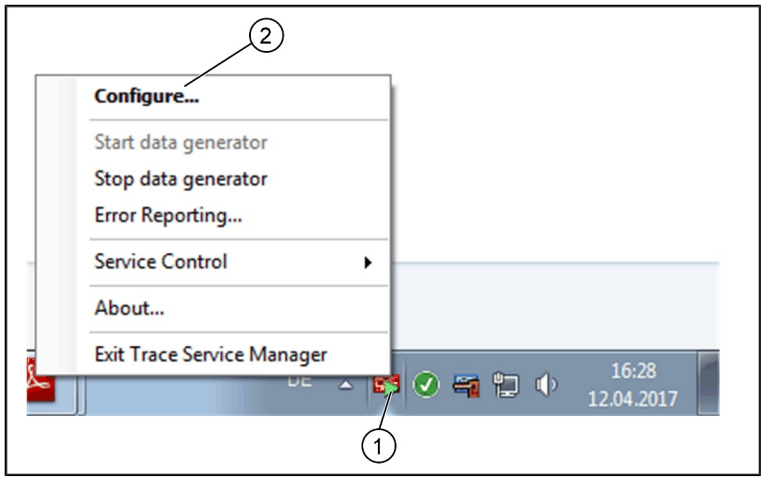
In accordance with the Trace Service settings, the traceability data and the database are transmitted to the file system or sent via OIB to a connected MES system (3.3.2 "Service Settings" [▶ 25]). If a configured target system is temporarily not available, the traceability data will be held in Trace Service until transmission is possible.

3.1 Starting the Trace Service Manager

The Trace Service runs as a Windows service and is automatically started each time your reboot your computer.

► To open the **Trace Service**, right-click with the mouse on the relevant icon (**1**) in the Windows toolbar.

⇨ The context menu (**2**) will open.



Context menu		
Configure...		The Trace Service is opened. The **Trace Service Properties** dialog box will open. See also 3.3 "Configuring Trace Service" [▶ 24].
Start data generator		The data generator is started.
Stop data generator		The data generator is stopped.
Error Reporting...		The Error Report for the **Trace Service** is started.
Service Control	**Start Windows Service**	If the Windows service **Trace Service** has been stopped, **Trace Service** can be restarted here.
	Stop Windows Service	The Windows service **Trace Service** is stopped.
	Restart Windows Service	An attempt will be made to start the Windows Service **Trace Service**. This can be necessary, for example, after changing the runtime configuration (e.g. logging settings).
About...		This shows information, such as the **Trace Service** version.
Exit Trace Service Manager		The **Trace Service Manager** will be closed.

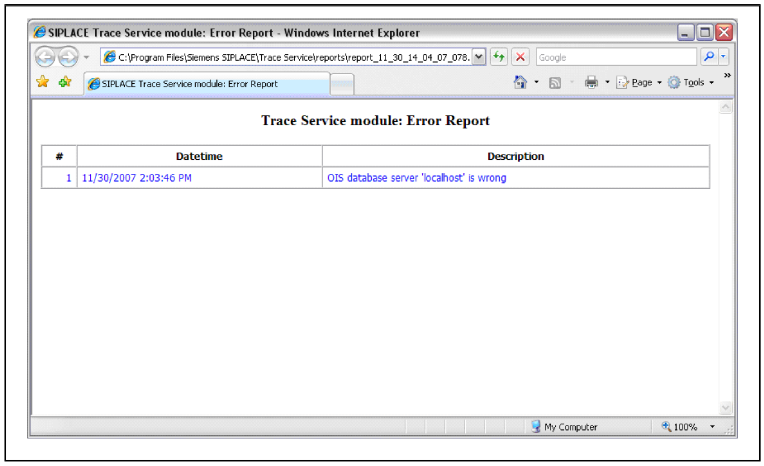
3.2 Trace Service Status Display

The status of the Trace Service can be viewed with the Windows console:

Windows console

The following states are possible:

- 
  - Trace Service is running. Data generator is running. No error.
- 
  - Trace Service is running. Data generator is stopped. No Trace data is generated. No error. Please note that the trace data at the station are not spooled while the data generator is stopped.
-  (blinking)
  - Trace Service is not started. No Trace data is generated.
-  (blinking)
  - Trace Service is running with some errors. You can click on the icon – an Internet Explorer window with error report will open:



-  (blinking)
  - Trace Service was stopped because of a severe error. No Trace data is generated.

3.3 Configuring Trace Service

Trace Service is configured with the parameters needed to generate traceability data in the Trace Service Properties.

Trace Service Properties consist of a dialog box with the following tabs:

Service Settings

This is where you can specify the database server for Setup Center and SIS plus the information required for transferring traceability files.

Trace Configuration

This is where you can define the operating options for line-based and station-based traceability.

Line Configuration

This is where you can define the traceability mode for individual lines and configured board bar-code readers.

Packaging Unit

This is where you choose how to specify the batch ID in the trace data.

Notifications

This is where you can define the notifications of error situations via e-mail.

Debug Settings

This is where you can define which levels are to be set for the log files.

3.3.1 Starting Trace Service

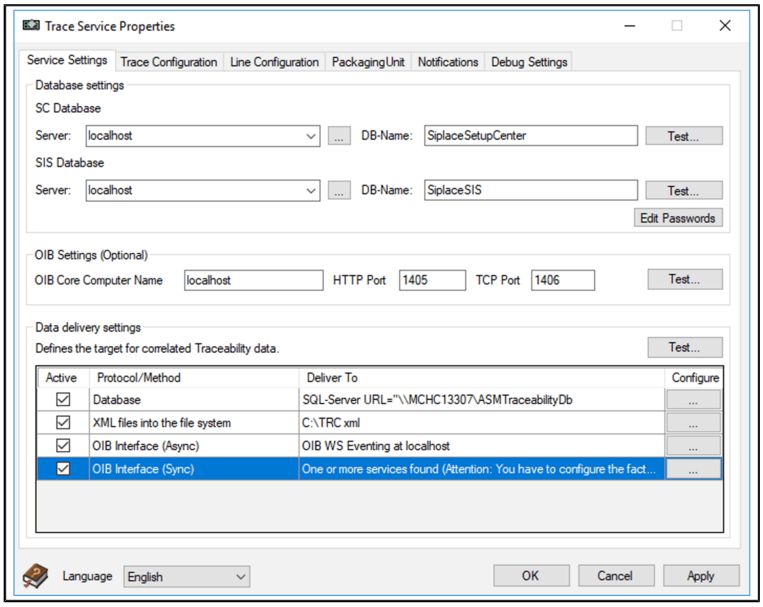
Trace Service is automatically started by Windows and does not need to be manually opened. Trace Service is configured in the **Trace Service Properties** dialog box.

► The **Trace Service Properties** dialog box can be opened by right-clicking on the "tray" icon in the toolbar  .

⇨ This opens the context menu.

► Select **Configure**.

⇨ The **Trace Service Properties** dialog box will open:



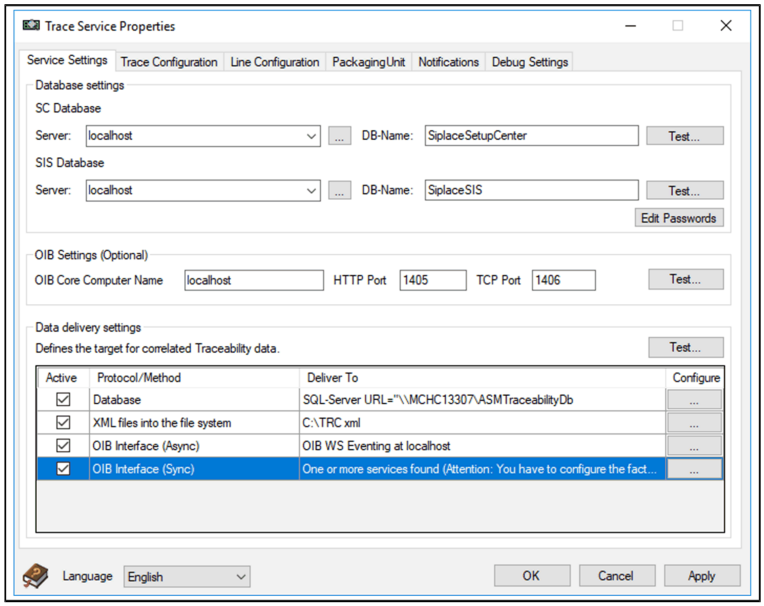
► Select the required tab for the parameters you want to configure.

►  **Help**
Opens the online help for Trace Service.

- **Language**
  Set the required language for further operation of the Trace Service. The following options are available:
  - ⇨ English
  - ⇨ German
- **OK**
  Each change will be applied to all tabs. The Trace Service Manager will be closed.
- **Cancel**
  Processing will be terminated. Changes will not be applied. The Trace Service Manager will be closed.
- **Apply**
  Each change will be applied to all tabs. The Trace Service Manager will not be closed.

3.3.2 Service Settings

- Go to the **Trace Service Properties** dialog box and select the **Service Settings** tab.



Database settings

Designation	
SC Database Server: / DB Name:	Enter the name of the Setup Center database server. After entering the database name, we strongly recommend that you check the connection to the database. Click **Test** to check whether the connection to the database can be established.
SIS Database Server: / DB Name:	Enter the name of the SIS database server. After entering the database name, we strongly recommend that you check the connection to the database. Click **Test** to check whether the connection to the database can be established.

Designation	
Edit Passwords	Opens the dialog box to enter the passwords for the SC database and for the SIS database. The complex ASM default passwords are entered by default. The **Test and Save** button checks these passwords, checks the connection to the databases and locally saves the passwords. If the fields are left empty, you can use the **Test** and **Save** button to test the default ASM passwords: the complex ones first and then the simple ones. If a test is successful, this password is saved locally. **NOTICE!** **Please note that you change passwords in the local password memory here but not the passwords in the database.**

NOTICE

SC and SIS database server

Only one SC and one SIS database server can be specified. When using multiple lines, the SIS server is installed as a default once per line, on the line PC. Trace Service then needs to be installed on all SIPLACE Pro computers.

OIB settings (optional)

If an OIB core is installed in the system, you have to configure the connection parameters to the OIB here. In this case, the OIB interfaces can be used to transfer the trace data. Trace Service is registered as a service in Health Manager and its status is monitored.

Button	
OIB core computer name	Name of the computer on which the OIB core is installed.
HTTP port	The HTTP port via which the OIB Core computer can be reached.
TCP port	The TCP port via which the OIB Core computer can be reached.
Test…	This button tests the connection to the OIB Core on the specified computer.

Transmitting trace data

Further processing of the correlated traceability data can be configured in the **Data delivery settings** section.

Button	
Test…	This button creates a traceability data record and sends this to all activated transmission options. These are test data which you may need to manually remove from your target system. Deletion of individual traceability information from a configured database is not supported. In this case, we recommend using test databases.

There are 4 interfaces for further processing of traceability data:

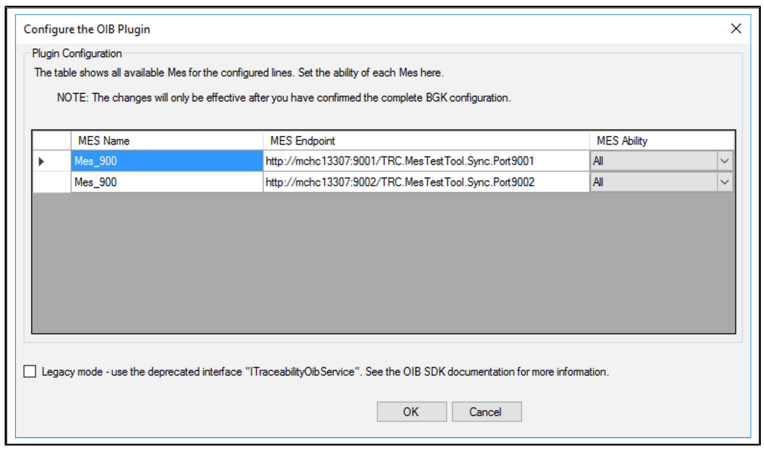
Interfaces	Description
Database	The traceability data is stored in a SQL database. This button tests the connection to the traceability database. The name of the traceability database will be shown. **NOTICE!** **The database is configured using the traceability installation and the Operation Manager.**

Interfaces	Description
Transferring XML files to the file system	This interface supplies external systems with traceability data via the file system. This type of data export can also be used for archiving purposes. Trace Service converts all traceability data to XML files which can then be imported. ► Here, you can define the folder in which Trace Service is to store the XML files. The same folder must then be configured as the data source for downstream processing systems.
OIB interface (asynchronous)	The **OIB interface (asynchronous)** is an asynchronous data interface, available via OIB Eventing. The traceability data can be transmitted to any number of target systems. OIB Eventing creates a query for each target system, to allow intermediate data storage should that system not be available and to then transmit the consolidated data once the system is available again. For more information about OIB Eventing and the traceability OIB interface, refer to the OIB product documentation. ► Select **OIB interface (asynchronous)**. ► Check if the configuration is correct by clicking the **"..."** button under **Config** and confirming the following dialog with **OK**.
OIB interface (synchronous)	The **OIB interface (synchronous)** is a synchronous data interface offering the following options: • Synchronous (real-time) traceability data supply • MES process regulation (Interlocking) ► Select **OIB interface (synchronous)**. ► Check if the configuration is correct by clicking the **"..."** button under **Config**. The Configure the OIB Plugin dialog box will open. See also 3.3.2.1 ""Configure the OIB Plugin" Dialog Box" [▶ 28] For a description of the OIB interface, refer to the OIB product information. Hints: • The "OIB interface (synchronous)" requires the following licenses per machine in the line: – MES Process Interlocking Machine Connection SIPLACE (placement machine) – MES Process Interlocking Machine Connection DEK (DEK printer) The following restrictions exist in the MES process regulation traceability: • PCB locking is supported only at the output belt of the station (station-based traceability) or at the rear barcode scanner (line-based traceability). • Only one MES system can accept the synchronous call (BoardGateKeeper: several MES systems are possible). • The PCB barcode cannot be changed by the MES system (which is supported by BoardGateKeeper, however, with restrictions).

3.3.2.1 "Configure the OIB Plugin" Dialog Box

The table shows all MES that are available for the configured lines. The MES abilities must be set.

- ► Go to the **Trace Service Properties** dialog box and select the **Service Settings** tab.
- ► Check if the configuration is correct by clicking the **"..."** button. The **Configure the OIB Plugin** dialog box will open.



Parameters/buttons	Description
MES Name	Shows the name of the MES.
MES Endpoint	Shows the endpoint of the MES.
MES Ability	Select which MES will receive which kind of data (e. g. printer, placement machine, or all...).
Legacy Mode	The discontinued "ITraceabilityOibService" interface is enabled instead of the new "ITraceablilityData" interface. Using the legacy mode is not recommended.

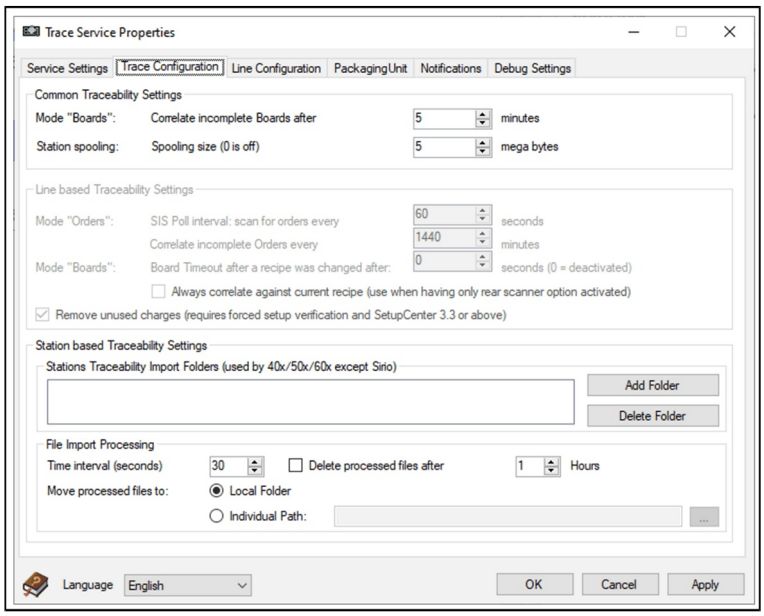
3.3.3 Trace Configuration

In this tab you can configure the settings for line-based and station-based traceability. In the **Line Configuration** tab (see next section) you will be able to define the traceability mode for a specific line.

NOTICE

Traceability operating modes

You can find a definition of the traceability modes in chapter 2.3 "Operation Modes" [► 14].



3.3.3.1 General Traceability Settings

"Board" mode

- **Correlate incomplete boards after ... minutes**

Traceability data for boards which have not yet been reported as complete are reported back after a certain time. Configure the time after which Trace Service should correlate incomplete boards (recommendation 60 minutes).

WARNING

Do not configure time intervals of less than 60 minutes

We do not recommend configuring shorter time intervals than 60 minutes. Shorter production delays (e.g. maintenance) could lead to unfinished boards (not completely placed) causing incomplete traceability data. In this event, the error code "DATA_LOST" will be issued.

Station spooling

- Spooling size (0 is off)

  If the station is temporarily unable to reach Trace Service, the trace data can be temporarily stored. This parameter specifies the size which the memory may reach before the station stops producing. Temporary storing is stopped when the value is 0 and the machine will then immediately stop producing.

3.3.3.2 Line based Traceability Settings

At the top of the screen you can configure the settings for line based traceability.

"Order" mode

- **SIS poll interval: Scan for orders every… seconds.**.

Here you can define how often Trace Service should query the SIS database (recommendation: 60 sec).

- **Correlate incomplete orders every… minutes.**

Define how often Trace Service should correlate incomplete orders (recommendation 1440 min).

"Board" mode

- **Board timeout after a recipe was changed after...seconds**

Traceability data for boards which have not yet been reported as complete will be reported back after a certain time, following a job change. Configure this time here. If you do not want to use this feature, enter "0" seconds (recommendation: 0 seconds).

- **Always correlate against the current recipe (use when having only the rear scanner option activated)**

If a line is equipped with a rear barcode only, the start of the placement is estimated by TraceService. As a result, it can happen that all production jobs which are produced in the detected time interval are assigned to the current board. By activating the **Always correlate against the current recipe** option, you ensure that only the currently valid production job is assigned based on the time of the scan at the rear barcode scanner.

Removing unused charges (batches)

Line based traceability has no direct feedback from the placement process, indicating which batches (charges) have been used. The respective board for all configured batches (charges) is therefore assigned according to certain rules. However, you can filter out any batches (charges) which were not picked up at the station, which is then recognized by the station according to component level changes. This option requires that Setup Center 3.3 or higher is installed and that the Setup Center GUI is running on the host on which Trace Service is installed.

► Check the box **Remove unused charges** for this (recommended setting).

NOTICE

Requires "forced setup verification"

Activation of this option requires "forced setup verification" for all lines for which line based traceability has been configured.

CAUTION

Restriction

It is recommended not to use the "Remove unused batches" options if the scanner is located directly after the last machine. This might lead to incorrect traceability data. If you want to use this option anyhow, place the scanner in such a way to ensure a delay of about 1 minute between the last placement and reading the barcode.

3.3.3.3 Station-based traceability - settings

In the lower part of the dialog you can configure settings for station-based traceability.

Station import folders

Here you specify the folders where stations with software versions 40x, 50x, 60x will save the XML files. You may use the buttons **Add folder** or **Delete folder** for this purpose.

Import of Trace XML files

Here you can define details for processing these XML files.

- **Time interval**

Define after how many seconds Trace Service shall look for new XML files (recommendation 30 sec).

- **Delete processed files after**

Configure after how many hours processed XML files shall be finally deleted. If you do not activate the checkbox, processed files will not be deleted and will remain in the defined folder. To avoid performance problems, it is recommended not to deactivate the delete option for a longer period of time.

- **Move processed files to**

Define the folder where processed files are to be shifted to. You can select the local folder which is shown in the dialog. Alternatively, you can define another folder. Please use the select button for this purpose.

3.3.4 Line Configuration

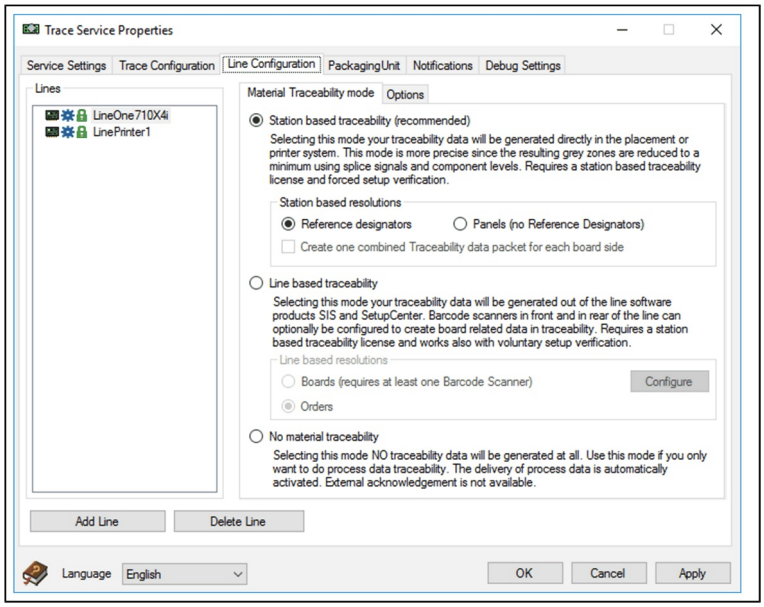
3.3.4.1 Line Configuration - Material Traceability Mode

In this tab you can configure the traceability production mode for each line.

NOTICE

Traceability operating modes

You can find a definition of the traceability modes in chapter 2.3 "Operation Modes" [▶ 14].



Lines

In the left part of the screen you can see all lines which are relevant for traceability. This list can be modified using the buttons **Add line** or **Delete Line**. If you want to add a line you will see a select list of all lines configured in SIPLACE Pro. In addition, you can exclude already configured lines from generating traceability data by right-clicking on the line name and selecting the option for **Stopping/Resuming[...]**. If you repeat this action on a deactivated line, the line will be reactivated for traceability.

Traceability Mode

In the right part of the screen you can set the traceability mode for the selected line.

Station-based traceability

- ► Select **Reference designator** if you want to get detailed traceability data for each placement position.
- ► Select **Panels** to get traceability data correlated to panels (single circuits). In this mode, no reference designators for the individual placement positions are generated.
- **Create one combined Traceability data packet for each board side**

The traceability data of all stations of a line are summarized per PCB and the consolidated data are sent to the target system. Each station of the line must transfer the PCB barcode to the traceability system to ensure an unambiguous identification. If the barcode is not read successfully or if the operator failed to enter the barcode correctly, the traceability data cannot be summarized for the respective PCB. In this case, each station individually transfers the data to the target system.

Line-based traceability

- ► Select **Board** if you want to configure line-based traceability with board correlated data. An external barcode reader in front of the line is required and an external barcode reader at the end of the line is optional. You can configure the connection to the external barcode readers with the “Configure” button (see below).
- ► Select **Order**, if you want to configure line-based traceability with order correlated data.

No material traceability



WARNING

No material traceability

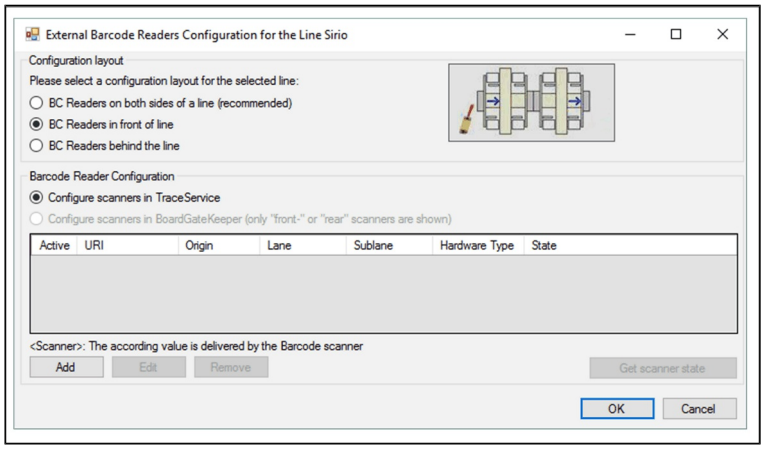
No material traceability data are generated.

This mode serves process data traceability only. The line option **Deliver Process Data** is automatically enabled.

If all lines are configured with this option, no material traceability tests (for example connection to the SIS DB) are performed when the configuration is confirmed.

Configuration of the barcode readers

Go to **Line-based traceability** and click the **Configure** button to open the following screen in which barcode readers can be configured:



Configuration layout

Go to **Configuration layout** and select the position of the external board barcode scanner in the line: **BC readers on both sides of a line (recommended)**, **BC readers in front of line** or **BC readers behind the line**. Greater accuracy of traceability data is achieved with a barcode reader at the rear and front of the line.

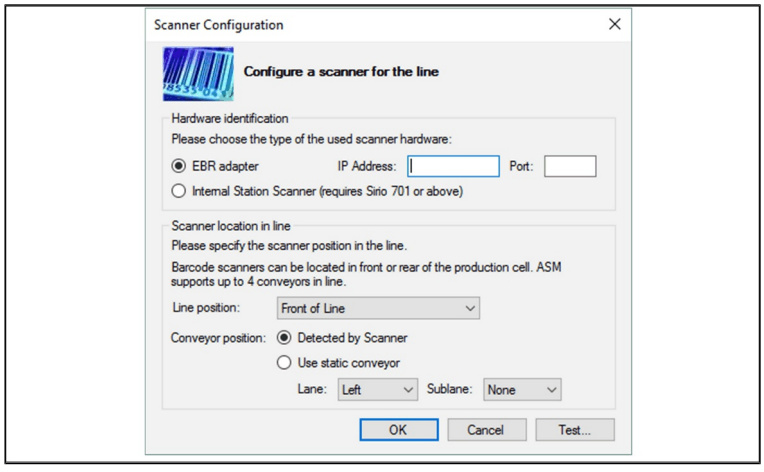
Barcode scanner configuration

In the **Barcode Scanner Configuration** you can specify the maximum number of boards which physically fit into the line. This quantity determines when a board must have left the line (configuration of scanner at front only) or when it was moved into the line (configuration scanner back only). In the mode with a reader both at the front and at the rear of the line, the number of possible boards between the front and the rear reader needs to be calculated. AOI systems or other equipment between the readers, which are part of the board transport, need to be taken into account during this calculation. If there is only one reader configured at the front of the line, you need to specify the number of possible boards between the front barcode reader and the last SIPLACE placement machine. If only one scanner is configured behind the line, you will need to specify the number of possible boards between the first SIPLACE station and the back scanner.

Configuring the scanner in TraceService: the scanner is configured in TraceService itself.

Configuring the scanner in BoardGateKeeper: select this mode if you have installed ASM BoardGateKeeper and you want to configure your scanner there. It is not possible to edit the scanner data in TraceService - this must be done with the BoardGateKeeper Manager tool.

► The table shows a list of the configured board barcode readers. The **Edit** button allows you to change the reader data, while **Remove** deletes reader data. **Get scanner state** checks the connection status to the reader (entry in last column). The **Add** button opens a screen in which reader data can be entered:



There are 2 types of external board barcode readers:

- Select **EBR Adapter** for scanners from other manufacturers. These scanners need to be connected to the SIPLACE line via the relevant interface for the external board barcode reader. This interface needs to be implemented by the customer to connect the scanner systems used in his production. Direct configuration of the hardware is not possible here. For further information about connection of scanner systems, contact your local customer support. Specify the IP address and the scanner port number for this.
- Select **Internal Station Scanner** to operate a SIPLACE board barcode reader as an external reader. This mode is only available for line-based traceability and requires station software 70x.

NOTICE

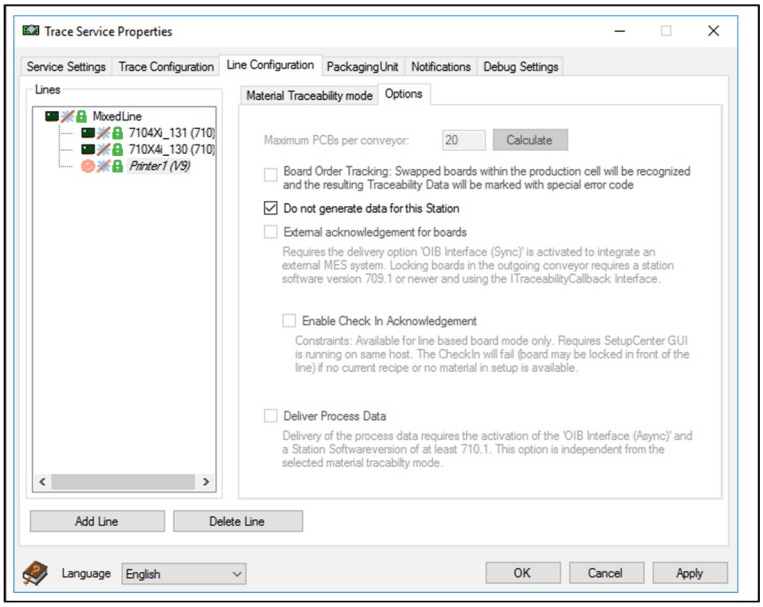
Scanner of the printer with non-line-based Traceability

The scanner of the printer cannot be configured for line-based Traceability. If you want to use the scanner of the printer, you must configure it using BoardGateKeeper.

- Now define the position of the reader in the line:
  - ⇨ Select either **in front of line** or **rear of line**.
  - ⇨ Select whether the conveyor lane is to be automatically detected by the reader (and reported via the interface) or manually enter which conveyor lane is to be used and whether a sublane is to be used (relevant for 4 lane option).

3.3.4.2 Line Configuration - Options

In the **Options** tab, you configure further line specific options.



Maximum PCBs per conveyor:

Click the **Calculate** button to see the proposed maximum number of boards per conveyor track.

Board order tracking

Swapped boards within the production cell will be recognized and the resulting traceability data will be marked with a special error code.

Do not generate data for this Station

No traceability data will be generated for this station and no license will be used.

External confirmation for boards

Allows external MES systems to suppress the further transportation of boards behind the line. Requires the integration of the MES system via the OIB interlocking interface as well as a station software 709.1 and higher.

When activated, an externally connected MES system can decide, based on the traceability event, if the currently produced PCB must be kept in the output conveyor for manual removal or if it can be moved to the next process step. Thus, an MES system can stop the production of a conveyor lane based on customer-specific quality analysis or process situations.

From Traceability 5.2 onwards, station-based traceability allows you to configure this option separately for each station. When using line-based traceability, this option can be separately configured for each printer and for the placement machine as a group.

Activating this option also requires an MES system that is configured and connected via the synchronous OIB interface (ITraceabilityCallback).

Enable Check In Acknowledgement

Allows external MES systems to suppress the further transportation of boards in front of the line. Requires the integration of the MES system via the OIB interlocking interface as well as the connection of barcode scanners in front of and behind the line. This mode is only possible for line-based traceability.

Deliver Process Data

Enables the additional generation and transmission of process data. Requires the integration of the MES system via the asynchronous OIB Eventing interface "IProcessData".

This option additionally requires the "Process Data Interface Machine Connection SIPLACE" license. One license per machine is required.

This option is forced in the **No material traceability** mode.

For further information on process interlocking with Trace Service, refer to the OIB SDK documentation.

3.3.4.3 Configuration Icons of a Line/Station

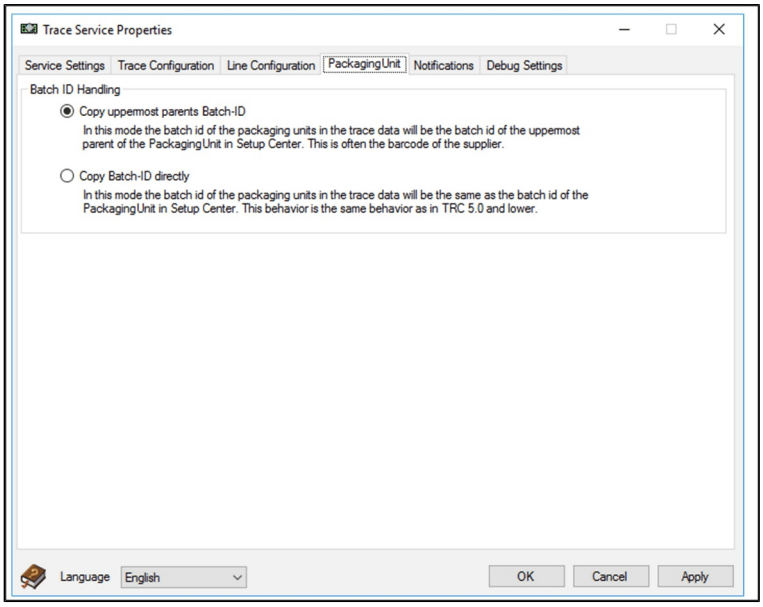
Icon	Meaning
	Line is in station-based material traceability mode.
	Line is in line-based material traceability mode (boards).
	Line is in line-based material traceability mode (jobs).
	No material traceability for this line.
	Process data are generated for this line/station.
	No process data are generated for this line/station.
	Locking enabled for this station/all stations of this line.
	Locking enabled only for some stations of this line.

Icon	Meaning
	Locking disabled for this station/all stations of this line.
	Line is disabled.

3.3.5 Packaging Unit

Here, you specify how the batch ID of the packaging unit is to be handled.

► Go to Trace Service Properties and select the **Packaging Unit** tab.



- **Copy uppermost parents batch ID**
  - In this mode, the batch ID of the packaging units in the trace data will be the batch ID of the uppermost parent of the packaging unit in Setup Center. This is often the barcode of the supplier.
- **Copy batch ID directly**
  - In this mode, the batch ID of the packaging units in the trace data will be the same as he batch ID of the packaging unit in Setup Center. This behavior is the same behavior as in TRC 5.0 and lower.

Example

A packaging unit is separated into packaging units with PUIDs "DMY1", "DMY2" and a batch ID "A123" or a scanned batch ID "SupplierBarcode123".

In Setup Center, the option **Copy the unique ID of the batch packaging unit into the batch field of the packaging unit** is activated under **Automatically individualize packaging units of a batch**.

- Result:

  With the option **Copy batch ID directly**, the following data will be generated:

  PUID= DMY1, batch ID = A123

  PUID= DMY2, batch ID = A123

  With the option **Copy uppermost parents batch ID**, the following data will be generated:

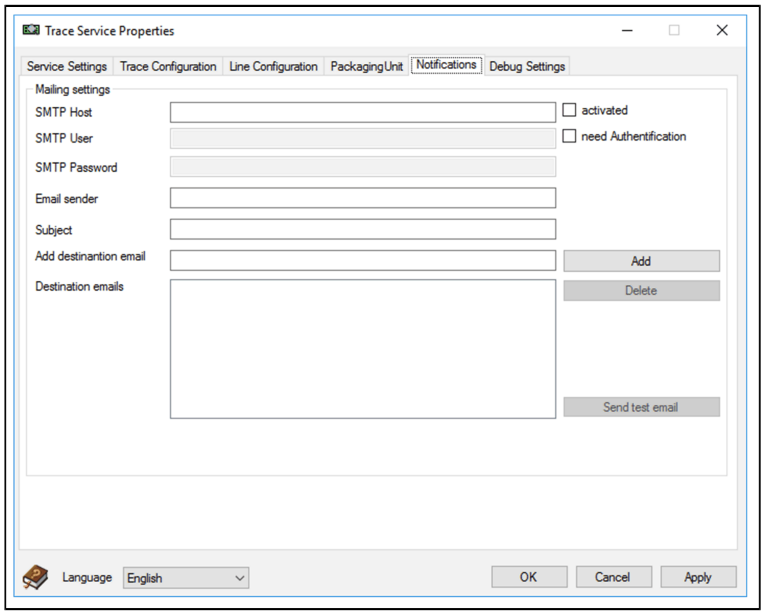
  PUID= DMY1, batch ID = supplierBarcode123

  PUID= DMY2, batch ID = supplierBarcode123

3.3.6 Notifications

► Go to **Trace Service Properties** and select the **Notifications** tab.

Traceability provides a function to send information about defect situations by email. Here you can configure the details.



► **SMTP Host**
Specify the SMTP host for your e-mail services. The “Activated” checkbox enables you to enable or disable an email address without deleting your settings.

► **Authentication**
If you tick this checkbox you can configure the details about the email authentication:

SMTP user

SMTP password

► **E-mail sender**
Enter the sender of the e-mail.

► **Subject**
Enter the text which shall be the subject of the E-mail.

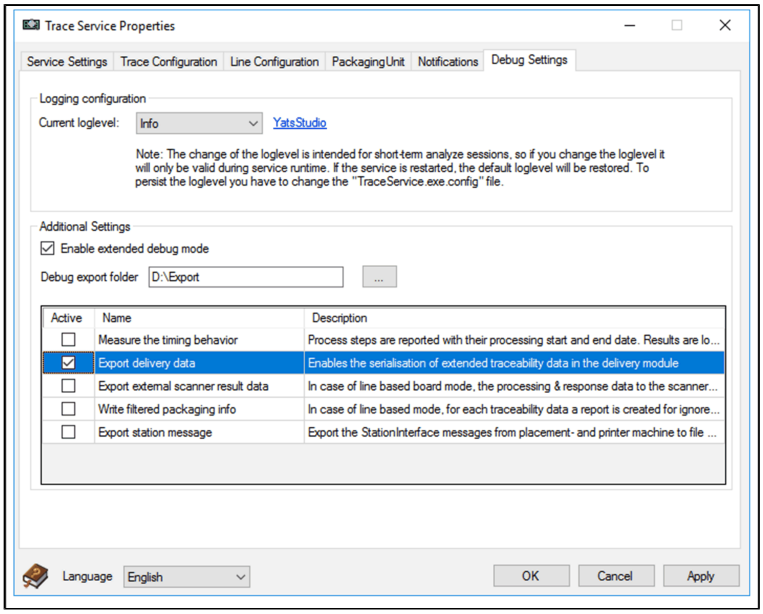
► **Add destination email**
Enter the email address of your destination. Click the button **Add** to add a destination to the list. If you want to delete a destination you can select it in the list and click **Delete**.

► **Destination emails**
Enter the text to be included in each notification e-mail, in addition to the automatically generated error information. Click on **Delete** if you want to delete this field.

► Once you have entered all data, click the **Send test email** button to check whether the e-mail is received by the recipient as required. Always send test e-mails to the address selected in the list of **Destination e-mails**.

3.3.7 Debug Settings

Go to Trace Service Properties and select the **Debug Settings** tab.



Logging Configuration

The following log levels can be preset: **Debug**, **Info**, **Warn**, **Error** and **Fatal**.

The **Debug** level includes all other level and thus is the most exact level. The exactness is constantly reduced from **Debug** level to **Fatal** level resulting in ever smaller sizes of the log files. The user can decide on a suitable level depending on the use case.

If you want to perform short-time analyses, you can adapt the log level here. During the runtime of the service, all its activities and events are recorded. Once the log is finalized, this recording can be visualized in YatsStudio. If you exit the service, the default level is set.

Additional Settings

You can define advanced settings. The data that are generated as a result can be stored in a separate **Debug export folder**.

- **Measure the timing behavior**
  - Process steps are recorded with their start and end time. These data are exported to the **Debug export folder**.
- **Export delivery data:**
  - Allows for serializing advanced traceability data in the delivery module.
- **Export external scanner result data**
  - In the line-based board mode, the production and response data including the time information to the scanner are serialized.
- **Write filtered packaging info**
  - In the line-based board mode, a record for ignored / filtered packaging data is created for each traceability file. This record is written into the default log.
- **Export station message**
  - Export the station interface messages of placement machines and printers to the file system.

4 Trace Monitor

4.1 General

The Trace Monitor is the interface with which the traceability data collected by the Trace Service Server can be visualized.

You can started specific data queries via the Trace Monitor. In doing so, it is possible to filter the search according to various criteria, e.g. according to produced boards, placed components etc. The data displayed can be saved as a CSV file for further processing in MS Excel.

The Trace Monitor is not dependent on the Trace Service Server. The data query performed in Trace Monitor need not be limited to the database used by the Trace Service Server to store processed data. You can switch databases by specifying the database server and name. Provided the database parameters are correct, even the data in the Trace database can be accessed.

NOTICE

ASM Studio

Trace Monitor is started from ASM Studio. For more information about ASM Studio, refer to the documentation or the online help of ASM Studio.

4.2 Starting and Operating the Trace Monitor

- ► Start *ASM Studio*.
- ► In ASM Studio, select the **Trace Monitor** module.

The **Traceability Monitor** view will open:

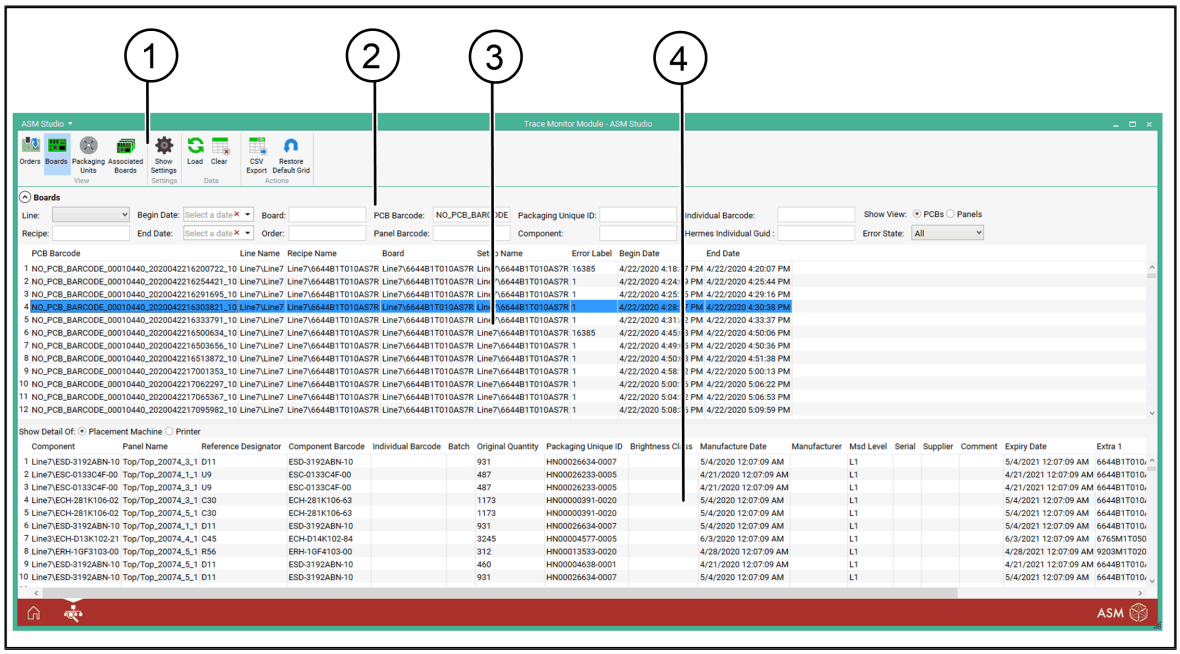


Fig. 1: View - Traceability Monitor

This view is divided into four main areas.

1. Toolbar
2. The filter area with the corresponding settings for each view
3. The results view for the data query
4. The detailed view for the results selected in the results view

Toolbar

You can select the various views, settings, filters and actions from the Trace Monitor toolbar.

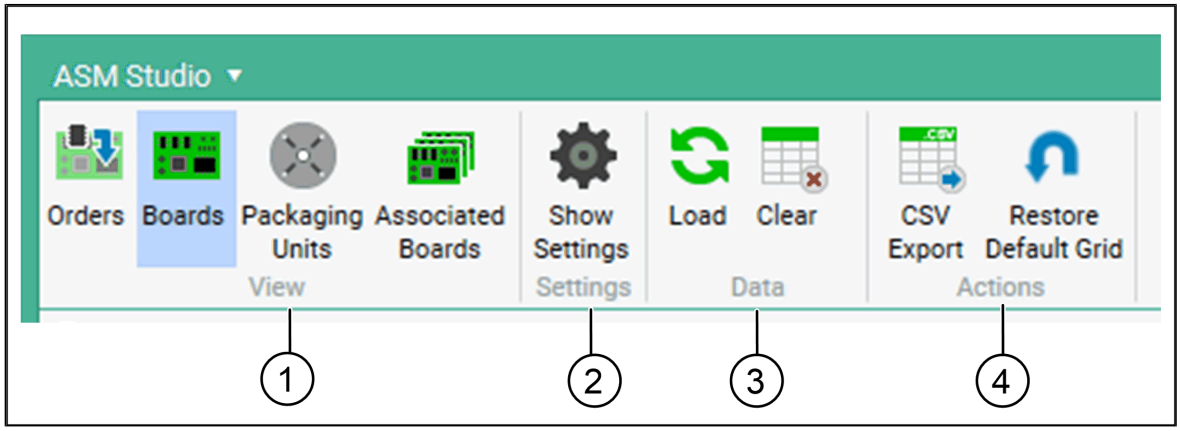


Fig. 2: Toolbar

1. Selecting a view
2. Opening the settings
3. Loading or clearing data
4. Carrying out actions

See also 4.2.1 "Toolbar Functions" [▶ 42].

Filter area

This where you specify the data to be used for the data query e.g. processed boards for a certain period.

The fields in this area differ according to the view selected.

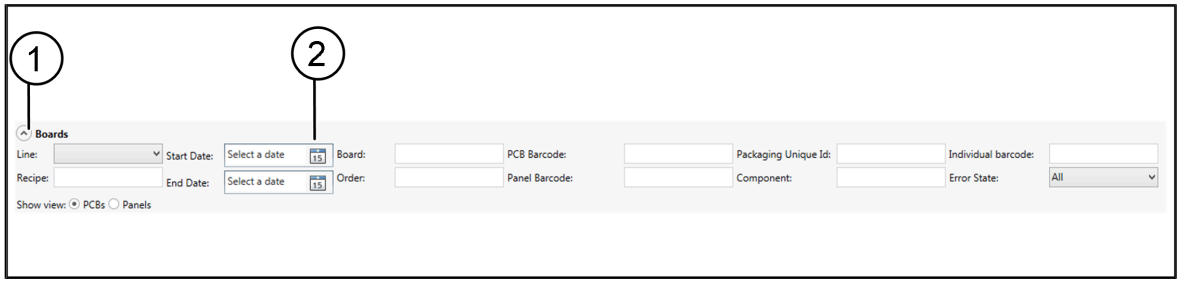


Fig. 3: Filter area - example for boards

1. Showing and hiding the filter area
2. Filter settings with calendar selected

The results view for the data query

This is where the queried data is shown in a table. The information is shown in individual columns, in accordance with the view selected and the current filter settings.

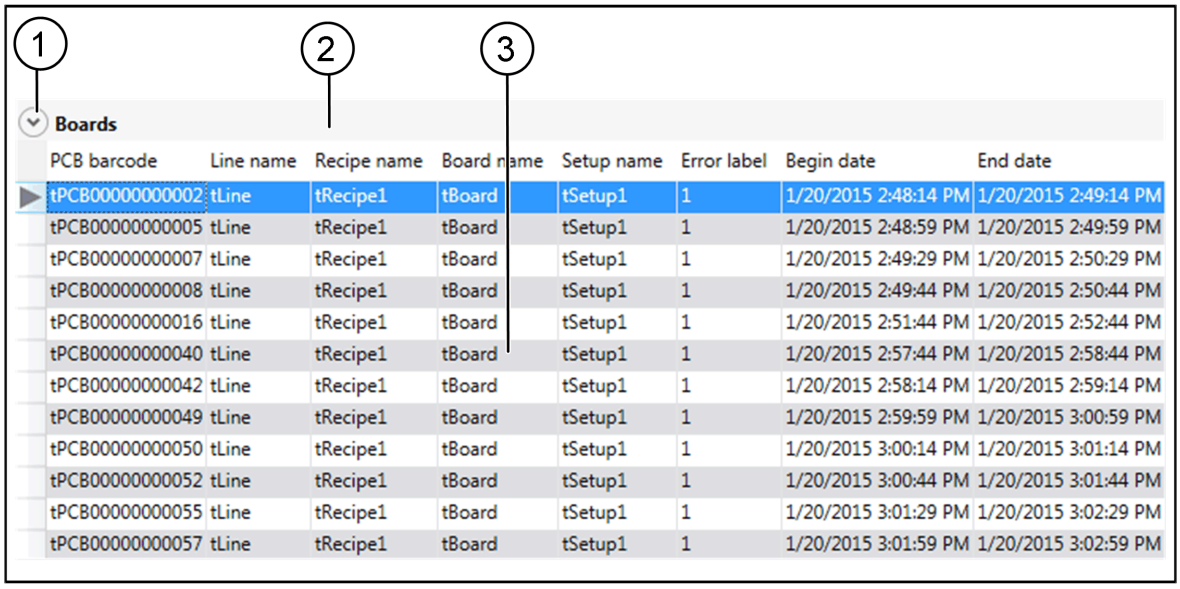


Fig. 4: Results view - example for boards

1. Showing and hiding the results view
2. Information in accordance with filter settings
3. Selection of individual results

Detailed view for selected results

If you mark an individual row in the results display, detailed information about this row will be shown here. The details view is either present or not, depending on which view is currently selected.

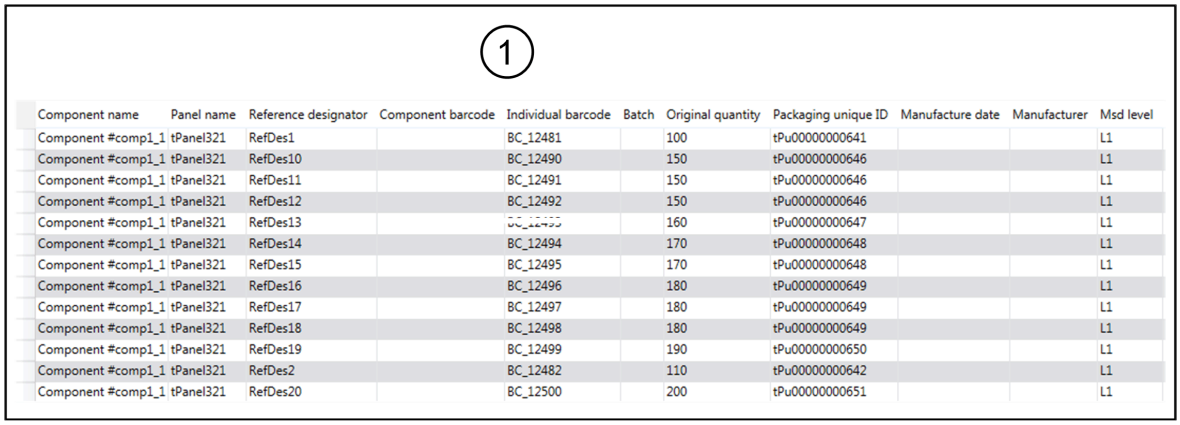


Fig. 5: Details view - example for a selected board

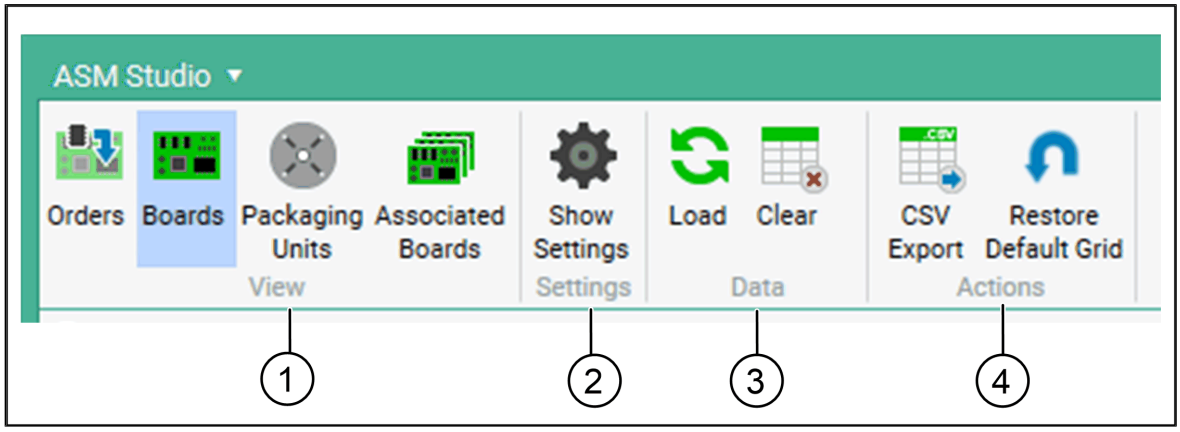
1. Showing and hiding the details view
2. Information about the selected entry in the results view

See also

4.2.1 "Toolbar Functions" [▶ 42]

4.2.1 Toolbar Functions

You can select the various views, settings, filters and actions from the Trace Monitor toolbar.



Views (1):

Icon	Designation	Meaning
	Order number	Data query for general information about the job orders. See also 4.4 "View - Orders" [▶ 45]
	Boards	Data query for general information about the produced boards or panels. Search for boards with details shown for the relevant placement (batch, placement position etc). See also 4.5 "View - PCBs" [▶ 46] See also 4.5.1 "View - Panels" [▶ 49]
	Packaging Units	Search for packaging units based on job and/or board information. See also 4.6 "View - Packaging Unit" [▶ 51]
	Associated PCBs	Search for boards or jobs (for configured job-related traceability only) in a certain period or for boards containing certain components/batches. See also 4.7 "View – Associated PCBs" [▶ 54]

Settings (2)

Icon	Designation	Meaning
	Show Settingsn	The 4.3 "Settings" [▶ 43] box will open. Select the Trace database and define the maximum number of results to be shown.

Data (3)

Icon	Designation	Meaning
	Load	This starts the data query for the current view and in accordance with the specified filter settings. The results will be shown as a table in the results view.
	Delete	This icon deletes the current result in the results view.

Actions (4)

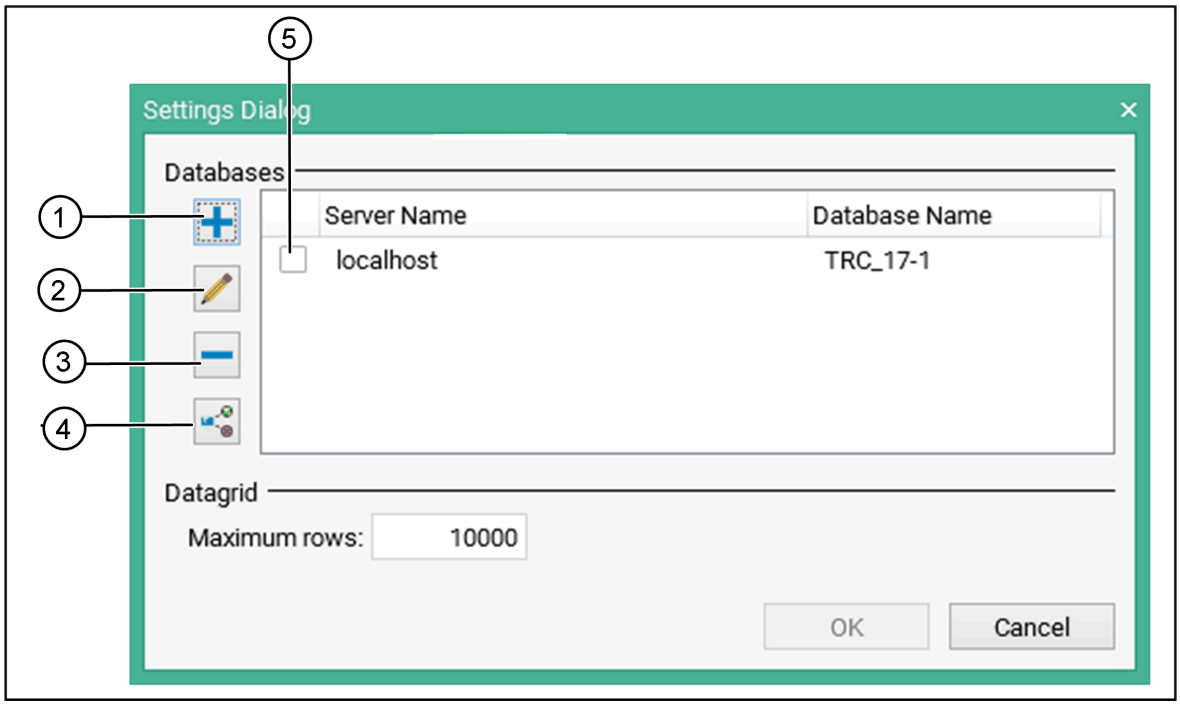
Icon	Function	Meaning
	CSV export	A CSV file is generated for the data shown.

Icon	Function	Meaning
	Restore default grid	If the sorting and order of columns in the tables has been changed, this option can restore the default setting.

4.3 Settings

► Select the icon  from the toolbar.

⇨ The **Settings Dialog** view will open.



Icon	Designation	Description
--	**Databases**	Specify the following database access parameters for the Trace database in which you want to search for data.
--	**Server Name**	The server on which the Trace database is located. Specify "computer name\database entity" here.
--	**Database name**	The name of the database.
1		The 4.3.1 "Showing and Searching the SQL Server" [▶ 44] dialog box will open. You can search for existing servers and databases.
2		The 4.3.1 "Showing and Searching the SQL Server" [▶ 44] dialog box will open. You can change the database.
3		You can terminate the connection with the selected database.
4		You can test the connection with the selected database.
5		You can activate the connection to the selected database. If a database was selected whose "monitor" password is not stored in the password store, the user can do so at a later point in time. A dialog box prompts you to enter the relevant password.
--	**Maximum rows**	In this field you can enter the maximum number of hits which the Trace Monitor should display. The more lines are to be displayed, the longer it takes to display the results.

Icon	Designation	Description
--	**Ok**	Confirm your choice.
--	**Cancel**	This closes the dialog box and your entries will be lost.

4.3.1 Showing and Searching the SQL Server

Enables you to search and change over servers and also to select a database.

NOTICE

Changing databases

You can only switch the Trace database if no database query has been started and no data is shown in the results view.

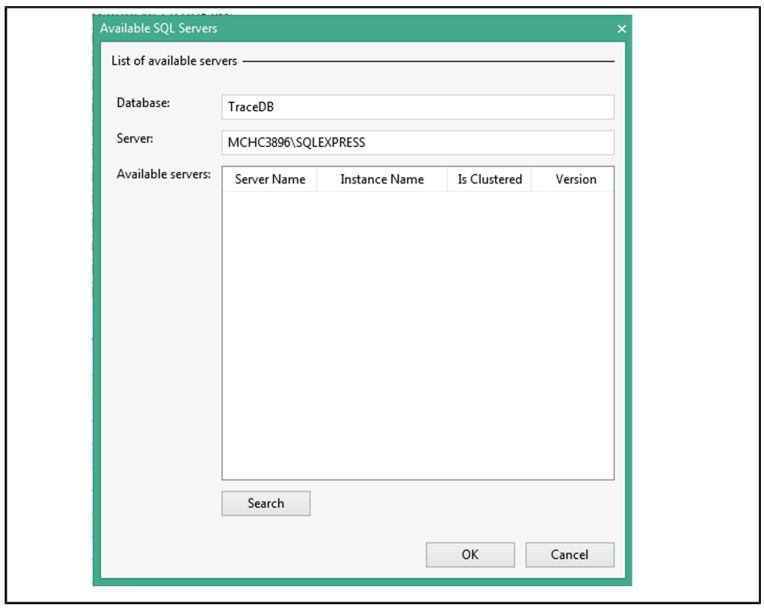


Fig. 6: Dialog box – available SQL servers

Designation	Description
Database	The database currently selected on the server.
Server	The server currently connected.
Available servers	Lists all the available servers found.
Search	Starts the search for all available servers.
Ok	Confirm your choice.
Cancel	This closes the dialog box and your entries will be lost.

4.4 View - Orders

► Select the icon  from the toolbar.

⇨ The **Order number** view will open.

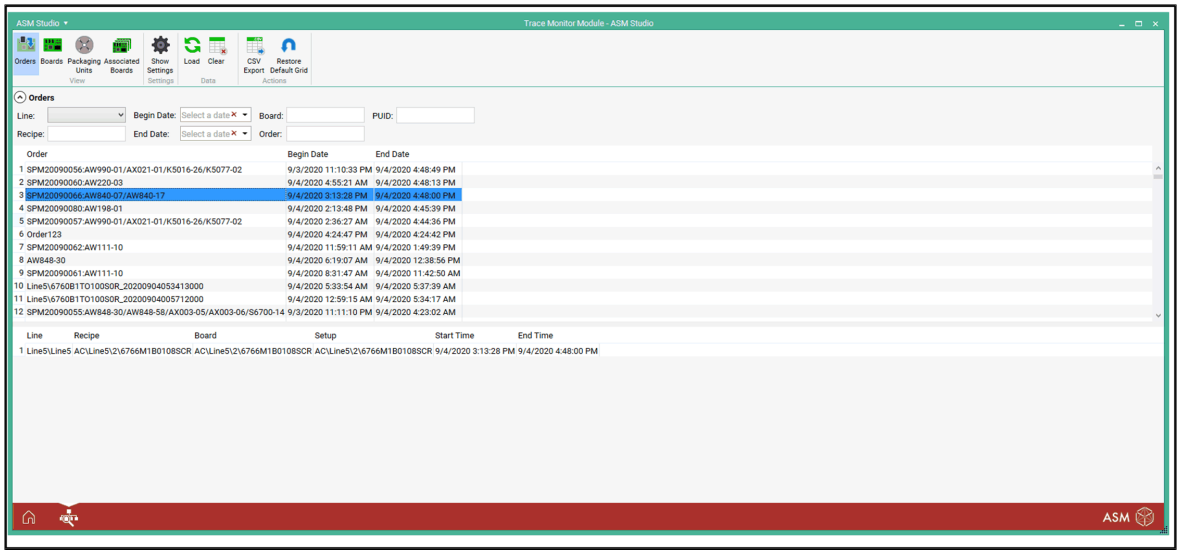


Fig. 7: View - Order number

In this view, you can, for example, view the traceability data of specific orders in a specific period.

The following filter options are available:

Filter	Description
Line	Select the required entry from the drop down menu.
Recipe	If you want to view the traceability data for a certain recipe, specify the recipe here.
Begin date	If you want to view the traceability data for a certain period, specify when the period starts. Click on the calendar icon and select the required date and time.
End date	If you want to view the traceability data for a certain period, specify when the period ends. Click on the calendar icon and select the required date and time.
Board	If you want to view the traceability data for a certain board, specify the board here.
Order	Number of the production job if specified when the job was commissioned.
PUID	If you want to view the traceability data for a certain packaging unit, specify the packaging ID here.
	Once you have entered all data for filtering the view, select the **Load** icon. This starts a data query, in line with the specified criteria. The results will be shown as a table in the results view.
	If you want to delete the result in the table, select the **Clear** icon. This deletes the current result in the table.
	You can export the data in *.csv format.
	If the sorting and order of columns in the tables has been changed, this option can restore the default setting.

Results view:

The following information is shown in this table:

- Order
- Begin date
- End date

► To view more details about a data record, select this specific data record. The data will then be shown in the details view.

Details view

The following information is shown in this table:

- Line
- Recipe
- Board
- Setup
- Start time
- End time

4.5 View - PCBs

► Select the icon  from the toolbar.

⇨ The **Boards** view opens.

► Go to **Show view** and select the **PCBs** option.

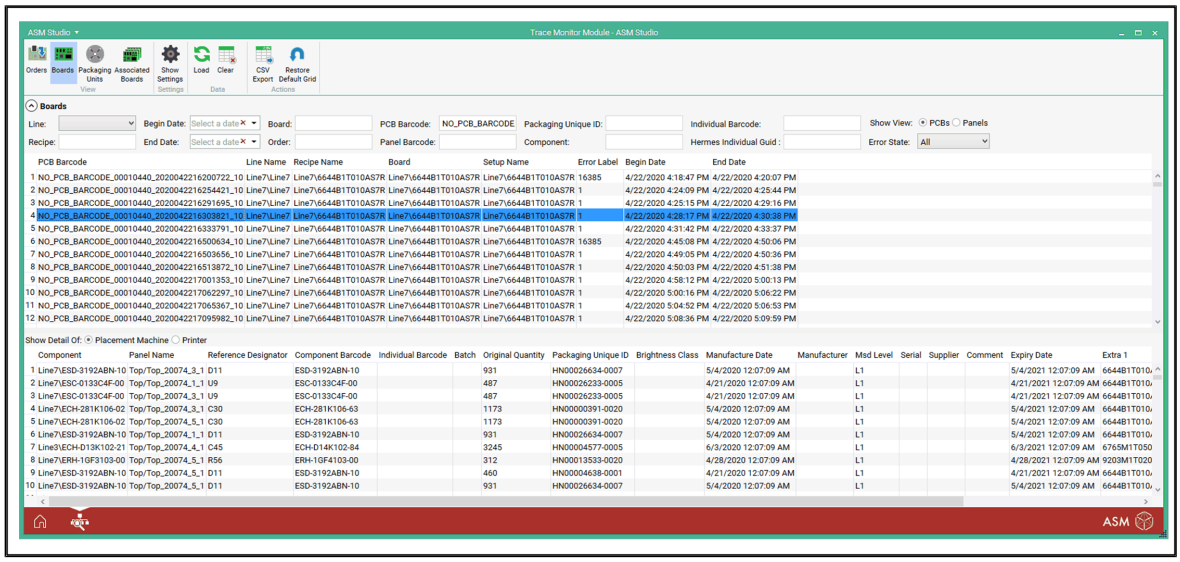


Fig. 8: View - Boards

For example, you can view the traceability data of specific recipes or boards in a specific period, and filter the search additionally by specifying a barcode.

The following filter options are available:

Filter	Description
Line	Select the required entry from the drop down menu.
Recipe	If you want to view the traceability data for a certain recipe, specify the recipe here.
Begin date	If you want to view the traceability data for a certain period, specify when the period starts. Click on the calendar icon and select the required date and time.

Filter	Description
End date	If you want to view the traceability data for a certain period, specify when the period ends. Click on the calendar icon and select the required date and time.
Board	If you want to view the traceability data for a certain board, specify the board here.
Order	Number of the production job if specified when the job was commissioned.
PCB barcode	If you want to view the traceability data for a certain PCB barcode, specify this PCB barcode here.
Panel barcode	If you want to view the traceability data for a certain panel barcode, specify this panel barcode here.
Packaging ID	If you want to view the traceability data for a certain packaging unit, specify the packaging ID here.
Component	If you want to view the traceability data for a certain component, specify the component here.
Individual barcode	Individual barcode that is printed on the component.
Hermes individual Guid	Hermes Individual Guid, created by the first machine in the Hermes communication chain.
Displays	The results view selected between **PCBs** and **Panels**.
Error state	Select the error code from the drop-down menu. **OK** means status "finished" (error code 1) **Not OK** means status "not finished" (error code <> 1). See also 9.1 "Traceability Data with Error Code >1" [▶ 109]
	Once you have entered all data for filtering the view, select the **Load** icon. This starts a data query, in line with the specified criteria. The results will be shown as a table in the results view.
	If you want to delete the result in the table, select the **Clear** icon. This deletes the current result in the table.
	You can export the data in *.csv format.
	If the sorting and order of columns in the tables has been changed, this option can restore the default setting.

Results view:

The following information is shown in this table:

- PCB barcode
- Line name
- Recipe name
- Board
- Setup name
- Error text
- Begin date
- End date

► To view more details about a data record, select this specific data record. The data will then be shown in the details view.

Show Detail Of

Select the relevant data for the **Placement machine** or **Printer**.

The following information can be found in this table for the **Placement machine**:

- Component name
- Panel name
- Reference designator
- Component barcode
- Individual barcode
- Batch
- Original quantity
- Packaging ID
- Brightness class
- Manufacture date
- Manufacturer
- MSD level
- Serial
- Supplier
- Comment
- Expiry date
- Start time
- End time

The following information can be found in this table for the **Printer**:

- Material type
- Barcode

If Hermes parameters are available, these will also be shown. The following information can be seen in this **Hermes Parameter** table:

- Individual Board GUID
- GUID Creator
- Failed State
- Flipped Board
- Product Type ID
- Top Barcode
- Bottom Barcode

4.5.1 View - Panels

► Select the icon  from the toolbar.

⇨ The **Boards** view opens.

► Go to **Show view** and select the **Panels** option.

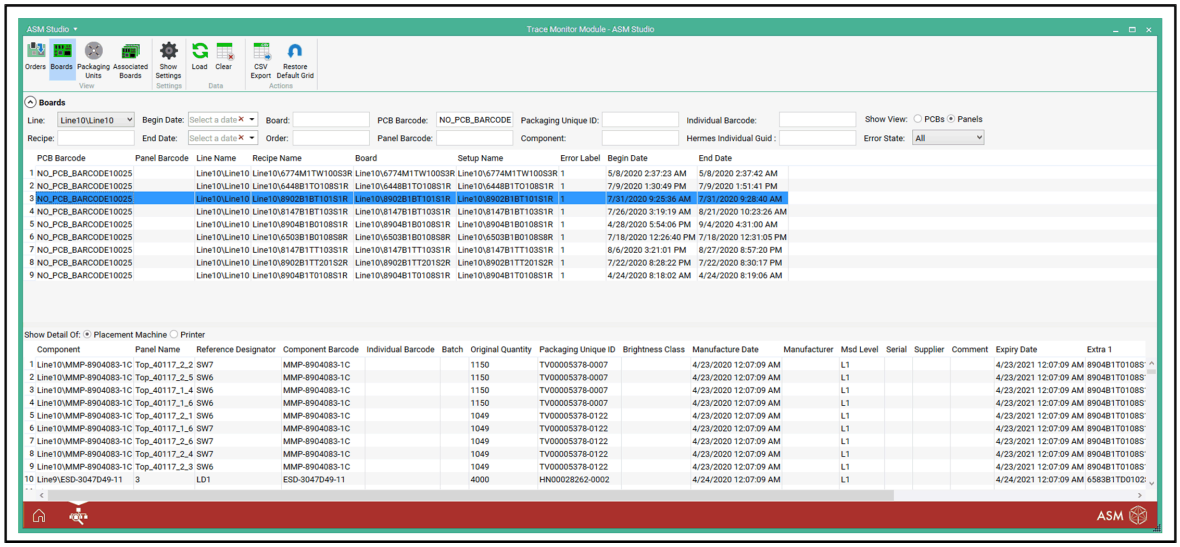


Fig. 9: View - Panels

For example, you can view the traceability data of specific recipes or boards in a specific period, and filter the search additionally by specifying a barcode.

The following filter options are available:

Filter	Description
Line	Select the required entry from the drop down menu.
Recipe	If you want to view the traceability data for a certain recipe, specify the recipe here.
Begin date	If you want to view the traceability data for a certain period, specify when the period starts. Click on the calendar icon and select the required date and time.
End date	If you want to view the traceability data for a certain period, specify when the period ends. Click on the calendar icon and select the required date and time.
Board	If you want to view the traceability data for a certain board, specify the board here.
Order	Number of the production job if specified when the job was commissioned.
PCB barcode	If you want to view the traceability data for a certain PCB barcode, specify this PCB barcode here.
Panel barcode	If you want to view the traceability data for a certain panel barcode, specify this panel barcode here.
Packaging Units	If you want to view the traceability data for a certain packaging unit, specify the packaging ID here.
Component	If you want to view the traceability data for a certain component, specify the component here.
Individual barcode	Individual barcode that is printed on the component.

Filter	Description
Error state	Select the error code from the drop-down menu. **OK** means status "finished" (error code 1) **Not OK** means status "not finished" (error code <> 1). See also 9.1 "Traceability Data with Error Code >1" [▶ 109]
Displays	The results view selected between **PCBs** and **Panels**.
	Once you have entered all data for filtering the view, select the **Load** icon. This starts a data query, in line with the specified criteria. The results will be shown as a table in the results view.
	If you want to delete the result in the table, select the **Clear** icon. This deletes the current result in the table.
	You can export the data in *.csv format.
	If the sorting and order of columns in the tables has been changed, this option can restore the default setting.

Results view:

The following information is shown in this table:

- Panel barcode
- Line name
- Recipe name
- Board name
- Setup name
- Error label
- Start time
- End time
- Navigation

► To view more details about a data record, select this specific data record. The data will then be shown in the details view.

Details view

The following information is shown in this table:

- Component name
- Panel name
- Reference designator
- Component barcode
- Individual barcode
- Batch
- Original quantity
- Packaging ID
- Manufacture date
- Manufacturer
- MSD level
- Serial
- Supplier
- Comment
- Expiry date

- Start time
- End time

4.6 View - Packaging Unit

► Select the icon  from the toolbar.

⇨ The **Packaging** view will open.

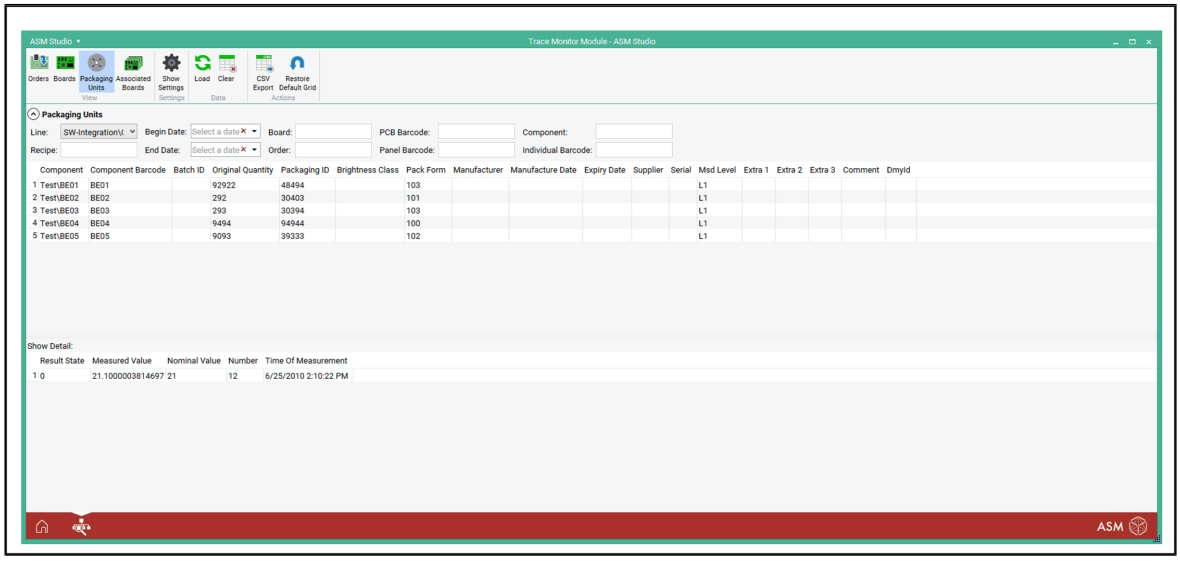


Fig. 10: View - packaging unit

For example, you can view the traceability data of specific recipes or boards in a specific period, and filter the search additionally by specifying a barcode.

The following filter options are available:

Filter	Description
Line	Select the required entry from the drop down menu.
Recipe	If you want to view the traceability data for a certain recipe, specify the recipe here.
Begin date	If you want to view the traceability data for a certain period, specify when the period starts. Click on the calendar icon and select the required date and time.
End date	If you want to view the traceability data for a certain period, specify when the period ends. Click on the calendar icon and select the required date and time.
Board	If you want to view the traceability data for a certain board, specify the board here.
Order	Number of the production job if specified when the job was commissioned.
PCB barcode	If you want to view the traceability data for a certain PCB barcode, specify this PCB barcode here.
Panel barcode	If you want to view the traceability data for a certain panel barcode, specify this panel barcode here.
Component	If you want to view the traceability data for a certain component, specify the component here.
Individual barcode	Individual barcode that is printed on the component.

Filter	Description
	Once you have entered all data for filtering the view, select the **Load** icon. This starts a data query, in line with the specified criteria. The results will be shown as a table in the results view.
	If you want to delete the result in the table, select the **Clear** icon. This deletes the current result in the table.
	You can export the data in *.csv format.
	If the sorting and order of columns in the tables has been changed, this option can restore the default setting.

Results view:

The following information is shown in this table:

- Component name
- Component barcode
- Batch ID
- Original quantity
- Packaging ID
- Brightness class
- Pack from
- Manufacturer
- Expiry date
- Supplier
- Serial
- MSD level
- Extra1
- Extra 2
- Extra 3
- Comment
- Bin code
- Map origin
- Notch orientation
- Map orientation
- Pickup position X
- Pickup position Y



NOTICE

Automatically separated packaging unit batches

If packaging unit batches are automatically separated (e.g. in the case of waffle pack trays which come from a package with multiple waffle pack trays), the Setup Center will create artificial dummy IDs for the individual waffle pack trays. These dummy IDs are located in the **Gebinde ID** field, the IDs for the whole package are in the **Batch ID**. If the customer searches the Traceability data for this package, he will need to search with the ID in the **Batch ID** and not in the **Gebinde ID**, as in the case of roles.

Details view

The following information is shown in this table:

- Status

- Measured Value
- Nominal Value
- Number
- Time Of Measurement

4.7 View – Associated PCBs

► Select the icon  from the toolbar.

⇨ The **Associated PCBs** view will open.

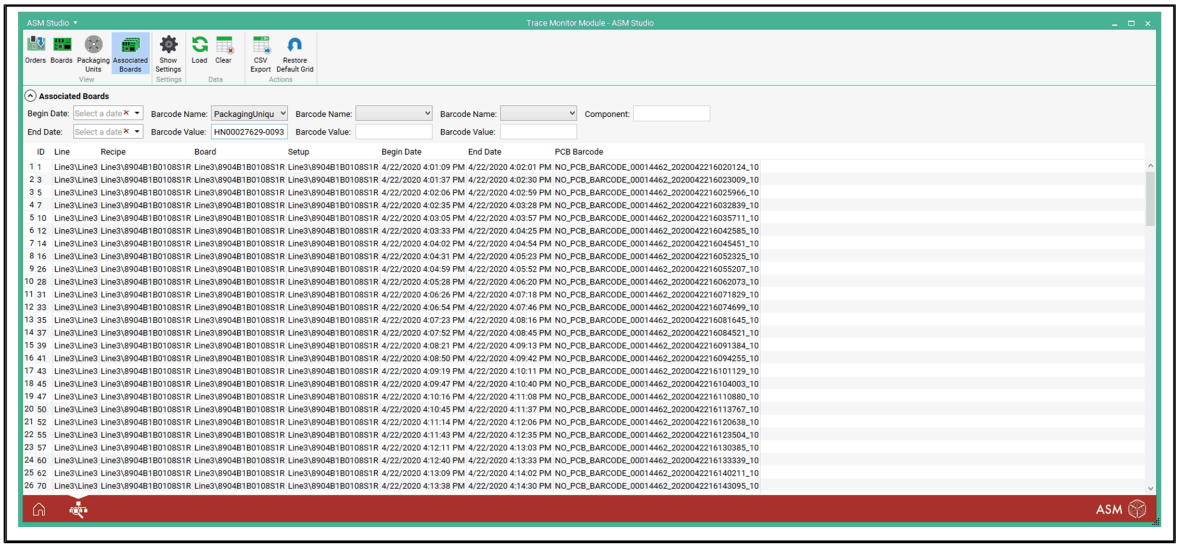


Fig. 11: View – Associated PCBs

In this view you can start the data query to find a specific component with specific barcodes. You can also view further details about the data queries which you executed in the **Packaging Unit** view.

The following filter options are available:

Filter	Description
Begin date	If you want to view the traceability data for a certain period, specify when the period starts. Click on the calendar icon and select the required date and time.
End date	If you want to view the traceability data for a certain period, specify when the period ends. Click on the calendar icon and select the required date and time.
Barcode Name	Use the drop down menu to select the barcode name for filtering the data query.
Barcode Value	This field is activated after you have selected a barcode name in the "Barcode Name" field.
Component	Specify the component name for filtering the data query.
	Once you have entered all data for filtering the view, select the **Load** icon. This starts a data query, in line with the specified criteria. The results will be shown as a table in the results view.
	If you want to delete the result in the table, select the **Clear** icon. This deletes the current result in the table.
	You can export the data in *.csv format.
	If the sorting and order of columns in the tables has been changed, this option can restore the default setting.

Results view:

The following information is shown in this table:

- Order

- Line
- Recipe
- Board
- Setup
- Start time
- End time
- PCB barcode
- Navigation

Details view

There is no detailed view in the **Associated PCBs** view.

4.8 Regular Expressions for Advanced Searches

Regular expressions are used for advance barcode searches during data queries in Trace Monitor. The key characters for barcode searches are explained below:

Syntax

Character	Description
%	Any character string with zero or more characters
?	Each individual character
[ ]	Each individual character in the prescribed area ([a-f]) or the quantity ([abcdef])
[^]	Each individual character outside the prescribed area ([^a-f]) or the quantity ([^abcdef])

Examples:

- Search for PCB barcodes "**%CODE%**" Finds all PCB barcodes with the word "CODE" somewhere in the barcode.
- Search for PCB barcodes "**?CODE**" Finds all barcodes which begin with any character and which end with the word "CODE" e.g. "1CODE", "ACODE" etc.
- Search for PCB barcodes "**[A-P]UK123**" Search for PCB barcodes "[A-P]UK123" finds all PCB barcodes which end with "UK123" and which begin with any character between "A" and "P" e.g. "AUK123 ", "BUK123" or "CUK123".
- Search for PCB barcodes "**[A1]234**" finds all PCB barcodes which end with "234" and begin with "A" or "1".
- Search for PCB barcodes "**de[^l]%**" finds all PCB barcodes which begin with "de" and for which the following character is not an "l".

5 Basic settings for working with traceability

This chapter provides detailed information about the settings to be defined in SIPLACE Pro and in the station software, in order for traceability to function. If you are using DEK software, you also need to define these settings.

5.1 Overview of the traceability licenses

The license which you need to install and register will depend on your individual scenario and the circumstances in your company. There are four different license types available:

Traceability (machine connection)

You need the relevant **Traceability machine connection** for each station in the line, for each scenario. Select the **Traceability (machine connection)** licenses that you want to use for the respective scenario.

Traceability server

You always need a **Traceability Server** license for each company, in order to operate the Traceability software.

Barcode with PCB camera (machine connection)

Scenarios in which the barcode is read in with the PCB camera require that you also have a **Barcode with PCB camera (machine connection)** license for each station in the line. Select the **Barcode with PCB camera (machine connection)** licenses which you want to use for the respective scenario.

Whispering Down the Line (WdtL)

Scenarios in which Whispering Down the Line (WdtL) is performed also require that you have one of the **Whispering Down the Line (WdtL)** licenses. Select the **Whispering Down the Line (WdtL)** licenses which you want to use for the respective scenario.

MES (machine connection)

Scenarios in which MES is to be used require that you also have one of the **MES (machine connection)** licenses for each station in the line. Select the MES (machine connection) licenses which you want to use for the respective scenarios.

Overview of license types and individual licenses

License	Item no.	Traceability (machine connection)	Traceability server	Barcode with PCB camera (machine connection)	Whispering Down the Line (WdtL)	MES (machine connection)
Barcode WdtL (MC) V1.0	116292-xx				X	
WDTL/WDTS inkspot 70x	116285-xx				X	
Reading barcode with PCB camera (MC)	116295-xx			X		
Traceability with PCB camera (MC) V1.0	116195-xx	X		X		
Line operations machine connection	116525-xx			X		
DEK traceability station connection V5.x	117315-xx	X				
SIPLACE traceability station connection	169578-xx	X				
Traceability basic license V5.x	117312-xx		X			
MES process interlocking SIPLACE MC	117946-xx					X
MES process interlocking DEK MC	117948-xx					X
SIPLACE Traceability with MES PI (MC)	117945-xx	X				X
DEK Traceability with MES PI (MC)	117947-xx	X				X
SIPLACE TRC, MES PI and PCB camera (MC)	117949-xx	X			X	X

5.2 Settings in SIPLACE Pro

The following settings can be defined in SIPLACE Pro (for detailed information about the settings, refer to the current SIPLACE Pro Online Help).

NOTICE

Station software 407

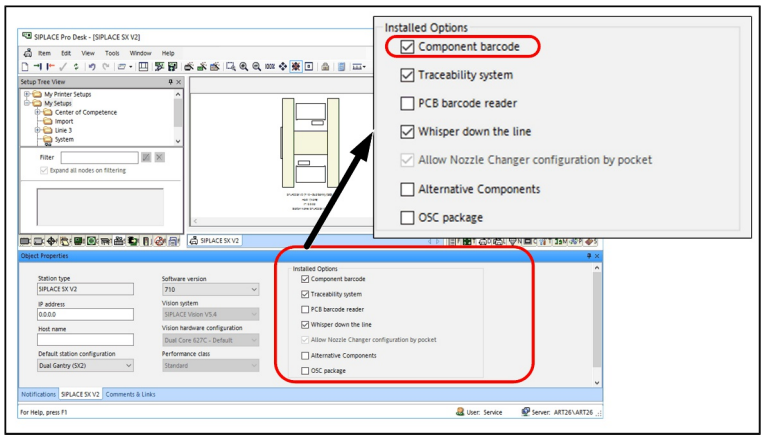
Automatically applied settings cannot be made to the programming system in the traceability option 407.

► For detailed information about the settings for station software 407, refer to section 6.3 "Station software settings" [▶ 67].

5.2.1 Enabling component barcodes at the station

In order to operate traceability, you need to enable the **Component barcode** option in the SIPLACE Pro Station Editor. Make sure that the barcode reader is parameterized in the programming system.

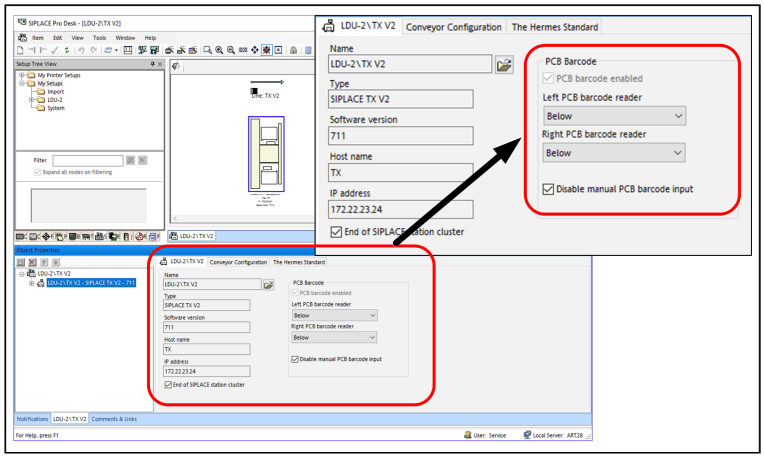
- ► Select the stations in the tree view **Devices of line** => **My stations**.
- ► Go to **Installed options** and enable the option **Component barcode**.



5.2.2 Enabling the board barcode reader (optional)

If you want to use SIPLACE barcode readers, you have to configure these in all stations.

- ► Open the Line Editor.



- ► Select the stations in the line.
- ► Go to **PCB barcode** and configure the barcode reader(s) for the stations.

Manual PCB barcode entry

- Stations with station software 40x, 50x and 60x:
  - On these stations, the manual barcode entry is activated or deactivated via the traceability installation medium. It is not possible in SIPLACE Pro.
- Stations with station software 7xx:

On 7xx stations, the manual barcode entry is enabled or disable in the SIPLACE Pro Line Editor at **PCB barcode** => **Disable manual PCB barcode input**.

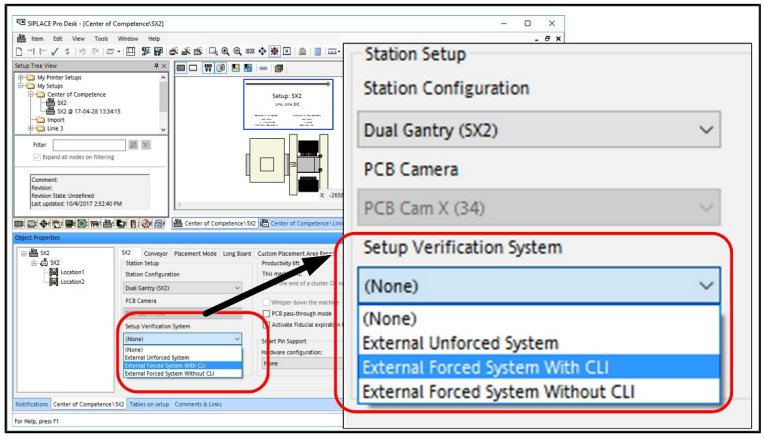
If a barcode cannot be read, a dialog is displayed at the station, offering the following choices:

- Entering the barcode manually
- Placement of PCB without barcode
- Abort PCB

5.2.3 Selecting the setup verification system for the station

In the target setup, select the installed setup verification system for each station setup.

- ► Open the Setup Editor for the target setup.
- ► Go to the **Object properties** for the target setup and select the **Station configuration**.
- ► Go to **Station configuration** => **Setup verification system** and select the option **External forced system with CLI**.



Setup verification system	Meaning
External Forced Setup Verification with CLI	ASM Setup Center with Component Level Indicator; i.e. with quantity barcode, splice sensor available as option.

NOTICE

Setup verification

- ► As station-based traceability only works together with forced setup verification, select the setup verification system **External forced system with CLI**when using station-based traceability.
- ► For line-based traceability, you can also choose unforced setup verification (**External unforced system**). However, in this case, the reliability of the traceability data depends on how exactly you execute the unforced setup verification.

5.2.4 Setting the feeder pitch (except 407)

The feeder pitch represents the step size for the feeder. The component level indicator of the station uses this feeder pitch, for example, to calculate how many components there are between a detected splice and the pickup position.

The feeder pitch can be set as follows:

- In SIPLACE Pro for a special component shape
- In SIPLACE Pro feeder type specific for a special component shape
- In SIPLACE Pro for a special component
- In SIPLACE Pro feeder type specific for a special component
- Directly at the feeder

The settings specified later on overwrite the settings specified earlier, in each case.

For more information about the settings options, refer to the product documentation for SIPLACE Pro and the X feeders.

5.2.5 Supplier Barcodes

If the component name and the corresponding barcode are identical, it is not necessary to make an entry in the barcode list. An entry is only needed if the component name differs from the supplier barcode.

NOTICE

Barcode list

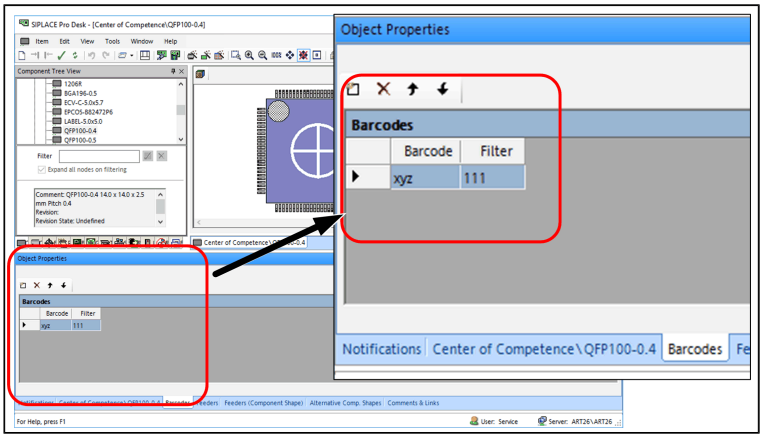
When using SIPLACE machines with station software 407, you need to enter a barcode list for the verified tracks into the programming system.

NOTICE

Supplier barcodes and alternative barcodes

Supplier barcodes can be used to offer alternative component barcodes when recording packaging units. They do NOT offer the option of defining alternative components for a component barcode.

- ► Open the Component Editor.
- ► Select the **Barcodes** tab and enter the relevant barcode(s).



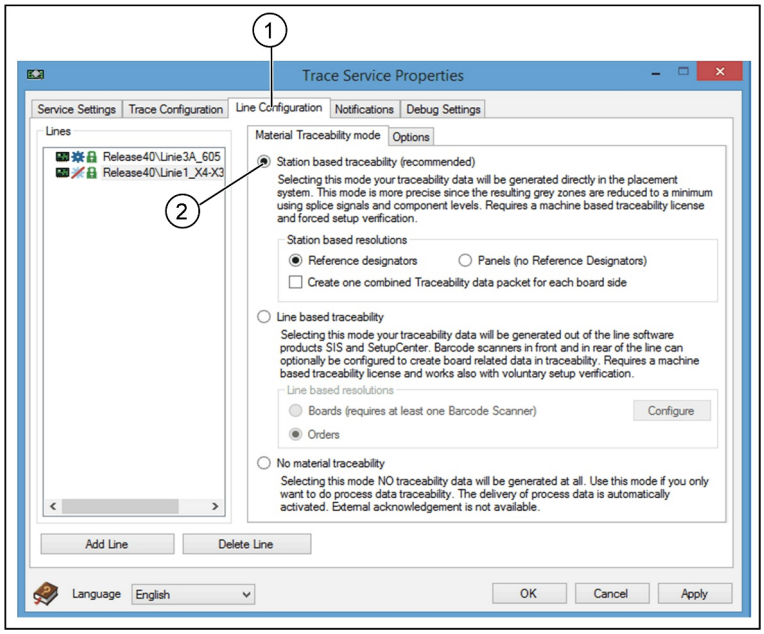
The barcode and the matching filter are defined here. The filter specifies which characters are to be evaluated (0=ignore / 1=character must be present).

6 Station-based traceability

6.1 Trace service settings

In order to run station-based traceability you need to enable this function in Trace Service.

► Start **Trace Service** and select **Configure…**



► Select the **Line configuration** (1) tab.

► Enable the option field **Station based traceability (recommended)** (2).

► For more settings, refer to chapter 3 "Trace Service" [▶ 21].

6.2 Settings in SIPLACE Pro

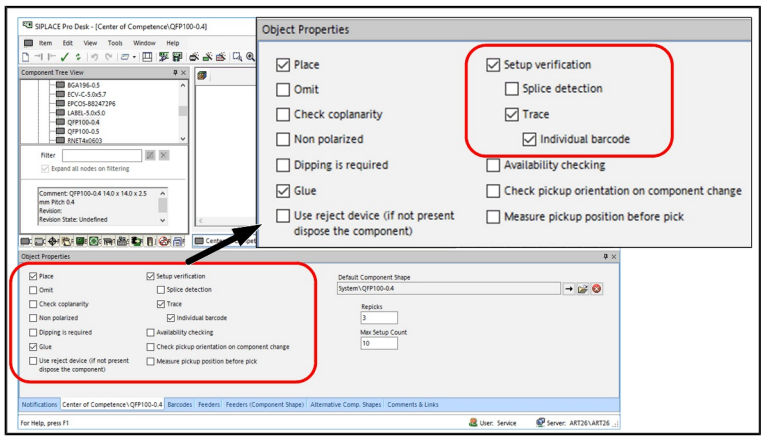
6.2.1 Component Editor

► Enable the setup verification for each component.

This setting specifies whether a component is subject to component verification. Setup verification must be activated separately for each component.

► Open the Component Editor.

► Go to the **Component** tab and enable the option **Setup verification** for the selected component.



Splice detection (except for 407)

This setting demands a splice detection sensor for the component concerned. If the auto detect function identifies that there is no splice detector (hardware) on the track concerned, the machine stops and an error message is issued.

NOTICE

Splice detection on critical tracks

This setting merely represents an additional safety check in order to force the use of a splice detector on critical tracks. If the option is not marked, the track concerned can still be operated with a splice detector. In this case, if no splice detection hardware is detected, the track is automatically monitored by the component level indicator.

Traceability

This setting specifies whether a component is relevant for capturing the traceability data. This allows non-critical components to be excluded, thus reducing the volume of data.

Tracks for which traceability is activated are always subject to setup verification.

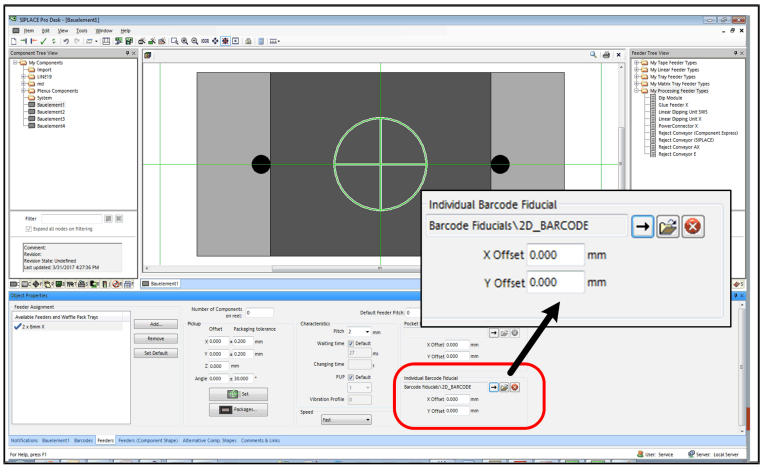
NOTICE

Station-based traceability

This setting is only necessary for station-based traceability.

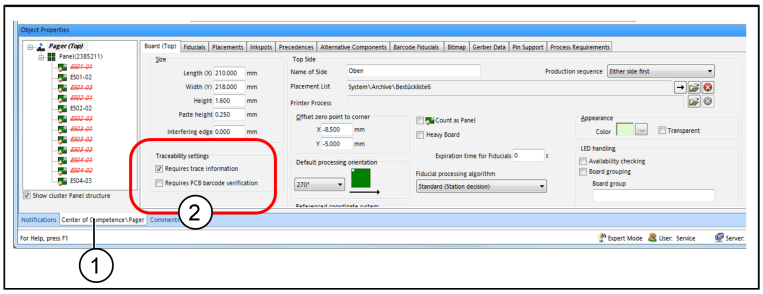
Traceability with component barcode

The **Individual Barcode** setting indicates that, for this component, the barcode for the traceability data is also relevant. Additionally, you have to set the **Individual Barcode Fiducial** in the **Feeder** tab for the component.



6.2.2 Board Editor

► Open the board in the SIPLACE Pro Board Editor (1).

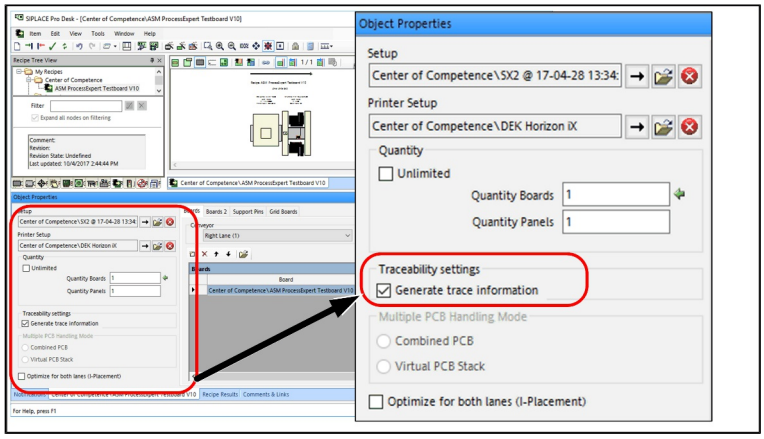


The following settings are required for this:

► Go to **Traceability settings** (2) and enable the option **Requires trace information**.

6.2.3 Recipe Editor

- ► Open the Recipe Editor.
- ► Go to **Setup** => **Traceability settings** and enable the option **Generate trace information**. This option must be enabled separately for each recipe. The boards for this recipe will be manufactured with the option "traceability".



NOTICE

Track information required

If the option **Requires trace information** is enabled for the board contained in the Board editor, this option will also be automatically enabled in the Recipe Editor for job optimization.

6.3 Station software settings

NOTICE

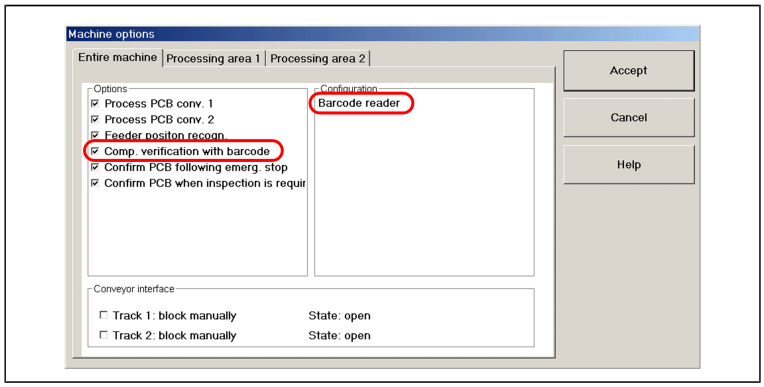
Station software

All settings described in the following chapter are only required for station-based traceability and only for the station versions 40x, 50x, 60x.

► No specific settings are required at the station for station software 70x.

6.3.1 Settings in the Machine Options

In order to operate station-based traceability or setup verification, you need to enable the setting "Setup verification with barcode" in the machine options. In order to do this, the operator level **Line Engineer** must be set.



NOTICE

New installation of station software

After a new installation of the station software, forced verification will always be disabled.

► Please activate it manually.

NOTICE

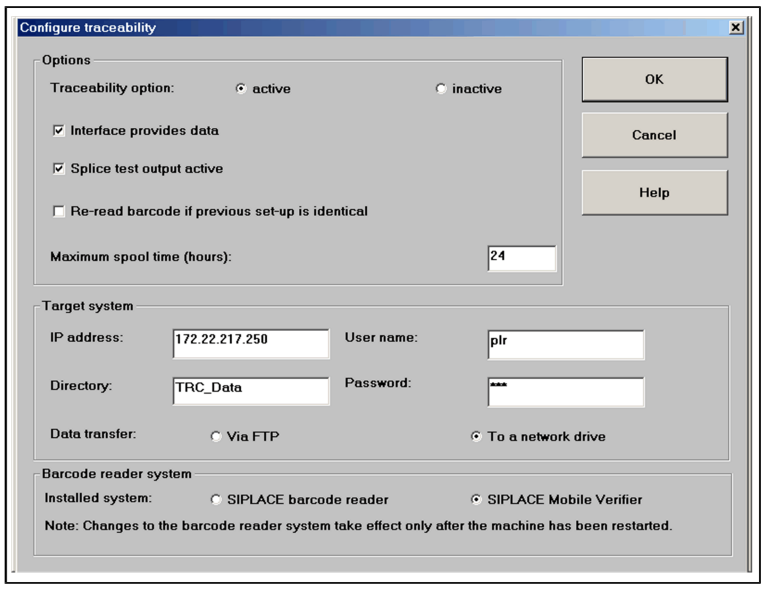
Checking the setup verification settings

The current version of Setup Center can not detect whether forced setup verification is switched on or off for the machine.
For this reason it is always necessary to ensure manually that the settings for setup verification are the same at the station and in Setup Center!

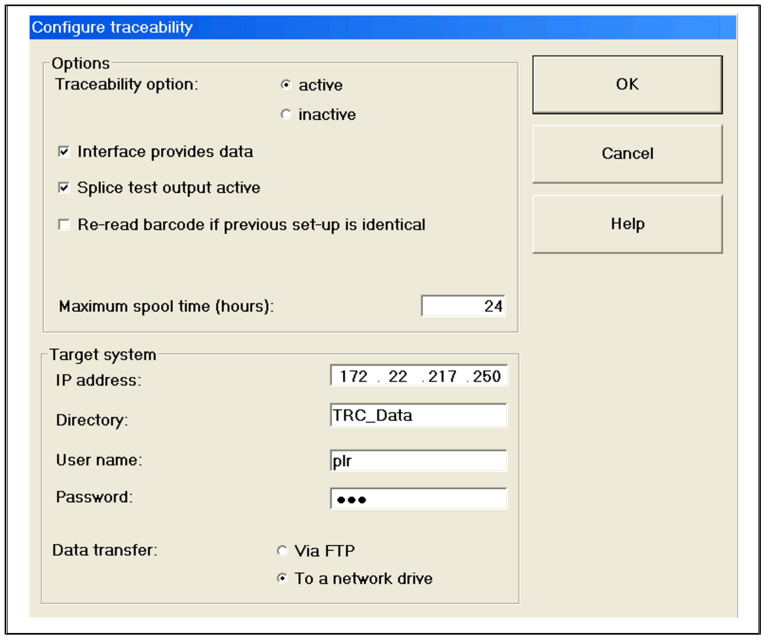
6.3.2 Settings in the 'Configure Traceability' Menu

Station-based traceability settings can be defined in the "Options - configure traceability" menu of the SIPLACE user interface:

For station software 407:



For station software 505:



For station software 60X

The traceability configuration window is identical to that for 505. The only change is that the option **Re-read barcode if previous setup is identical** is no longer present in the user interface.

6.3.2.1 Traceability Option Active/Inactive

The traceability function can be activated or deactivated. Deactivation works immediately and can, for example, make it possible to continue production if technical problems (e.g. spool full) occur.

Activation only takes effect after the machine has been rebooted. You will be asked to restart the machine, if you enable this option.

6.3.2.2 Interface Provides Data

This setting enables you to freeze the standard traceability interface, while you are still recording traceability data.

The deactivation of the traceability data interface is needed if traceability data is to be made available via the GEM interface only. Traceability data should not be generated, to prevent the file system from running full.

6.3.2.3 Splice Test Output Active

If the test output is enabled, an operator message will be issued at the user interface for each splice point detected (Error menu - Track errors). This enables you to check the function of the splice sensor.

6.3.2.4 Re-Read Barcode if Previous Setup is Identical

This parameter specifies whether packaging unit information from an identical predecessor setup is to be adopted or not. This helps you to avoid superfluous scan runs but can affect the safety. The system assumes that the same packaging unit is still present at all feeders.

The barcodes for an identical predecessor setup may only be automatically accepted if you are sure that no changes have been made to the setup.

NOTICE

Configuring the setup verification settings

If the setup verification is performed by ASM Setup Center, this setting must only be configured in Setup Center (delta setup).

► For detailed information, refer to the ASM Setup Center User Guide.

6.3.2.5 Setting the Connection to the Target System - Spool Mechanism

The traceability data must be stored in a target system. You can specify whether the data is to be transferred via FTP or via a shared network drive.

The target folder must be shared. Enter the name of the shared folder in the "Folder" field.

If the connection to the target system can be established and if traceability is active, the Traceability icon on the user interface will be green. If a connection cannot be established, the data will be stored locally, in the spool folder (C:\traceability\tracespool).

As soon as the connection can be established, the data will be transferred from the spool folder to the target folder.

If the maximum spool time is exceeded, the tracks relevant for traceability will be

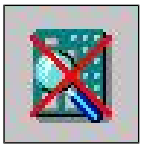
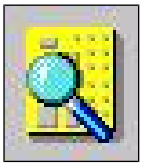
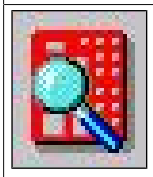
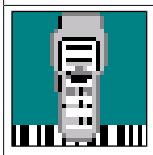
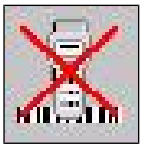
disabled. The machine will stop. You now need to establish a connection to the target system

or disable Traceability.

6.3.2.6 Type of Component Barcode Reader System (407 Only)

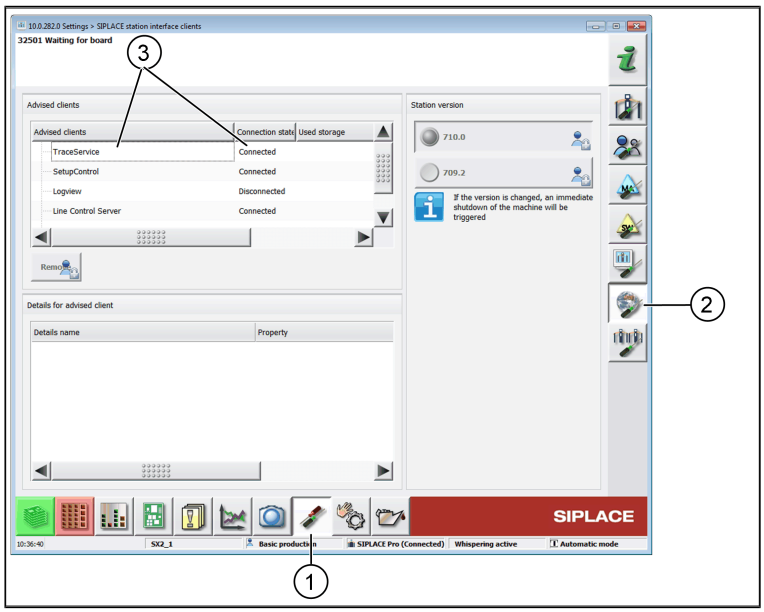
Select the component barcode reader type **SIPLACE Mobile Verifier (SMV)**.

6.3.3 Traceability Icon Status

Icon	Status	Meaning
	No icon	The traceability function is not installed.
	Gray icon	The traceability function is installed but not active. Data will not be generated. Advanced setup verification is active.
	Green icon	The traceability function is installed and active. The data will be stored in the target system.
	Green icon, canceled	Traceability is installed on the system but has been deactivated. Data will not be generated. The icon will disappear when you reboot the machine.
	Yellow icon	The traceability function is active and generating data. However, the data will be spooled, as the target folder is not available.
	Red icon	The setting "Interface Provides Data" is inactive and a severe error has occurred.
	Reader icon	If a connection to the SMV interface has been established, this icon will be visible.
	Reader icon, canceled	If no connection to the SC Station or SMV interface exists, this icon will appear with the reader crossed out.

6.3.4 Checking the the connection to Trace Service

The connection between the SIPLACE station and Trace Service can be checked in the station software.



- ► In the station software, open the **Settings** (1) view.
- ► Select **Station interface clients** (2).
  - ⇨ At (3) you can see the connection status for **Trace Service** => **Connected**.

6.4 Settings in the DEK software

6.4.1 Recording Data

Before you can use station-based traceability, you have to active this option in the setup of the DEK software.

- ► In **Maintenance**=>**Machine Setup**, activate **Verification** and **Traceability**. Set both options to **Enabled**.
- ► Open **Setup** => **Options**=> **Traceability**.
  - ⇨ The **Traceability** dialog box opens.



- ► In the **Traceability** dialog, check the Verification Data option (green tick) to activate the recording of verified material.

To additionally recorded process and SPC data, activate these options with a green tick.

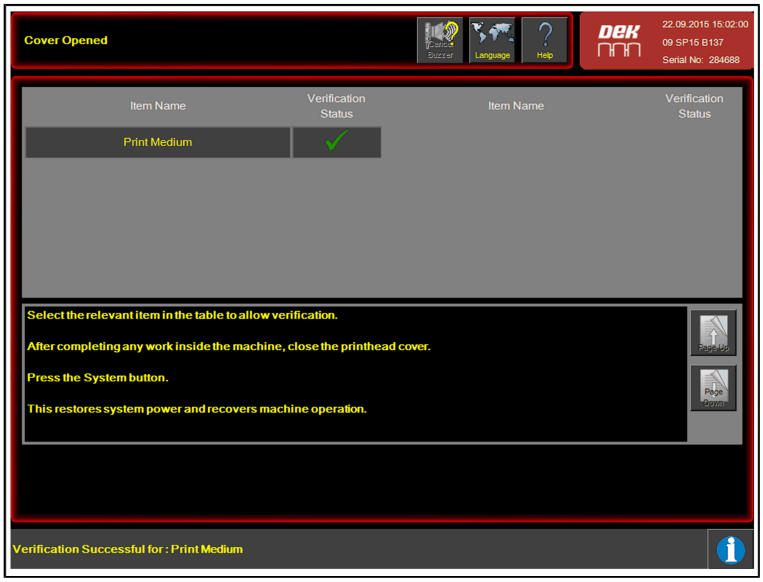
► Open **Setup** => **Options**=> **Verification**.

⇨ The **Verification** dialog box opens.



► In the **Verification** dialog, define the material for the allowed barcodes of the relevant material to be able to verify the material at the printer with a barcode scanner.

► After that, verify the material at the printer using your hand-held scanner.

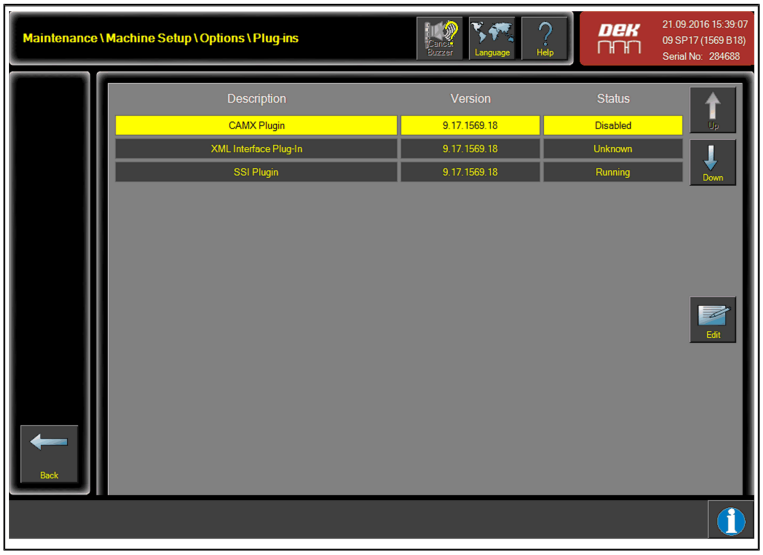


6.4.2 External Confirmation

To activate the external confirmation at the printer through Trace Service and BoardGateKeeper, proceed as follows:

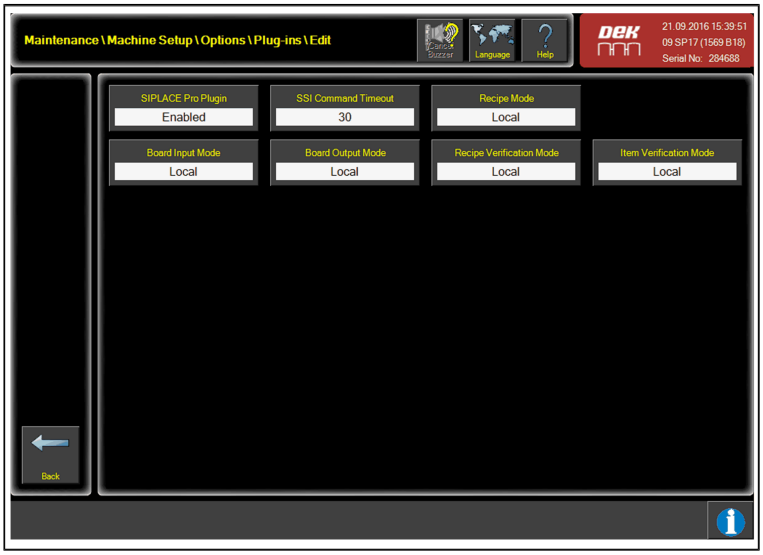
► Open **Maintenance => Machine Setup => Options => Plug-Ins.**

⇨ The **Plug-Ins** dialog box opens.



► Select **SSI Plugin** and confirm with the **Edit** icon.

⇨ The **Edit** dialog box opens.



► To lock the input side (front) with BoardGateKeeper, set the **Board Input Mode** to **Remote**. To lock the output side (rear) with TraceService, set the **Board Output Mode** to **Remote**.

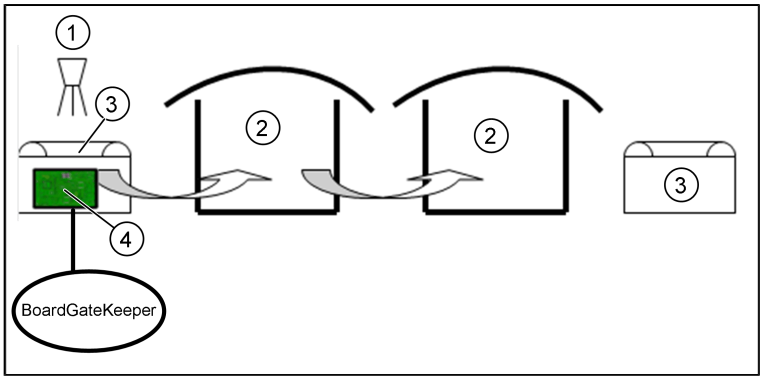
6.5 Scenario 1 - external scanner with EBR or Product Flow Controller (PFC)

6.5.1 Overview

NOTICE

For station software 7xx.x only

This scenario is only possible for stations with station software 7xx.x.



1. Barcode scanner
2. SIPLACE station (7xx.x)
3. External device (e.g. conveyor)
4. Board barcode which can be read

This scenario describes the use of an external scanner. The barcode information about the board present is sent to the BoardGateKeeper via the EBR interface, the Moxa Box or the CogiScan Product Flow Controller (PFC).

NOTICE

User manual for BoardGateKeeper

Refer to the user manual for the BoardGateKeeper.

The barcode information is send via "Whispering Down the Line" (WdtL) from the scanner to the first station and is then "whispered" on to the next station.

Licenses required

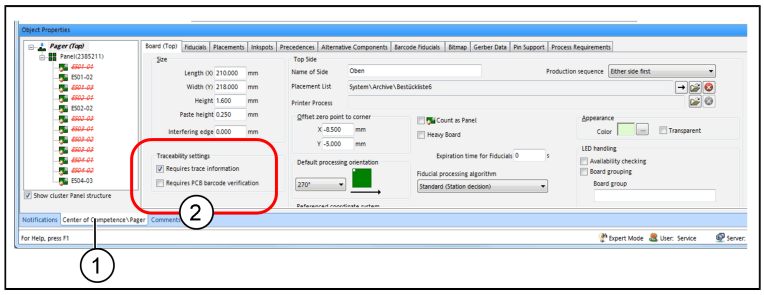
- **Traceability machine connection)** for each station in the line
- 1 x **Traceability server** for your company
- **Whispering Down The Line (WdtL)** for each station in the line

► See also 5.1 "Overview of the traceability licenses" [▶ 57].

6.5.2 Settings

6.5.2.1 SIPLACE Pro – Board Editor

► Open the board in the SIPLACE Pro Board Editor (1).



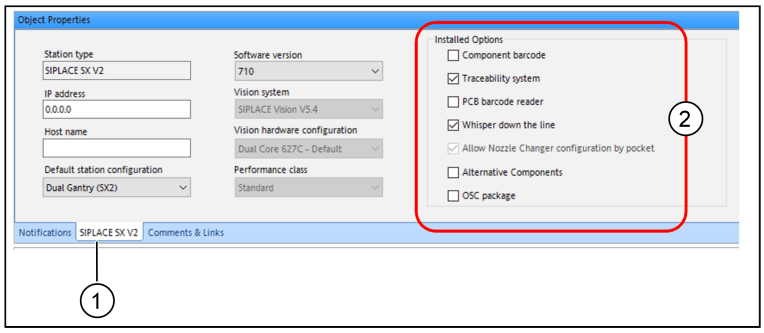
The following settings are required for this:

► Go to **Traceability settings** (2) and enable the option **Requires trace information**.

► Go to **Traceability settings** (2) and disable the option **Requires PCB barcode verification**.

6.5.2.2 SIPLACE Pro - Station Editor

► Open the stations for the line in the SIPLACE Pro Station Editor (1).



The following settings need to be defined for all stations in the line:

► Go to **Installed option** (2) and disable the option **Component barcode**.

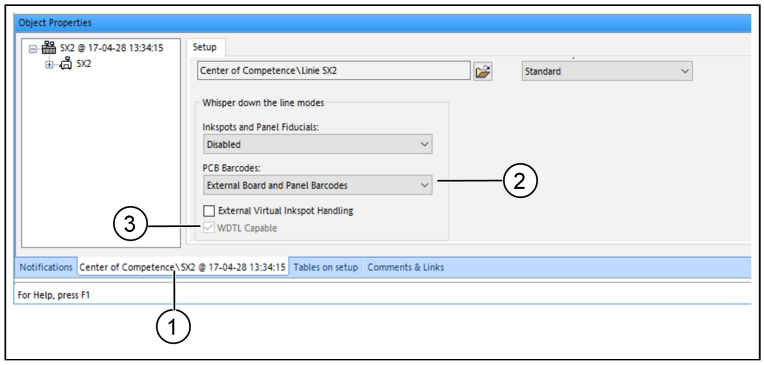
► Go to **Installed option** (2) and enable the option **Traceability-System**.

► Go to **Installed option** (2) and disable the option **PCB barcode reader**.

► Go to **Installed option** (2) and enable the option **Whisper Down the Line**.

6.5.2.3 SIPLACE Pro - Setup Editor

► Open the setup in the SIPLACE Pro Setup Editor (1).



The following settings are required for this:

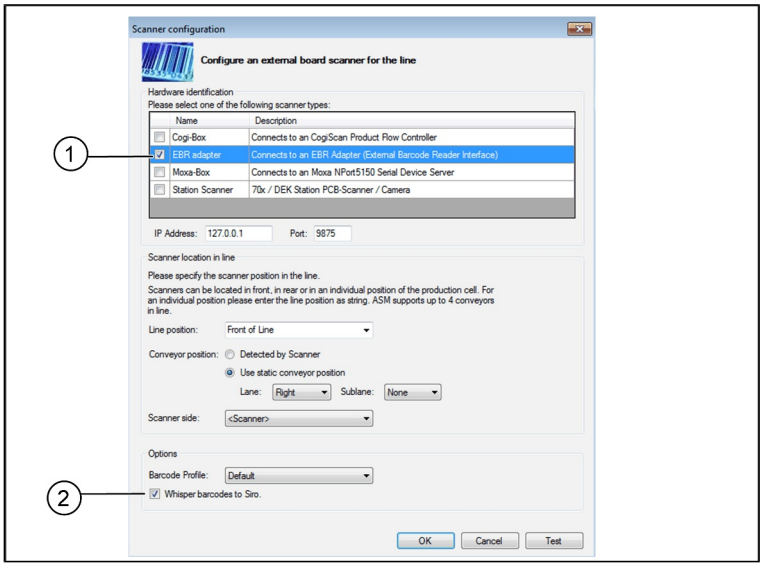
► Go to **PCB-Barcodes** and select the option **External Board and Panel Barcodes** (2).

► The option **WDTL Capable** (3) has been enabled. This must be already defined in the Station Editor, for the stations in the line. See 6.5.2.2 "SIPLACE Pro - Station Editor" [▶ 75].

6.5.2.4 BoardGateKeeper

Configuring an external board scanner for the line

- ► Start the BoardGateKeeper Manager.
- ► Select the **Line settings** tab.
- ► Select the line for which you want to add or edit a scanner hardware.
- ► Go to **Configured Scanner Hardware** and select a scanner type from the list and then click on the **Edit** button. Or select the **Add** button to create a new scanner.
  - ⇨ The **Scanner Configuration** dialog box will open.



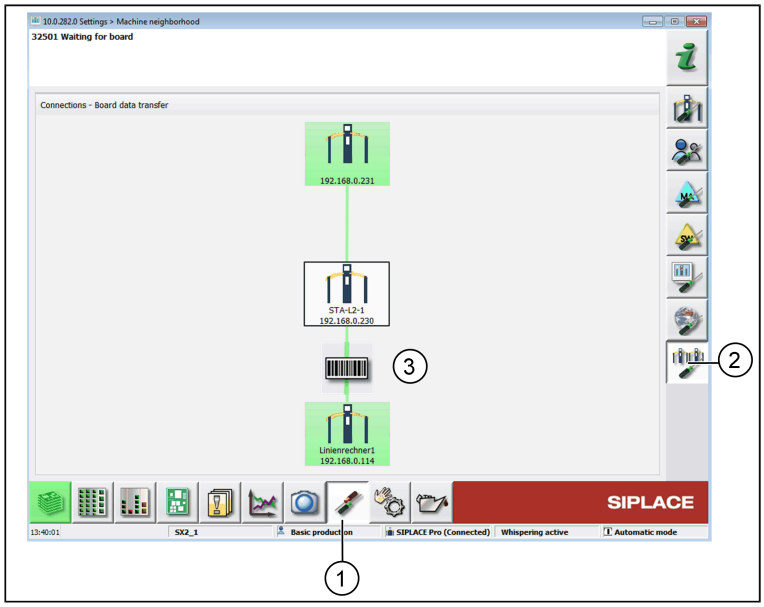
The following settings are required for this:

- ► Enable the scanner type **EBR Adapter** (1).
- ► Enable **Whisper barcodes to Sirio** (2).

For details about the settings, refer to the user manual for **ASM BoardGateKeeper**.

6.5.3 Results on the station software

In the station software, you can control whether the identification of boards is performed using the PCB barcode and whether the barcode data entered should be "whispered" to the stations in the line.



- ► In the station software, open the **Settings** (1) view.
- ► Select **Station interface clients** (2).
  - ⇨ The icon (3) shows that the board identification is conducted using the PCB barcode.

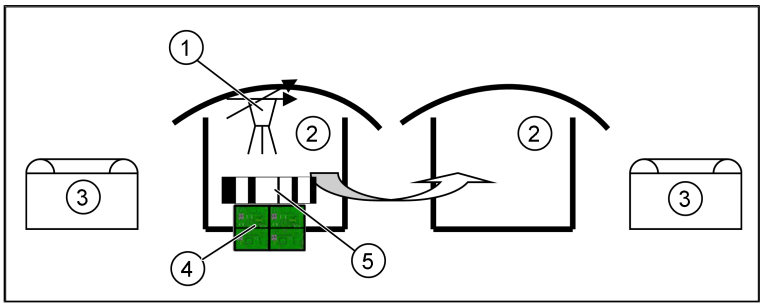
6.6 Scenario 2 - placing the barcode and reading with the PCB camera

6.6.1 Overview

NOTICE

For station software 7xx.x only

This scenario is only possible for stations with station software 7xx.x.



1. Read barcode with the PCB camera
2. SIPLACE station (7xx.x)
3. External device (e.g. conveyor)
4. Panel barcode which can be read
5. Placing the barcode

This scenario describes the use of a PCB camera for reading the barcodes. The barcode to be read is placed by the SIPLACE station and then read by the PCB camera. You need additional hardware for this: a label feeder, which can place the barcode labels on the board.

The barcode information is send via "Whispering Down the Line" (WdtL) from the scanner to the first station and is then "whispered" on to the next station.

Licenses required

- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company
- **Whispering Down The Line (WdtL)** for each station in the line
- **Read barcode with PCB camera** for each station in the line

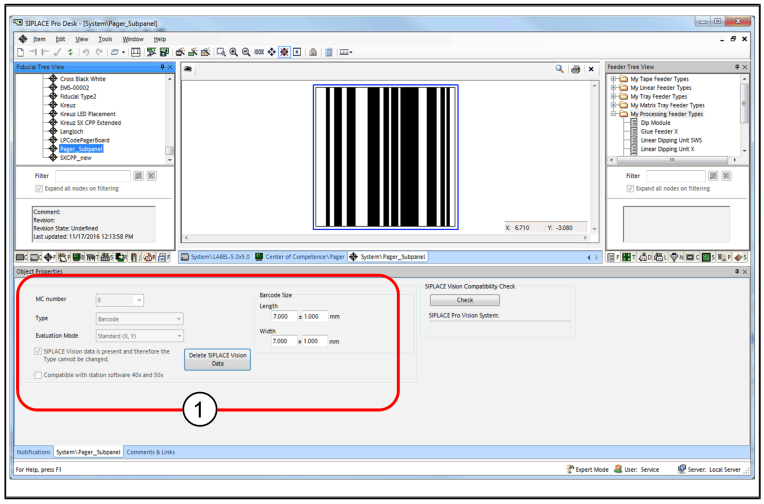
► See also 5.1 "Overview of the traceability licenses" [▶ 57].

6.6.2 Settings

6.6.2.1 SIPLACE Pro – Fiducial Editor

In order for the barcode to be placed, you need to create a fiducial of the type "barcode" in SIPLACE Pro.

► Open the Fiducial Editor in SIPLACE Pro.



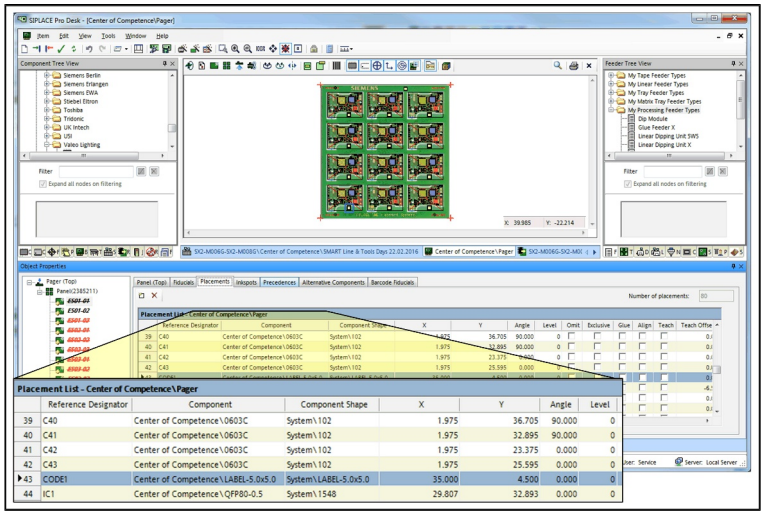
► Go to **Type** and select => **Barcode**.

► Enter the barcode dimensions at **Barcode size**.

6.6.2.2 SIPLACE Pro - Placement List Editor

In order for the barcode to be placed, the barcode label must have been created as a component/ component shape. This component/component shape is then defined in the placement list for the relevant placement positions.

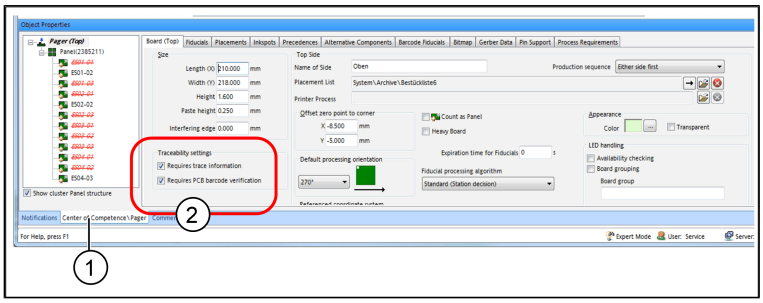
► Open the placement list in the SIPLACE Pro Placement List Editor.



In the example, number 43 shows the position designator **CODE1** with the corresponding parameters for the placement position.

6.6.2.3 SIPLACE Pro – Board Editor

► Open the board in the SIPLACE Pro Board Editor (1).



The following settings are required for this:

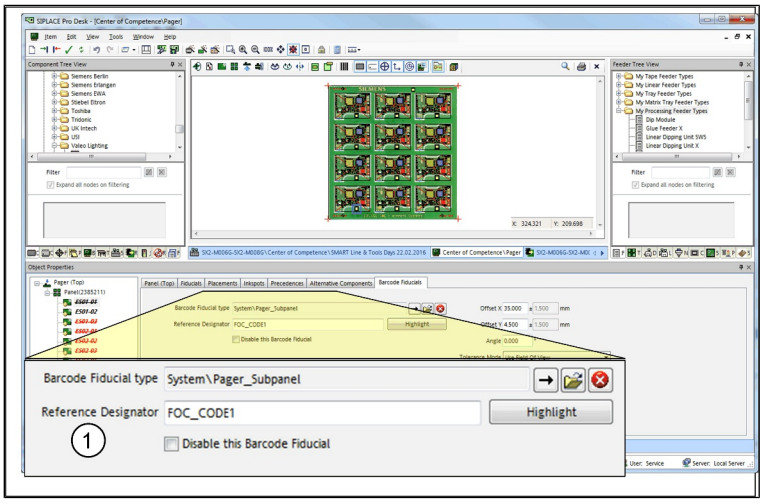
► Go to **Traceability settings** (2) and enable the option **Requires trace information**.

► Go to **Traceability settings** (2) and enable the option **Requires PCB barcode verification**.

6.6.2.4 SIPLACE Pro - Board Editor - barcode fiducials

Define the position designation and the position of the barcode on the board.

► Open the board in the SIPLACE Pro Board Editor.



► Enter the name of the position designation (1).

WARNING

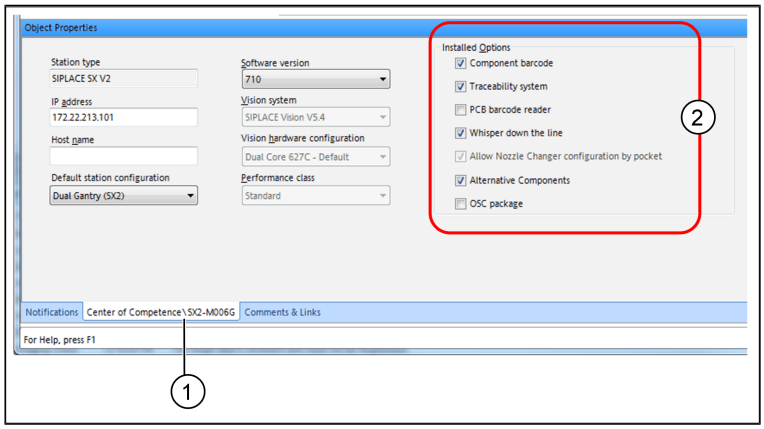
Define the position designator with the suffix FOC_

The position designator for the barcode must have the prefix **FOC_** at the front. The example shows FOC_CODE1. FOC stands for Fiducial On Component and defines that the fiducials is to be placed as a barcode label.

6.6.2.5 SIPLACE Pro - Station Editor

In order for the barcode to be read with the PCB camera, you need to define the following settings in the Station Editor.

► Open the stations for the line in the SIPLACE Pro Station Editor (1).



The following settings need to be defined for all stations in the line:

► Go to **Installed options** (2) and enable the option **Component barcode**.

► Go to **Installed options** (2) and enable the option **Traceability-System**.

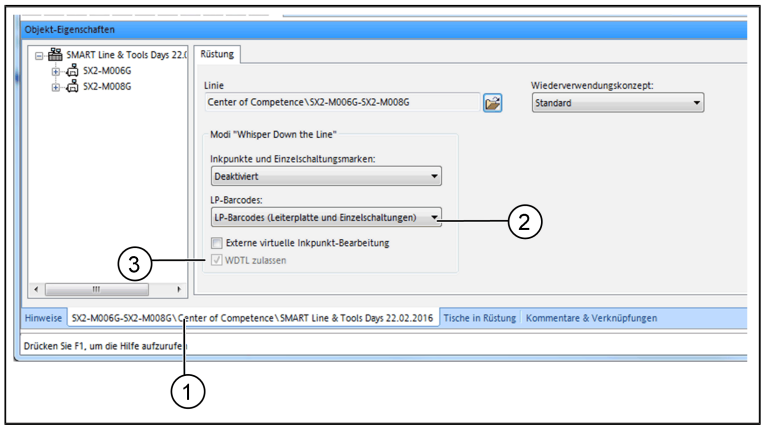
► Go to **Installed options** (2) and disable the option **PCB barcode reader**.

► Go to **Installed options** (2) and enable the option **Whisper Down The Line**.

6.6.2.6 SIPLACE Pro - Setup Editor

In order for the barcode to be read with the PCB camera, you need to define the following settings in the Setup Editor.

► Open the setup in the SIPLACE Pro Setup Editor (1).



► Go to **PCB barcodes** and select the option **PCB barcodes (board and panels)** (2).

► The option **WDTL Capable** (3) has been enabled. This must be already defined in the Station Editor, for the stations in the line. See 6.5.2.2 "SIPLACE Pro - Station Editor" [▶ 75].

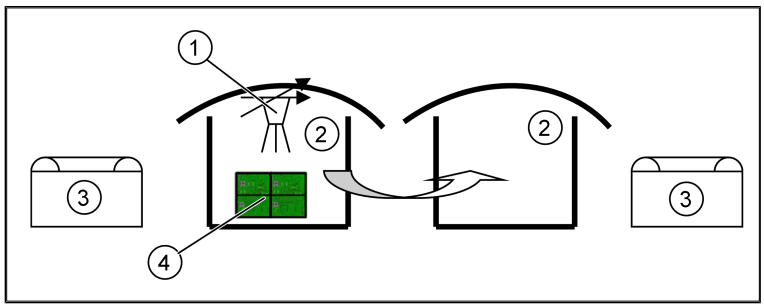
6.7 Scenario 3 - read barcode with the PCB camera

6.7.1 Overview

NOTICE

For station software 7xx.x only

This scenario is only possible for stations with station software 7xx.x.



1. Read barcode with the PCB camera
2. SIPLACE station (7xx.x)
3. External device (e.g. conveyor)
4. Panel barcode which can be read

This scenario describes the use of a PCB camera for reading the barcodes. The barcode information is send via "Whispering Down the Line" (WdtL) from the scanner to the first station and is then "whispered" on to the next station.

Licenses required

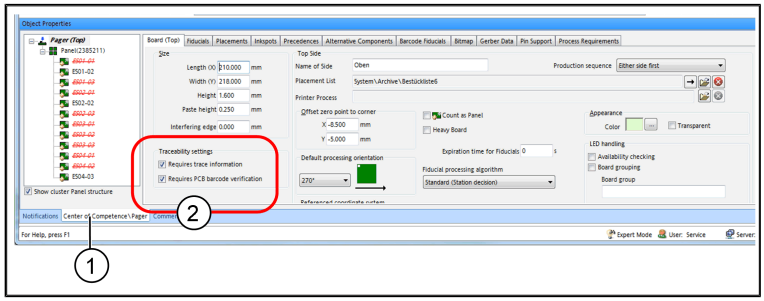
- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company
- **Whispering Down The Line (WdtL)** for each station in the line
- **Read barcode with PCB camera** for each station in the line

► See also 5.1 "Overview of the traceability licenses" [▶ 57].

6.7.2 Settings

6.7.2.1 SIPLACE Pro – Board Editor

► Open the board in the SIPLACE Pro Board Editor (1).



The following settings are required for this:

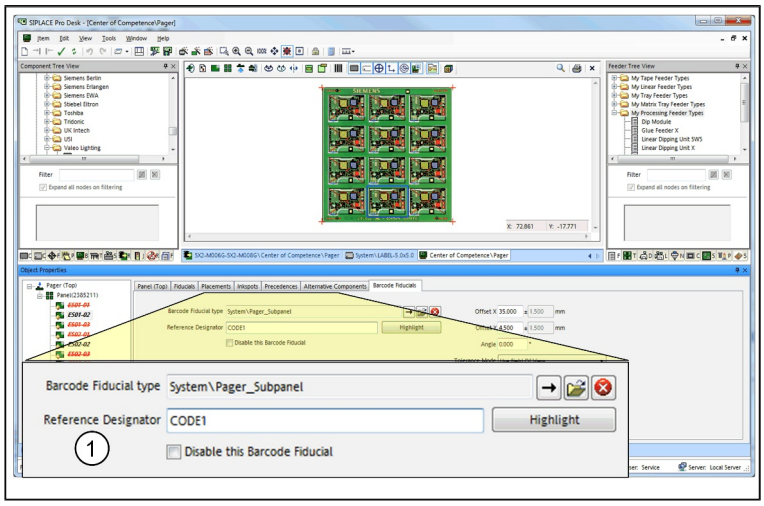
► Go to **Traceability settings** (2) and enable the option **Requires trace information**.

► Go to **Traceability settings** (2) and enable the option **Requires PCB barcode verification**.

6.7.2.2 SIPLACE Pro - Board Editor - barcode fiducials

Define the position designation and the position of the barcode on the board.

► Open the board in the SIPLACE Pro Board Editor.

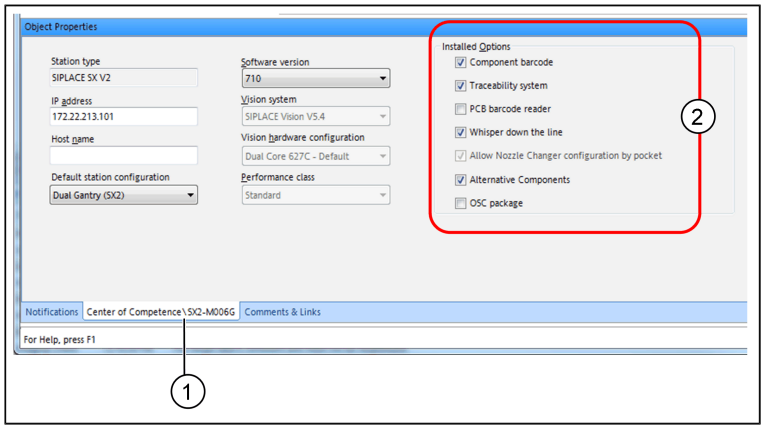


► Enter the name of the position designator (1). In this example it is **CODE1**.

6.7.2.3 SIPLACE Pro - Station Editor

In order for the barcode to be read with the PCB camera, you need to define the following settings in the Station Editor.

► Open the stations for the line in the SIPLACE Pro Station Editor (1).



The following settings need to be defined for all stations in the line:

► Go to **Installed options** (2) and enable the option **Component barcode**.

► Go to **Installed options** (2) and enable the option **Traceability-System**.

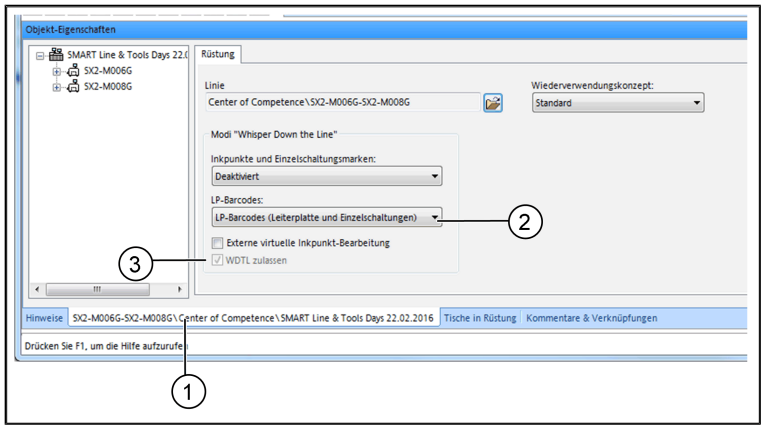
► Go to **Installed options** (2) and disable the option **PCB barcode reader**.

► Go to **Installed options** (2) and enable the option **Whisper Down The Line**.

6.7.2.4 SIPLACE Pro - Setup Editor

In order for the barcode to be read with the PCB camera, you need to define the following settings in the Setup Editor.

► Open the setup in the SIPLACE Pro Setup Editor (1).



► Go to **PCB barcodes** and select the option **PCB barcodes (board and panels)** (2).

► The option **WDTL Capable** (3) has been enabled. This must be already defined in the Station Editor, for the stations in the line. See 6.5.2.2 "SIPLACE Pro - Station Editor" [▶ 75].

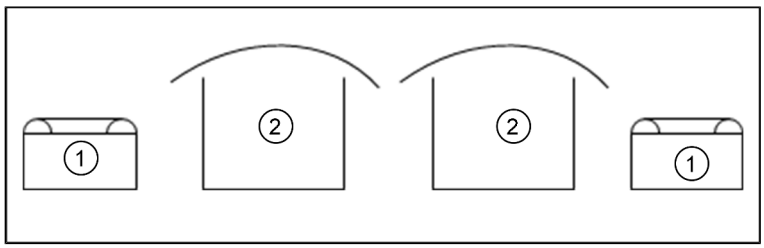
6.8 Scenario 4 - without barcode scanner

6.8.1 Overview

NOTICE

For station software 407.x/505.x/605.x/7xx.x

This scenario is possible for stations with all software versions (407.x/505.x/605.x/7xx.x).



1. External device (e.g. conveyor)
2. SIPLACE station (each SIPLACE – all versions)

This scenario describes the use of traceability without a barcode scanner. A "barcode" is generated in stead of the scanned barcode:

- Example
  - B_NO_PCB_BARCODE_00000011_2017081607350203_20-20170816073502030

Licenses required

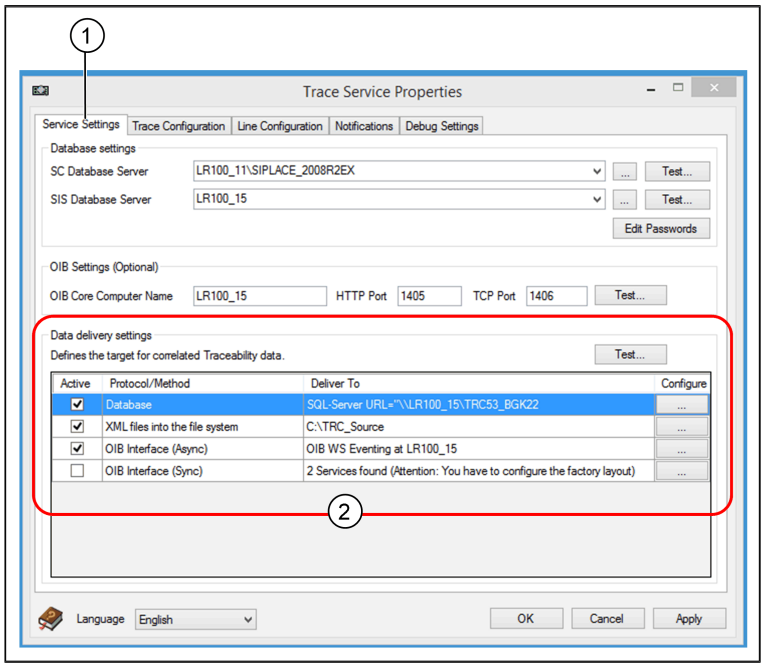
- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company

► See also 5.1 "Overview of the traceability licenses" [▶ 57].

6.8.2 Settings

6.8.2.1 Trace Service - service settings

► Start **Trace Service** and select **Configure…**.



► Select the **Service Settings** tab (1).

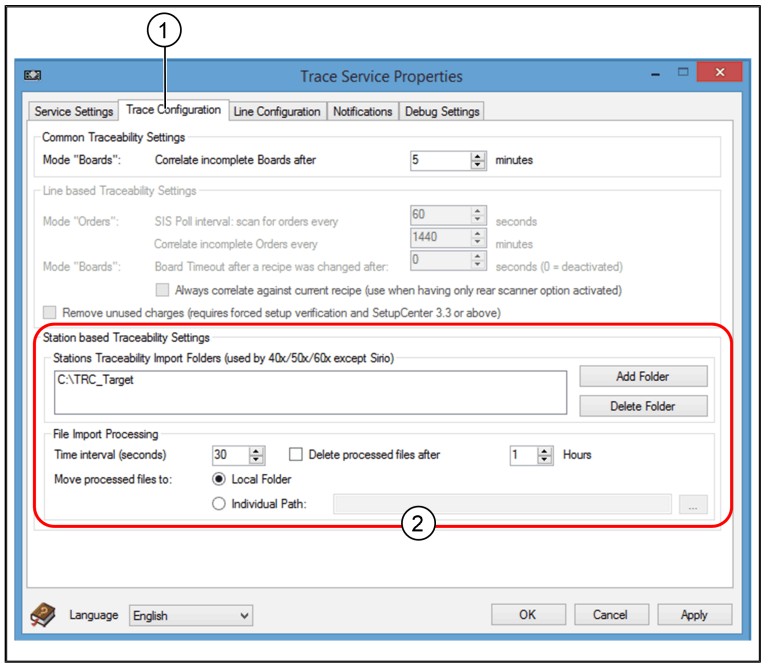
► Go to **Data delivery settings** (2) and enable and configure the following options:

- **Database**
- **Transferring XML files to the file system**
- **OIB interface (asynchronous)**

► For details, refer to chapter 3.3.2 "Service Settings" [▶ 25].

6.8.2.2 Trace Service - trace configuration

► Start **Trace Service** and select **Configure…**.



► Select the **Trace configuration** (1) tab.

► Go to **Station-based traceability settings** (2) and add a folder in which the traceability data can be stored.

► For details, refer to chapter 3.3.3 "Trace Configuration" [▶ 28].

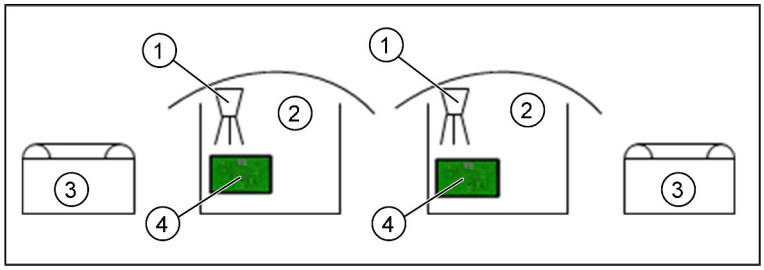
6.9 Scenario 5 - standard scenario (classic)

6.9.1 Overview

NOTICE

For station software 407.x/505.x/605.x/7xx.x

This scenario is possible for stations with all software versions (407.x/505.x/605.x/7xx.x).



1. Barcode scanner
2. SIPLACE station (each SIPLACE – all versions)
3. External device (e.g. conveyor)
4. Board barcode which can be read

This scenario describes the use of traceability with one barcode scanner in each case. Each station has a barcode scanner which scans the board barcode.

Licenses required

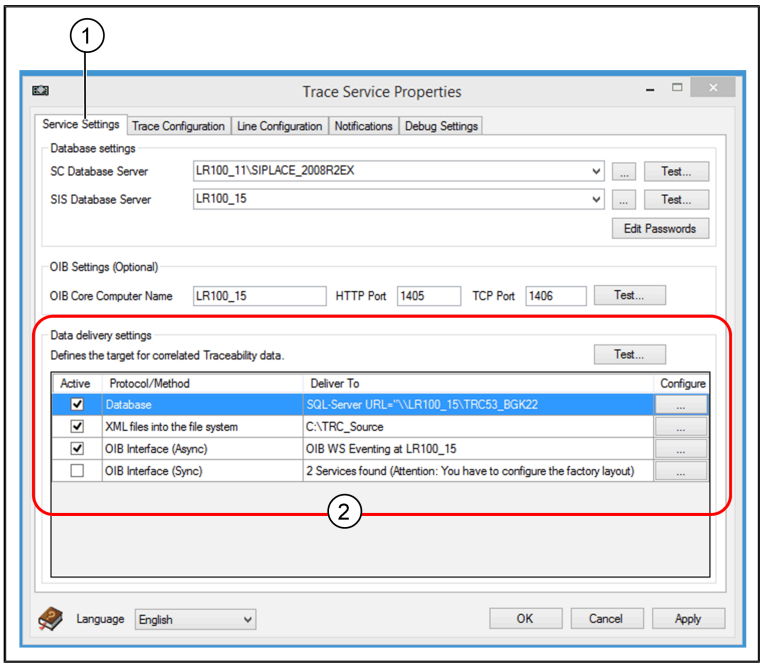
- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company

► See also 5.1 "Overview of the traceability licenses" [▶ 57].

6.9.2 Settings

6.9.2.1 Trace Service - service settings

► Start **Trace Service** and select **Configure…**.



► Select the **Service Settings** tab (1).

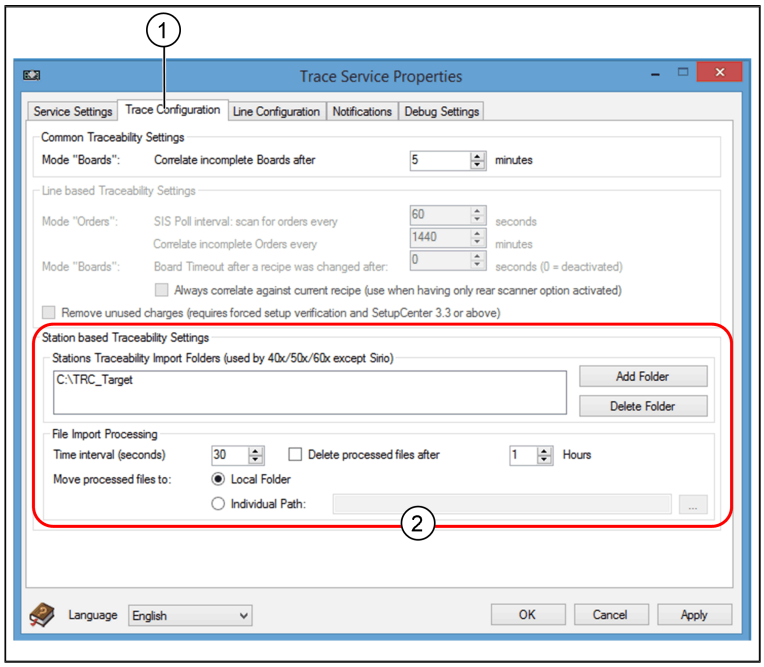
► Go to **Data delivery settings** (2) and enable and configure the following options:

- **Database**
- **Transferring XML files to the file system**
- **OIB interface (asynchronous)**

► For details, refer to chapter 3.3.2 "Service Settings" [▶ 25].

6.9.2.2 Trace Service - trace configuration

► Start **Trace Service** and select **Configure…**.



► Select the **Trace configuration** (1) tab.

► Go to **Station-based traceability settings** (2) and add a folder in which the traceability data can be stored.

► For details, refer to chapter 3.3.3 "Trace Configuration" [▶ 28].

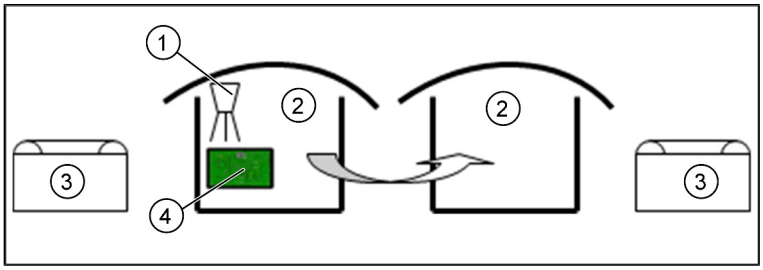
6.10 Scenario 6 - standard scenario (classic+WdtL)

6.10.1 Overview

NOTICE

For station software 7xx.x only

This scenario is only possible for stations with station software 7xx.x.



1. Barcode scanner
2. SIPLACE station (7xx.x)
3. External device (e.g. conveyor)
4. Board barcode which can be read

This scenario describes the reading of the barocde with a barcode scanner at the first station in the line. The barcode information is sent via "Whispering Down the Line" (WdtL) from the scanner to the first station and is then "whispered" on to the next station.

Licenses required

- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company
- **Whispering Down The Line (WdtL)** for each station in the line

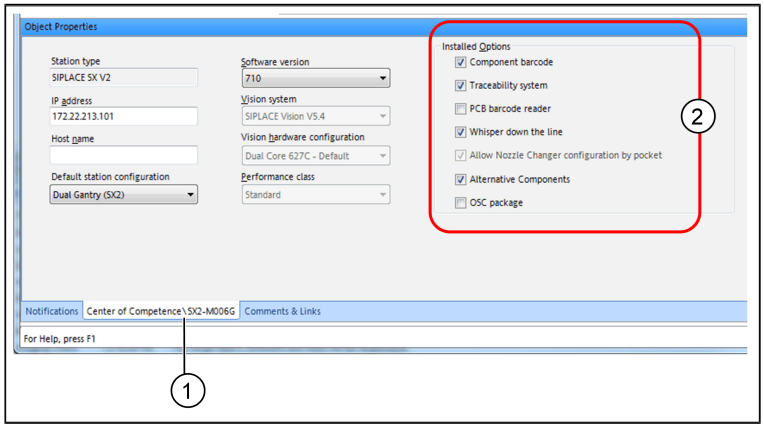
► See also 5.1 "Overview of the traceability licenses" [▶ 57].

6.10.2 Settings

6.10.2.1 SIPLACE Pro - Station Editor

In order for the barcode to be read with the PCB camera, you need to define the following settings in the Station Editor.

► Open the stations for the line in the SIPLACE Pro Station Editor (1).



The following settings need to be defined for all stations in the line:

► Go to **Installed options** (2) and enable the option **Component barcode**.

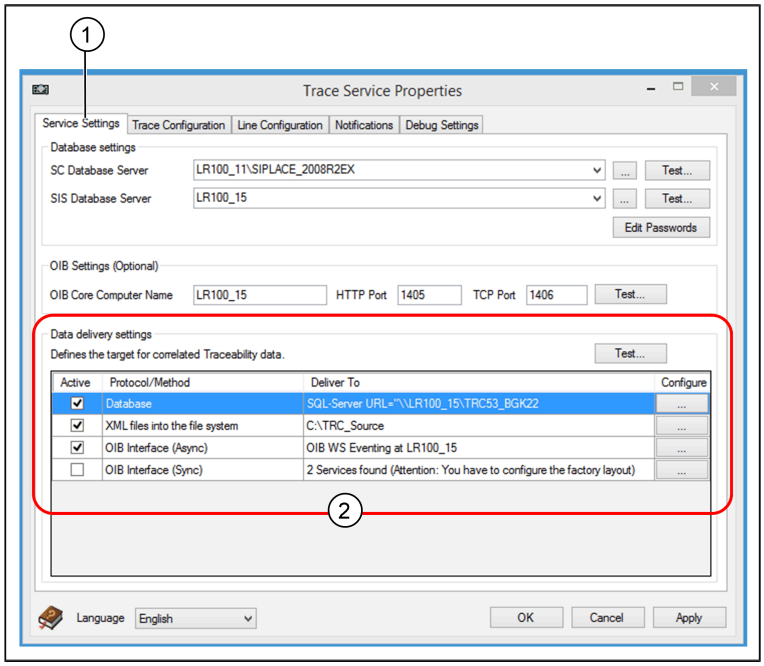
► Go to **Installed options** (2) and enable the option **Traceability-System**.

► Go to **Installed options** (2) and disable the option **PCB barcode reader**.

► Go to **Installed options** (2) and enable the option **Whisper Down The Line**.

6.10.2.2 Trace Service - service settings

► Start **Trace Service** and select **Configure…**.



► Select the **Service Settings** tab (1).

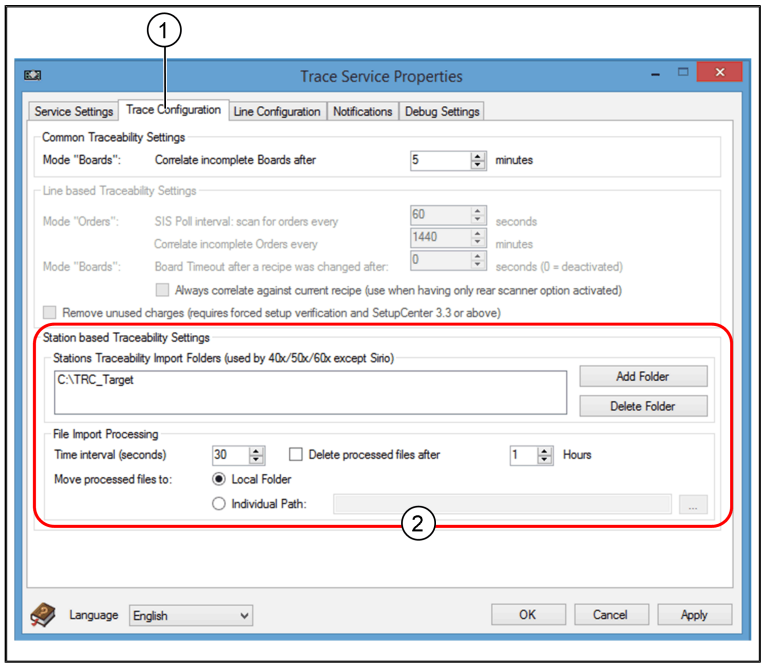
► Go to **Data delivery settings** (2) and enable and configure the following options:

- **Database**
- **Transferring XML files to the file system**
- **OIB interface (asynchronous)**

► For details, refer to chapter 3.3.2 "Service Settings" [▶ 25].

6.10.2.3 Trace Service - trace configuration

► Start **Trace Service** and select **Configure…**.



► Select the **Trace configuration** (1) tab.

► Go to **Station-based traceability settings** (2) and add a folder in which the traceability data can be stored.

► For details, refer to chapter 3.3.3 "Trace Configuration" [▶ 28].

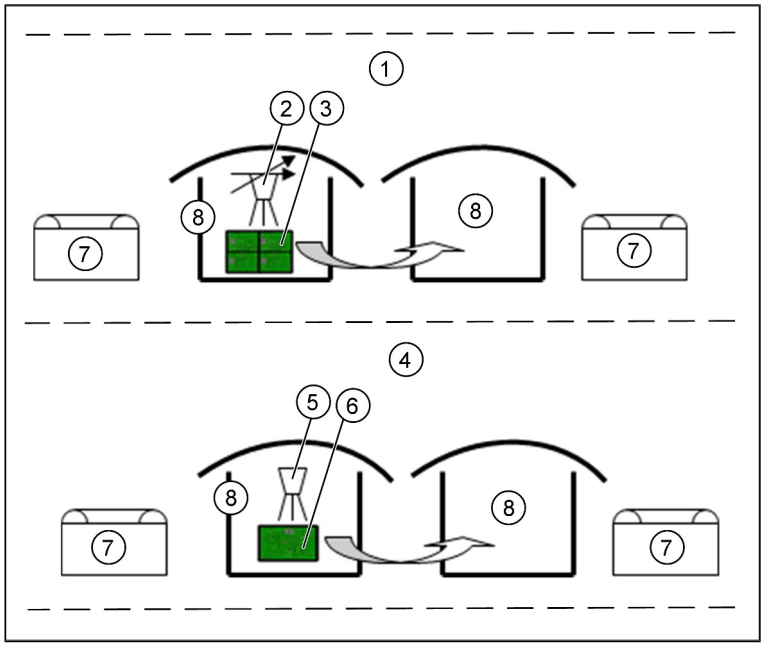
6.11 Best practice scenario

Overview

NOTICE

For station software 7xx.x only

This scenario is only possible for stations with station software 7xx.x.



1. Production of board top side
2. Read barcode with the PCB camera
3. Panel barcode which can be read
4. Production of board bottom side
5. Barcode scanner
6. Panel barcode which can be read
7. External device (e.g. conveyor)
8. SIPLACE station (7xx.x)

The top barcode of a double-sided board should be read with the PCB camera. The bottom barcode can then only be read with a barcode reader which is at the bottom of the input conveyor.

These two data sets then need to be merged externally.

Scenario example

- Top side of board
  - The master barcode and the four panel barcodes are located on the top of the board.
  - All five barcodes are read with the PCB camera.
- Bottom side of board
  - The master barcode is read from below with a barcode scanner.
  - It is not possible to read the panel barcodes.

Two traceability data sets are created for the top and bottom of the board.

Problem

How can one merge the contents of these two traceability data sets?

Solution

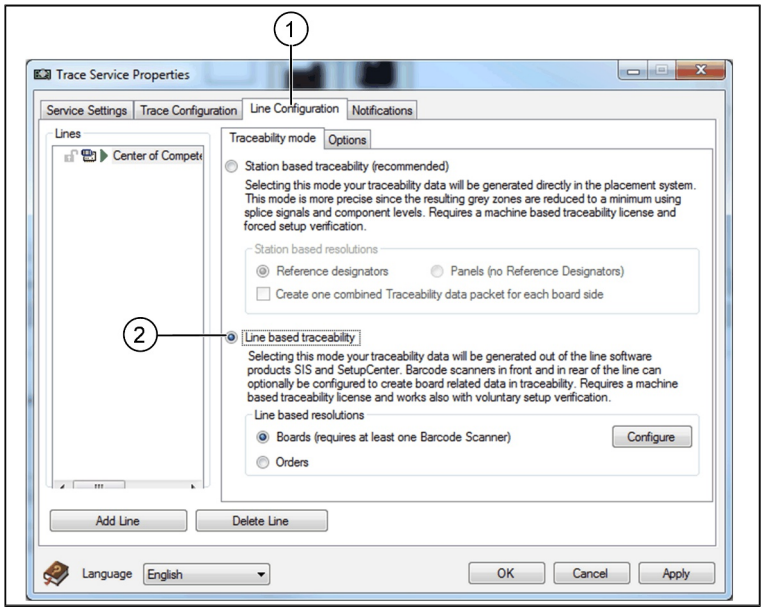
- You need to select the same name in SIPLACE Pro for the panels on the top and bottom of the board.
  - The data sets for top and bottom are accessible via the master barcode.
  - The panel name can be used to extract the traceability data (including the panel barcode) for the panel.

7 Line-based traceability

7.1 Trace service settings

In order to run line-based traceability you need to enable this function in the Trace Service.

► Start **Trace Service** and select **Configure…**



► Select the **Line configuration** (1) tab.

► Enable the option field **Line-based traceability** (2).

► For more settings, refer to chapter 3 "Trace Service" [▶ 21].

7.2 Settings in SIPLACE Pro

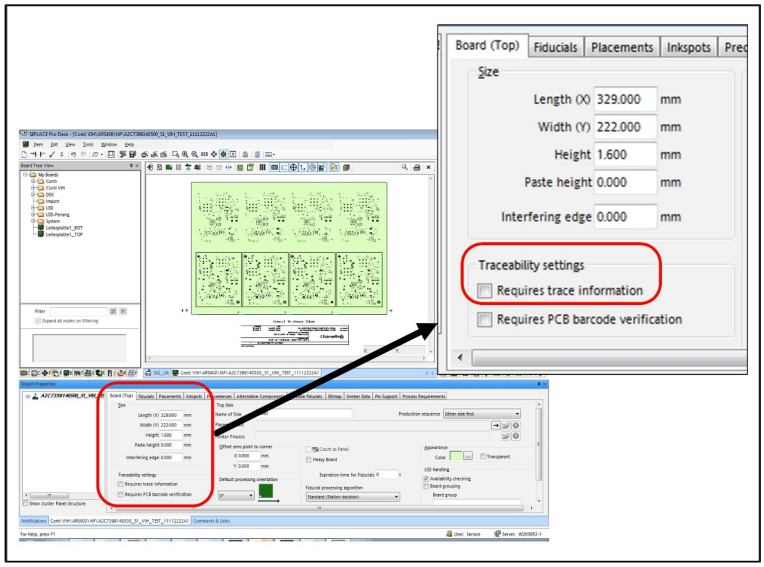
NOTICE

Disabling the traceability option

In line-based traceability, you should disable the following option in SIPLACE Pro for license reasons: **Traceability** (in the **Station Editor**, **Board Editor**and **Recipe Editor**). Otherwise you would consume the licenses for machine connections twice (1 license for machine connection station-based traceability and one license for machine connection line-based traceability).

7.2.1 Board Editor

► Open the board in the SIPLACE Pro Board Editor.

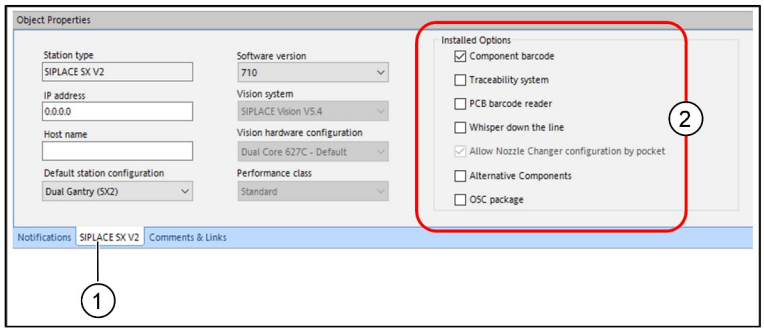


► Go to **Traceability settings** and disable the option **Requires trace information**.

You need to disable the setting **Requires trace information** for each board.

7.2.2 Station Editor

► Open the stations for the line in the SIPLACE Pro Station Editor (1).



The following settings need to be defined for all stations in the line:

► Go to **Installed option** (2) and enable the option **Component barcode**.

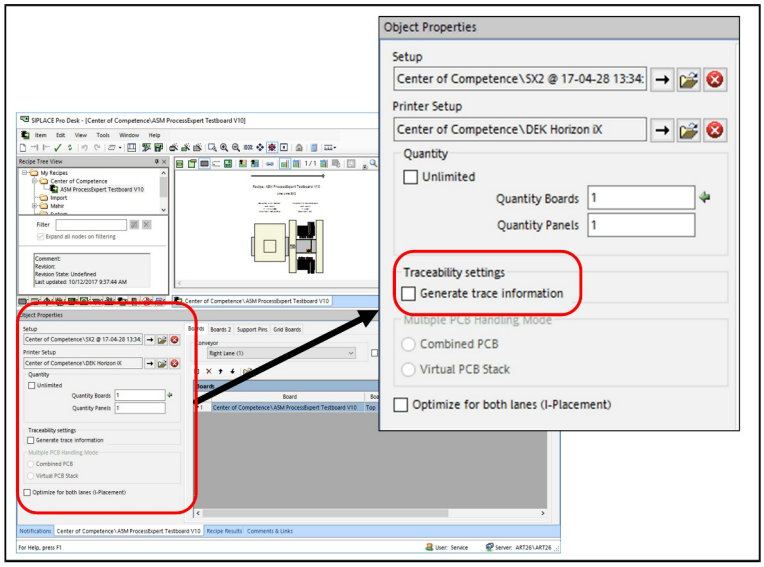
► Go to **Installed option** (2) and disable the option **Traceability system**.

► Go to **Installed option** (2) and disable the option **PCB barcode reader**.

► Go to **Installed option** (2) and disable the option **Whisper down the line**.

7.2.3 Recipe Editor

► Open the Recipe Editor.



► Go to **Setup** => **Traceability settings** and disable the option **Generate trace information**. This option must be disabled separately for each recipe.

NOTICE

Track information required

If the option **Requires trace information** is disabled for the board contained in the Board editor, this option will also be automatically enabled in the Recipe Editor for job optimization.

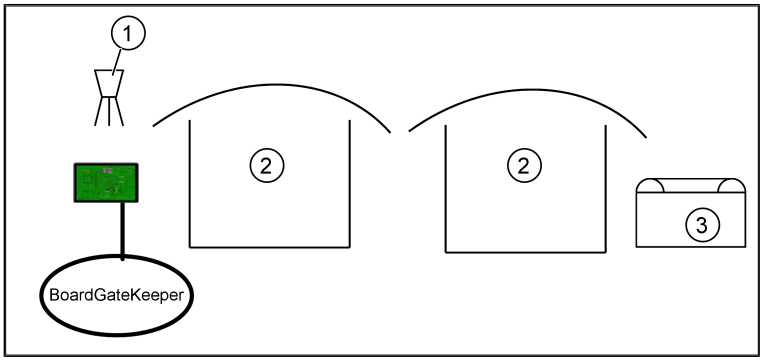
7.3 Scenario 1 - external scanner with EBR and PFC

7.3.1 Overview

NOTICE

For station software 407.x/505.x/605.x/7xx.x

This scenario is possible for stations with all software versions (407.x/505.x/605.x/7xx.x).



1. Barcode scanner
2. SIPLACE station (each SIPLACE – all versions)
3. External device (e.g. conveyor)

This scenario describes the use of an external scanner before the line. The barcode information about the board present is sent to the BoardGateKeeper via the EBR interface, the Moxa Box or the CogiScan Product Flow Controller (PFC).

NOTICE

User manual for BoardGateKeeper

Refer to the user manual for the BoardGateKeeper.

Licenses required

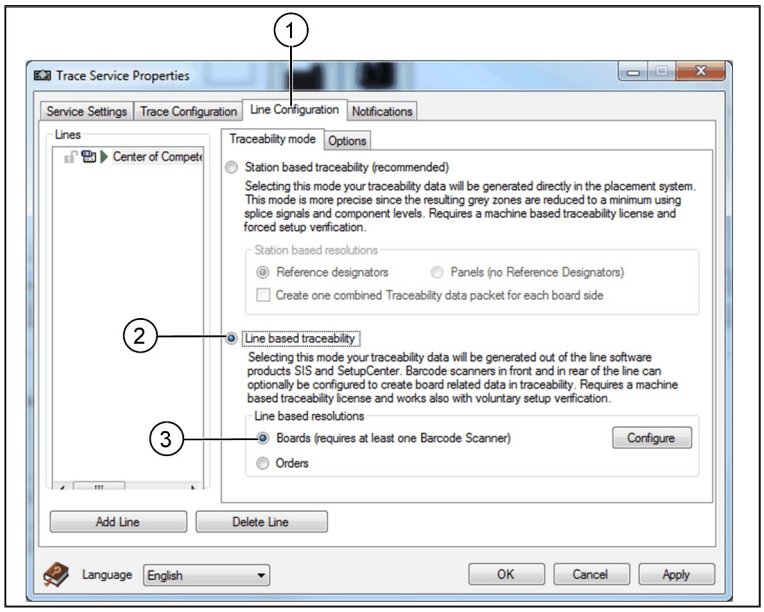
- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company

► See also 5.1 "Overview of the traceability licenses" [▶ 57].

7.3.2 Settings

7.3.2.1 Trace service - line configuration

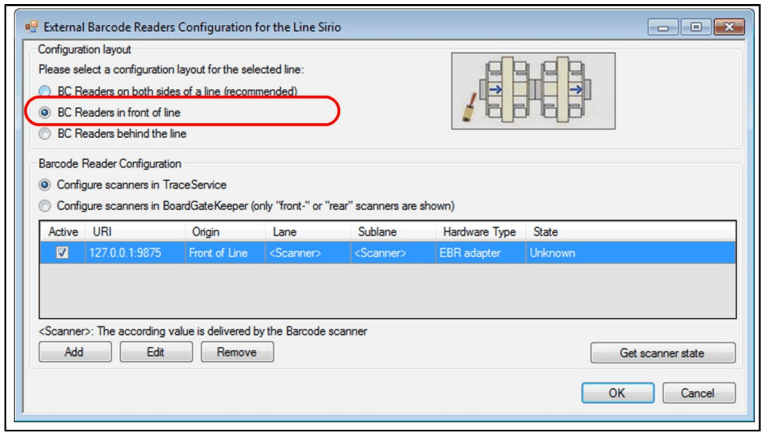
Start **Trace Service** and select **Configure…**



- ► Select the **Line configuration** (1) tab.
- ► Enable the option field **Line-based traceability** (2).
- ► Enable the option field **Boards (requires at least one barcode scanner)(3)**.

Trace service - configuring the barcode before the line

- ► Go to the **Line configuration** tab, select **Line-based traceability** and click on the **Configure..**button.
  - ⇨ The **External barcode readers configuration for the line Sirio** dialog box will open.



- ► Enable the option field **BC readers in front of line**.
- ► Use the **Add** button to add the relevant barcode scanner to the list and then enable the checkbox.

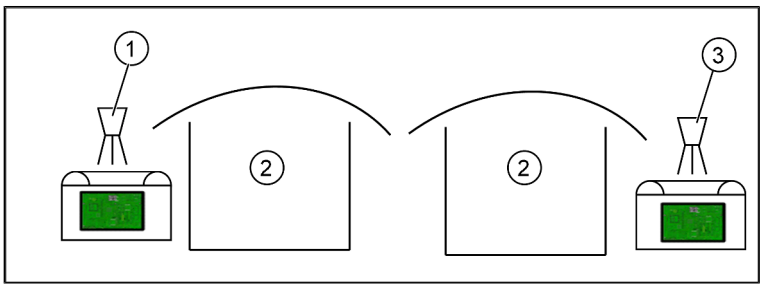
7.4 Scenario 2 - scanner before and behind the line

7.4.1 Overview

NOTICE

For station software 407.x/505.x/605.x/7xx.x

This scenario is possible for stations with all software versions (407.x/505.x/605.x/7xx.x).



1. Barcode scanner – before the line
2. SIPLACE station (each SIPLACE – all versions)
3. Barcode scanner – behind the line

This scenario describes the use of a scanner before and behind the line.

Licenses required

- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company

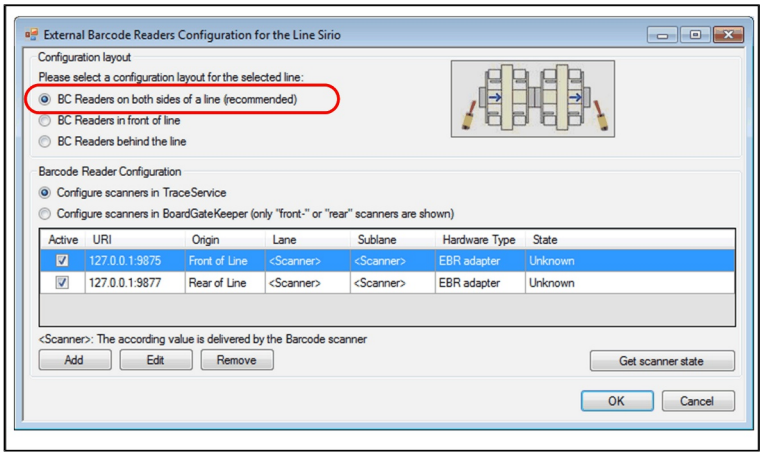
► See also 5.1 "Overview of the traceability licenses" [▶ 57].

7.4.2 Settings

7.4.2.1 Trace service - configuring the barcode before and behind the line

► Go to the **Line configuration** tab, select **Line-based traceability** and click on the **Configure..**button.

⇨ The **External barcode readers configuration for the line Sirio** dialog box will open.



► Enable the option field **BC readers on both sides of a line (recommended)**.

► Use the **Add** button to add the relevant barcode scanner to the list and then enable the checkbox.

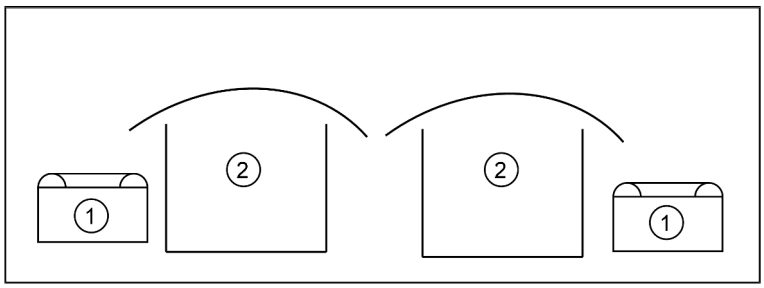
7.5 Scenario 3 - job-related - without barcode scanner

7.5.1 Overview

NOTICE

For station software 407.x/505.x/605.x/7xx.x

This scenario is possible for stations with all software versions (407.x/505.x/605.x/7xx.x).



1. External device (e.g. conveyor)
2. SIPLACE station (each SIPLACE – all versions)

This scenario describes the use of traceability without a barcode scanner. The board is identified using a job number.

Licenses required

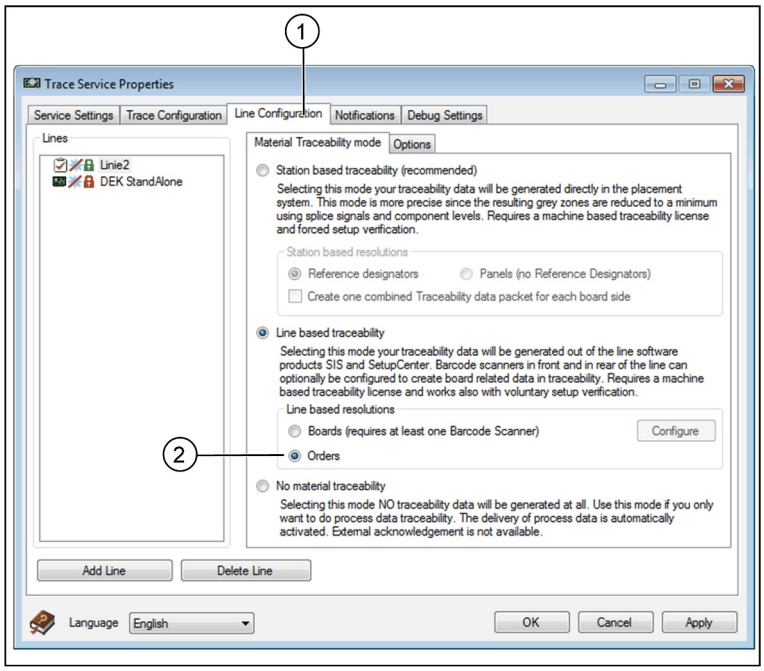
- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company

► See also 5.1 "Overview of the traceability licenses" [▶ 57].

7.5.2 Settings

7.5.2.1 Trace service - line settings

► Start Trace Service and select **Configure…**



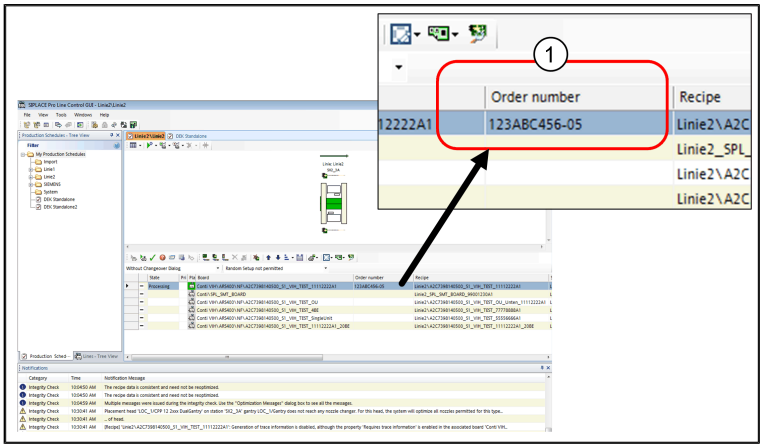
► Select the **Line configuration** (1) tab.

► Go to **Line-based resolutions** and enable the option field **Orders** (2).

► For more settings, refer to chapter 3 "Trace Service" [▶ 21].

7.5.2.2 Line Control GUI - production schedule

► Open the production schedule in the Line Control GUI.



You need to enter an **Order number** (1) for the boards in the recipe.

NOTICE

Production order number in the production schedule

If the **Order number** column is not shown, enable it at **Tools** => **Configuration** => **Application** => **Columns**.

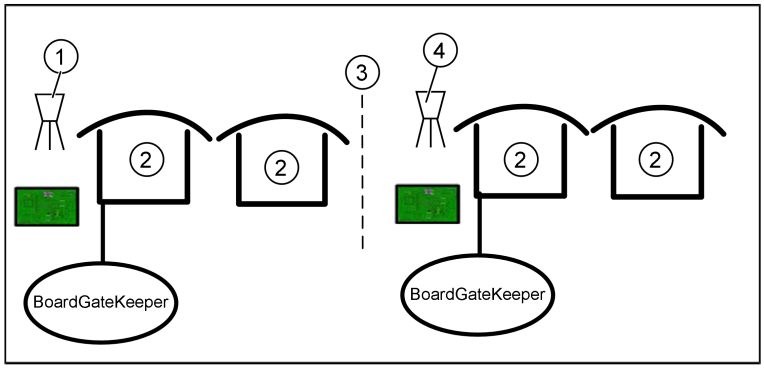
7.6 Scenario 4 - scanner in the center of the line

7.6.1 Overview

NOTICE

For station software 7xx.x only

This scenario is only possible for stations with station software 7xx.x.



1. Barcode scanner in front of the line
2. SIPLACE station (7xx.x)
3. Interrupted line
4. Barcode scanner in the middle of the line

This scenario describes the positioning of the barcode scanner at any place in the line.

The barcode scanner can be placed at an arbitrary position of the line in front of a specified station. If a line is interrupted at a position, the barcode can be re-read and whispered to the first station of a cluster. By means of an additional scanner, the barcode can be whispered to the first station of a cluster in the part of line succeeding the interruption.

Licenses required

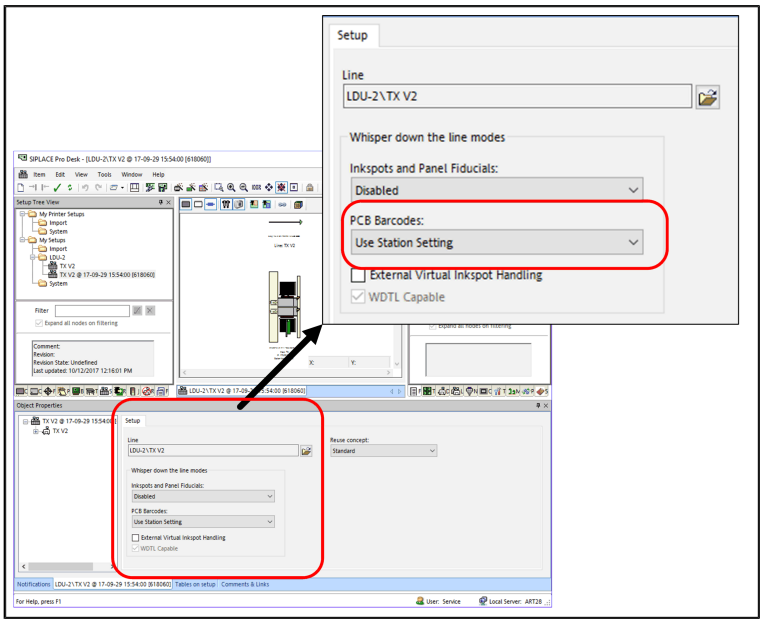
- **Traceability (machine connection)** for each station in the line
- 1 x **Traceability server** for your company

► See also 5.1 "Overview of the traceability licenses" [▶ 57]

7.6.2 Settings

7.6.2.1 SIPLACE Pro - Setup Editor

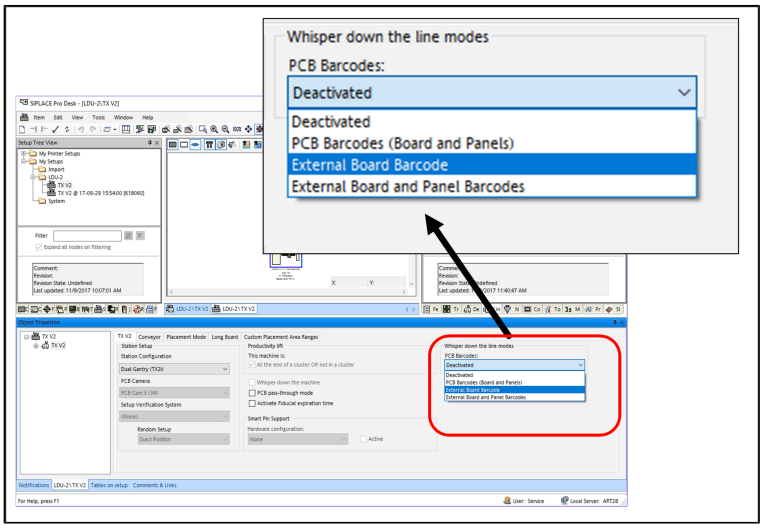
► Open the relevant setup in the SIPLACE Pro Setup Editor.



► Go to **PCB barcodes** and select the option **Use station settings**.

This setting defines that, these barcodes are to be received from an external device via the whisper interface, for each station in the Setup Editor, **Station configuration** tab.

► Switch over to the station configuration in the Setup Editor.

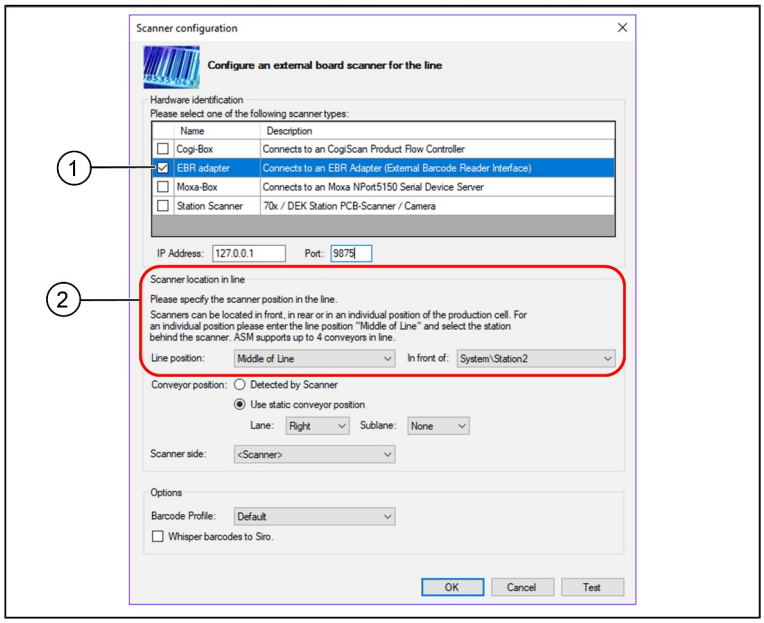


NOTICE

The setting **Disabled** is not permitted.

7.6.2.2 BoardGateKeeper

- ► Start the BoardGateKeeper Manager.
- ► Select the **Line settings** tab.
- ► Select the line for which you want to add or edit a scanner hardware.
- ► Go to **Configured Scanner Hardware** and select a scanner type from the list and then click on the **Edit** button. Or select the **Add** button to create a new scanner.
  - ⇨ The **Scanner Configuration** dialog box will open.



Define the following setting:

- ► Select, for example, the scanner type **EBR Adapter** (1).
- ► Go to **Scanner location in line** => **Line position:** and select the option **Middle of line**.
- ► In the selection field **In front of:**, select the station in front of which there is a barcode scanner.
- ► For details about the settings, refer to the user manual for **ASM BoardGateKeeper**.

8 Traceability Interfaces

The generated traceability data are made available via the OIB interfaces.

The interfaces are documented in the product OIB. A Traceability OIB tutorial can be found in the OIB SDK documentation.

Traceability XML interface

The traceability data are exported to the file system in XML format.

Database view interface

This interface consists of several database views via which the required data can be accessed from the Trace database.



NOTICE

Interfaces

- ► If you need more information about the database view interface or the XML interfaces, register yourself at http://www.asm-smt.com and request the document package.
- ► For more information about the OIB interface, refer to the OIB SDK documentation.

9 Annex

9.1 Traceability Data with Error Code >1

Traceability data with error code = 1 are reliably generated data. If irregularities occur during placement, an error code >1 will be set.

Error codes already defined for station-based traceability:

Error definitions in the station software	Error code (decimal)	Error definitions in the documentation	407 .xx	505 .xx	605 .xx	70x .xx	Sta-tionb. TRC	Lineb. TRC
DEFAULT_ER-ROR	0	Station based traceability: PCB left the machine before reaching the output con-veyor Line-based traceability: Unfinished job, no error	x	x	x	x	x	x
FINISHED	1	Traceability data were gen-erated without error	x	x	x	x	x	x
DUPLIC-ATE_PCB_IN_P ROCESS	2	Board was detected several times at different places	x	x	x	x	x	
BAR-CODE_DOUBLE T	4	2 boards with identical bar-code detected						x
INPRO-CESSIBLE_ USER_ACTION	8	Splice without following batch could not be pro-cessed. Gray zone already active or no component available. It is checked if there is a valid packaging unit on this track.	x	x	x		x	
TRACEAB-ILTY_ON_ DURING_PRO-CESS	16	Traceability was activated during placement	x	x	x		x	
TRACEAB-ILTY_OFF_ DURING_PRO-CESS	32	Traceability was deactiv-ated during placement	x	x	x		x	
CANCELLED	64	Placement of the PCB was canceled - * or components were skipped	x*	x		x	x	
DATA_LOST	128	Traceability data could not be completely recorded for a produced PCB.	x	x	x	x	x	
ORPHAN	256	The board was only pro-cessed in one processing area	x	x	x	x	x	
INTERNAL_ER-ROR	512	Traceability data could not be assigned within the machine	x	x	x	x	x	

Error definitions in the station software	Error code (decimal)	Error definitions in the documentation	407 .xx	505 .xx	605 .xx	70x .xx	Sta-tionb. TRC	Lineb. TRC
RECOV-ERY_DATA_ LOST	1024	PCB was processed with recovery function, possible loss of data.		x	x	x	x	
ONLINE_EDIT-ING	2048	Possible change of fiducials or pickup positions during placement of the board.		x	x	x	x	
HAND_LPBC	4096	PCB barcode manually entered; only available in Traceability XML data ver-sion 1.1 and 2.1.	x	x	x	x	x	
PRO-CESSING_AREA _ DISABLED	8192	The processing in one or more processing areas was deactivated by the operator.			x		x	
MANUAL_ INSPECTION REQUIRED	16384	The operator was requested to manually inspect the board in the output con-veyor. Some components may not have been placed correctly.			x	X	x	
MISS-ING_BACK_ BARCODE	65536	Board was recorded by the front but not the rear bar-code reader						X
MISS-ING_FRONT_ BARCODE	131072	Board was recorded by the rear but not the front bar-code reader						X
CORRELATION_ INCOMPLETE	524288	The correlation process failed or is incomplete						X
CORRELATION_ FAILED	1048576	Correlation of traceability data failed						X
PARTIALLY_ CORRELATED	2097152	In line based mode only: traceability information of an uncompleted production job. Job related queries in Trace Monitor result in in-complete traceability data as long as the job is not completed.						X
PREVI-OUS_BOARD_M ISSING	4194304	Order of boards in the line was changed i.e. the board predecessor at entry into the line is not the same as the predecessor at exit from the line.						X
MISSING_INDI-VIDUAL_COM-PONENT_BAR-CODE	32768	An individual component barcode was expected. There is non for the place-ment position.				X	X	

NOTICE

Bit encoding

The error codes are bit based. A value combination is possible e.g. error code 16385 is a composition of 16384 + 1
(**MANUAL_INSPECTION_REQUIRED + FINISHED**).

9.2 Barcode reading with SIPLACE Vision

9.2.1 Overview

This document describes barcode reading with SIPLACE Vision algorithms and SIPLACE cameras. It does not cover all topics of the official standards for barcodes such as quiet zones etc., but only topics that are specific to barcode reading with SIPLACE Vision.

If not stated otherwise, all features are supported by Station Software-Version 707.1 or above. An overview of required versions for new features is listed in 9.2.6 "Version overview" [▶ 123].

The document is divided into 3 main sections:

- 9.2.2 "Barcode algorithms" [▶ 112] describes the general behavior of the SIPLACE Vision barcode algorithms, independently of a specific camera system.
- 9.2.3 "Reading barcodes on SIPLACE machines" [▶ 117] describes specific aspects for barcode reading on SIPLACE machines, using SIPLACE Pro and Sirio station software
- 9.2.4 "Specifications for SIPLACE cameras" [▶ 120] gives the specifications for symbol sizes etc. for all supported camera types.

In the following specifications, the term "ROI" (Region of interest) is used to describe the area in which the barcode is searched. The ROI may by a full camera image, a rectangular subarea within a camera image or even a large ROI composed of several camera images.

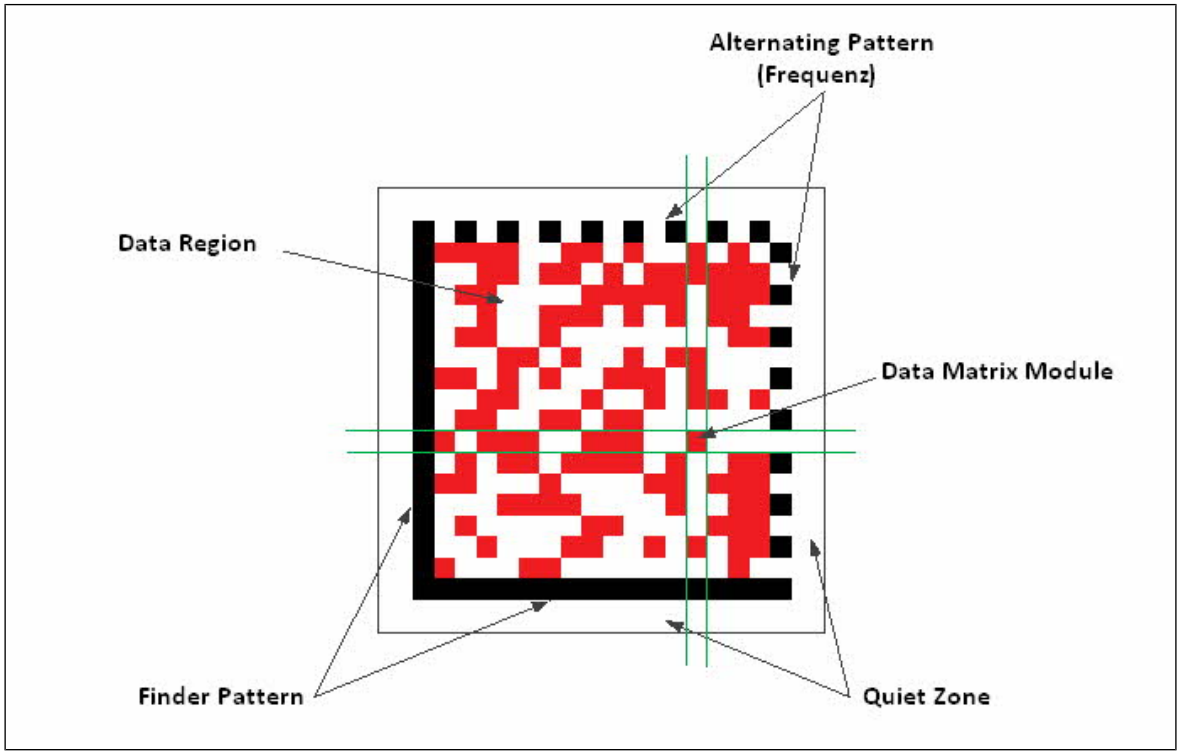
9.2.2 Barcode algorithms

9.2.2.1 General remarks on all barcode types:

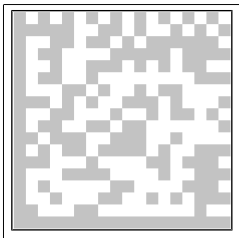
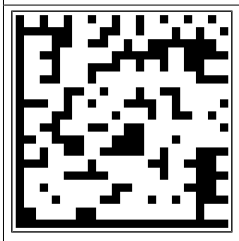
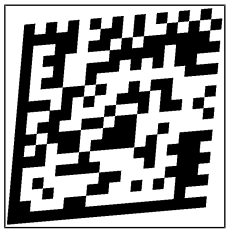
Ambiguity	There are no read errors, if more than one valid barcode symbol can be found in the ROI. It is not specified, which of the valid barcode symbols is read if more than one valid barcode symbol could be found! Ambiguity can be avoided by • using an appropriate ROI size • using "Conditions" for valid barcodes, see below
ASCII Null character	Barcode data should **not** contain ASCII Null characters (ASCII NULL), because the 0-byte is interpreted in the software as termination character.

9.2.2.2 Data matrix codes

Code type	Only data matrix ECC 200 is supported
Numbers of rows / columns	All combinations of numbers of rows / columns as defined in the standard are accepted, including rectangle symbols and large symbols with several data regions.
Minimal dot size	5 pixels
Symbol angle	All angles of the symbol are accepted
Inverse symbols	Inverse symbols (light modules on a dark background) are accepted
Mirrored symbols	Mirrored Symbols are accepted (Station software version 707.1 SP2 or above required)
Ratio of column width to row height	1/2 <= (column width) / (row height) <= 2 (see "axial non-uniformity" below)
Region of interest (ROI)	The area in which the barcode is searched (ROI), should not exceed the following values: • ROI width <= 6 * Barcode symbol width • ROI height <= 6 * Barcode symbol height



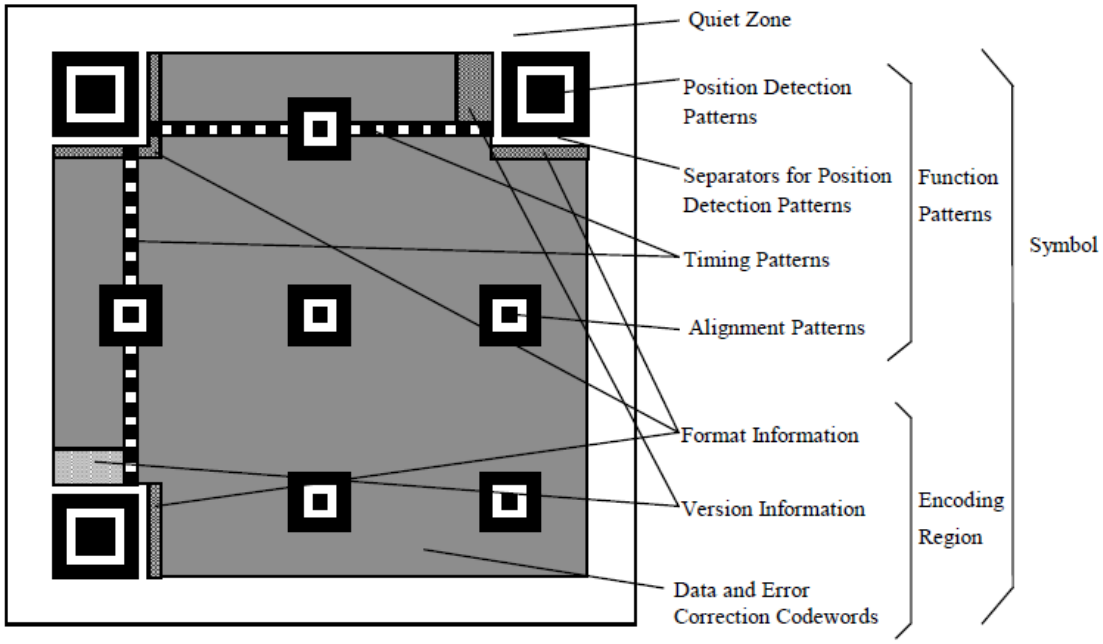
- In the data matrix code, the solid "L" shaped finder or alignment pattern must be one module wide.
- In the data matrix code, the Quiet Zone must be one module wide one all four sides. Within the data matrix Quiet Zone, the gray value of each pixel must not vary more than 10 gray values from the mean gray value in the Quiet Zone.

	**Contrast:** The module contrast is the difference between the gray values of a bright module to an adjacent dark module. That contrast must be at least 20 gray values. Also the contrast between the finder pattern and the quiet zone must be at least 20 gray values.
	**Print growth:** The print growth is a value for the module sizes. With a too strong adjusted laser (or printer) it can happen, that a module is too big for the data matrix grid cell. With a too weak adjusted laser (or printer) it can happen, that a module is smaller than the data matrix grid cell. A module must be at least 1/3 of a data matrix grid cell size. A module must be at most 5/3 of a data matrix grid cell size.
	**Grid non-uniformity:** A uniform, right-angled grid is used to decode. Any non-rectangular deformation is not allowed. A barcode symbol must be rectangular.
	**Axial non-uniformity:** Axial non-uniformity is a measure for the ratio of the column width to the row height. The following must be valid for the ratio: 1/2 <= (column width) / (row height) <= 2

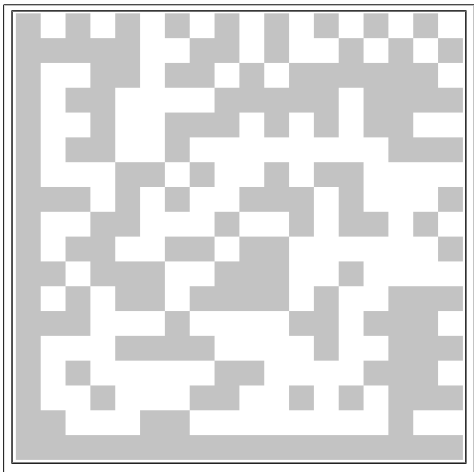
9.2.2.3 QR Codes

Station software version 711.0 or above required!

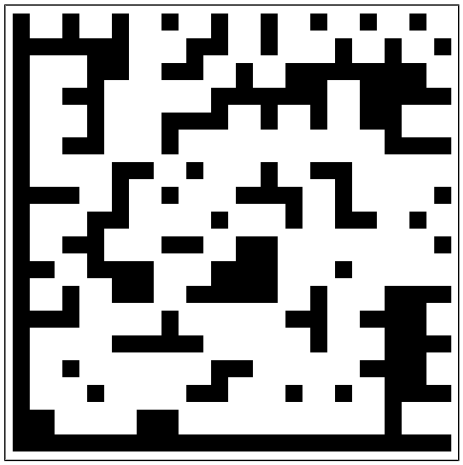
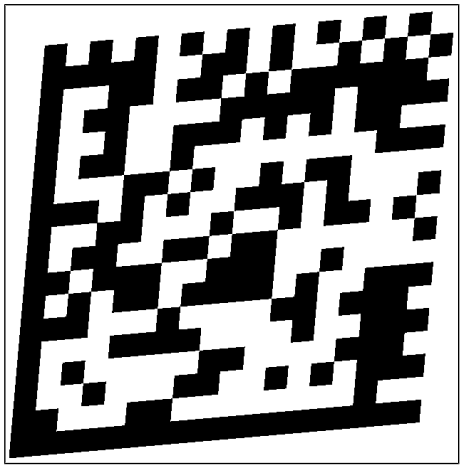
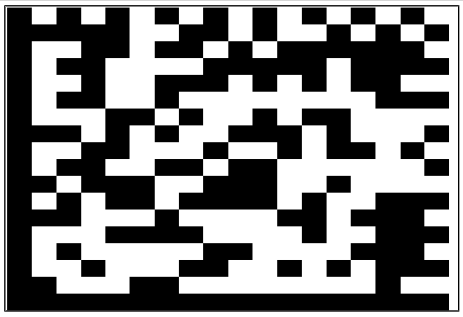
Code type	QR Code according to ISO/IEC 18004 Model 2
Numbers of rows / columns	All versions (i.e. numbers of rows / columns) as defined in the standard are accepted.
Minimal dot size	5 pixels
Symbol angle	All angles of the symbol are accepted.
Inverse symbols	Inverse symbols (light modules on a dark background) are accepted.
Mirrored symbols	Mirrored symbols are accepted.
Ratio of column width to row height	1/2 <= (column width) / (row height) <= 2 (see "Axial non-uniformity" below)
Region of interest (ROI)	The area in which the barcode is searched (ROI), should not exceed the following values: ROI width <= 6 * Barcode symbol width ROI height <= 6 * Barcode symbol height



- QR-Barcode Quiet Zone must be one module wide one all four sides. Within the QR-Barcode Quiet Zone the gray value of each pixel must not vary more than 10 gray values from the mean gray value in the Quiet Zone.



Contrast: The module contrast is the difference between the gray values of a bright module to an adjacent dark module. That contrast must be at least 20 gray values. Also the contrast between the finder pattern and the quiet zone must be at least 20 gray values.

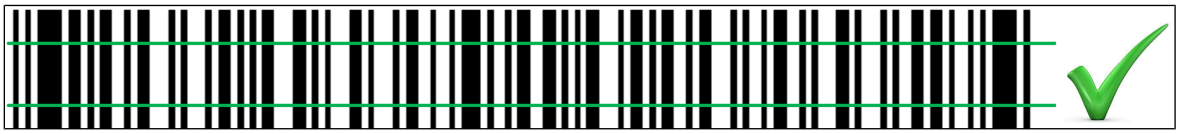
	**Print growth:** The print growth is a value for the module sizes. With a too strong adjusted laser (or printer) it can happen, that a module is too big for the data matrix grid cell. With a too weak adjusted laser (or printer) it can happen, that a module is smaller than the data matrix grid cell. A module must be at least 1/3 of a data matrix grid cell size. A module must be at most 5/3 of a data matrix grid cell size.
	**Grid non-uniformity:** A uniform, right-angled grid is used to decode. Any non-rectangular deformation is not allowed. A barcode symbol must be rectangular.
	**Axial non-uniformity:** Axial non-uniformity is a measure for the ratio of the column width to the row height. The following must be valid for the ratio: 1/2 <= (column width) / (row height) <= 2

9.2.2.4 1 D barcodes

Code types	Code 39 Code 93 Code 128 EAN-8 (AddOn 2, AddOn 5), EAN-13 (AddOn 2, AddOn 5) Interleaved 2 of 5 (Station software version 709.1 or above required!)
Minimal width of bars	5 pixels The width can be reduced up to 3 pixels with appropriate marking quality.
Minimal height of symbol	5% of length of the complete symbol
Symbol angle	All angles of the symbol are accepted.
Inverse symbols	Light bars on a dark background are accepted. (Station software version 708.0 or above required)
Mirrored symbols	Mirrored symbols are accepted. (for 1D codes this is equivalent to a 180 degree rotation)
Region of interest (ROI)	The area on the PCB in which the barcode is searched (ROI), should not exceed the following values: ROI (reading direction) <= 3 * symbol width ROI (perpendicular to reading direction) <= 10 * symbol height

ROI of 1D barcodes

It is accepted if the ROI covers only a slice of the symbol:



However, it is not accepted, if the barcode angle differs too much from the slice angle, because the system always tries to read the complete code in a single slice along the symbol angle:



Uniformity of bars and spaces

For all implemented code types, each bar and space represents 1 to 4 bits, where the number of bits corresponds to the width of the bar/space. When reading barcodes with PCB camera, it is assumed that the code is flat. Therefore, 1D Barcode reading with PCB camera applies the same nominal width to all bars and spaces in the complete symbol. Consequently, barcodes are not readable, if a bar/space with n bits appears smaller than another bar/space with n -1 bits.

9.2.2.5 Conditions for valid barcodes

This requires station software version 711.0 or higher.

To avoid ambiguous read of barcodes or partial reads of 1D barcodes, some conditions can be defined for the algorithm:

1. Minimum code length (i.e. minimum number of bytes in ASCII / UTF-8 data)
2. Maximum code length (i.e. maximum number of bytes in ASCII / UTF-8 data)
3. A regular expression (using ECMAScript syntax, details see Appendix). The complete code content must match the regular expression.

If a read barcode does not fulfill all conditions, the algorithm continues to look for another valid code in the region of interest (It's not necessary to acquire another image).

Conditions can be used for all supported 1D and 2D barcode types.

9.2.3 Reading barcodes on SIPLACE machines

The following definitions apply to all code types (2D and 1D):

Barcode data	The SIPLACE software displays the decode result as ASCII string. Non-ASCII characters are displayed according to the Unicode UTF-8 code page.
Barcode conditions	During teaching with SIPLACE Vision, the user can define "Barcode conditions" to avoid ambiguous or partial reads. See below
Output format	During teaching with SIPLACE Vision, the user can define "Barcode filters" and an "Output format" to return only certain parts of the data as measurement result. For example, the user can configure that only the first 20 characters of barcode data are returned as measurement result. Details see below.
Maximum data length	Data length is not limited by the vision software. However, further data processing of the measurement result may restrict length of the measurement result after output formatting has been applied. By default, the maximum number of characters in the measurement result is set to 40. Depending on the current configuration, SIPLACE Pro can set another limitation on the number of characters.

9.2.3.1 Barcode conditions

The user can define up to three different conditions for valid barcodes while teaching the barcode with SIPLACE Vision. Conditions are saved as part of the vision data of the "barcode fiducial" in SIPLACE Pro. Conditions are checked BEFORE output formatting is applied!

A barcode is valid if all conditions are valid:

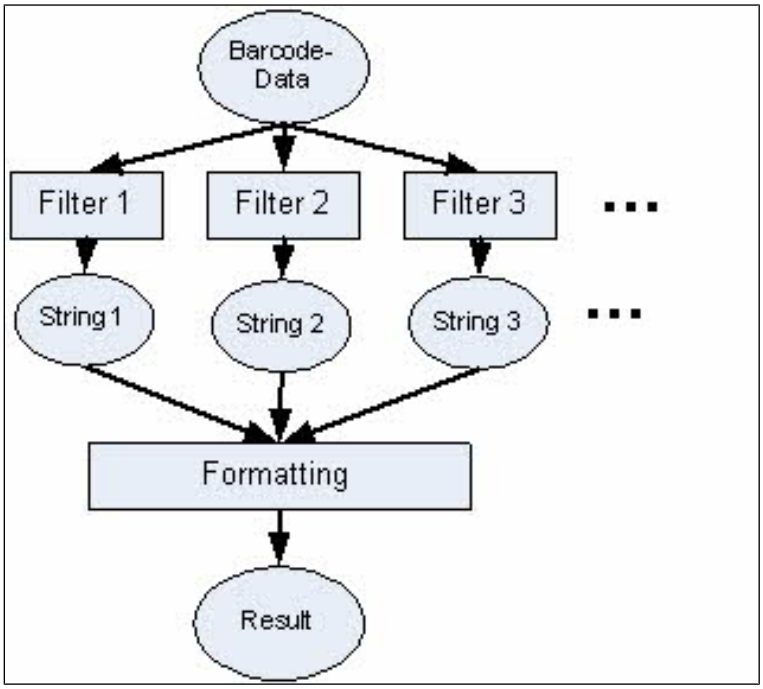
1. Minimum code length (i.e. minimum number of bytes in ASCII / UTF-8 data)
2. Maximum code length (i.e. maximum number of bytes in ASCII / UTF-8 data)
3. Text condition, either using wildcards or regular expressions:

   Code content using wildcards (* and ?)
   ? matches any single character,
   * matches zero or more of any character
   Example: A?X* matches all codes where first character is ‘A’ and third character is ‘X’

   Code content in a regular expression (in ECMAScript syntax, refer to the appendix for details).
   The whole content of the code must match the regular expression.
   If an invalid regular expression is defined, e.g. A{2}, the result is always "no match".

9.2.3.2 Output format

It is possible to format the reading result so that it meets certain requirements for traceability data. Up to 6 filters can be used to select a certain part of the barcode content, for example the first three characters. A "format" string can be used to combine the filtered content to a formatted string:



Each filter has the following optional attributes:

- "Start": Start sequence
- "Offset": Offset
- "Length": Length
- "Stop": End sequence

Start sequence and Offset of each filter are used to compute the **Start position** of the filter result. In a first step, the start sequence is searched in the barcode data. The start position is the first character after the start sequence. Then the start position is shifted by the given Offset.

If there is no Start sequence, then the Start position is equal to the offset.

If neither a Start sequence nor Offset is defined, the Start position is the first character of the barcode data.

Length and End sequence are used to compute the **End position** of the filter result:

- If only the length is given, this means: End position = Start position + Length -1
- If only the End sequence is given, then the End sequence is searched in the barcode data beginning from the start position. The End Position is the character before the End sequence.
- If the length as well as the End sequence is given, the End position is the minimum of both cases above.
- If neither length nor End sequence is given, the End position is the last character of the barcode data.

In the Start and Stop sequence, the character "*" can be used as wildcard for an arbitrary substring.

Special case: It is still possible to search for a "*" in the barcode data by using "**" in the search string. For example, "A**B" can be used to search for "A*B".

The format string defines how the filter results are combined to the measurement result. The following replacements are performed:

- %1 … %n is replaced by the respective result of filter 1 … n, if 0<n<= Number of filters
- %n is replaced by an empty string, if 0<n<=9 and n>Number of filters
- %% is replaced by %

Examples

1. Code content: 12345678-ABCD

   Requirement: Filter out the first 6 characters

   Filter 1: Length = 6

   Format: %1

   Result : 123456

2. Code content: S/N-12345-Date-01/31/2015

   Requirement: Filter out the serial number (number after "S/N-" and before "-Date"):

   Filter 1: Start text=" S/N-" End text="-Date"

   Format: %1

   Result: 12345

3. Code content: 1234\ABCD

   Requirement: Replace "\" by a "-" and exchange content before and after "\":
   Filter 1: End text="\"
   Filter 2: Start text="\"
   Format: %2-%1
   Result: ABCD-1234

9.2.4 Specifications for SIPLACE cameras

9.2.4.1 Reading barcodes with PCB camera

For the PCB camera (SST 34), there are the following limits:

Minimal dot size of data matrix and QR codes	50 µ
Minimal width of bars of 1D codes	50 µ Width can be reduced up to 30 µ with appropriate marking quality.

The following definitions apply to all code types (2D and 1D):

Maximum symbol size	Symbol size and size tolerance can be defined in SIPLACE Pro Desk and during barcode teaching at the machine. Size x/y + size tolerance x/y should be within one of the following limits: • X <= 30 mm and Y <= 30 mm or • X <= 10 mm and Y <= 100 mm or • X <= 100 mm and Y <= 10 mm
Region of interest (ROI)	The region of interest is the area in which the barcode is searched. The center of the ROI is always the "offset" of the barcode fiducial in the SIPLACE Pro description. There are two different modes for computing the ROI size. The mode can be selected in SIPLACE Pro Desk. If **Use field of view** is selected in SIPLACE Pro (Default), the ROI size is equal to size of the camera's field of view (5.8x5.8mm). Exception: If symbol size + tolerance is larger than the camera's field of view, the ROI size is equal to the symbol size + size tolerances. If **Limited tolerances** is selected in SIPLACE Pro, the ROI size is equal to the symbol size + size tolerances + position tolerance as defined in SIPLACE Pro. Using "Limited tolerances", the ROI can be defined smaller than the camera's field of view, for example to avoid ambiguity (see below).
Multiple measurement	If the ROI is larger than the field of view of the PCB camera (5.8 x 5.8 mm), the camera executes a multiple measurement and the individual images are combined to a single "virtual image".

9.2.4.2 Reading barcodes on the bottom side of components

This requires station software version 712.0 or higher.

Barcodes on the bottom side of a component can be read during the component measurement with the component camera. This can be done by adding a "barcode group" to the component shape. Barcode reading for traceability can be activated for individual components in SPLACE Pro.

Remarks

- The barcode group should be completely inside the component shape body.
- It is not possible to have more than one barcode group in a component shape.
- The barcode measurement is only applied, if position measurement was successful.
- The ROI is derived from the actual measured component position and the position as defined by the barcode group in the component shape.
- The user can optionally activate a separate illumination for the barcode measurement
- It is not possible to use the barcode orientation for orientation detection, because barcode algorithms don't report an orientation
- If the component shape wizard is used to teach the component shape, the data for the barcode on bottom side is deleted and has to be taught again.

Minimum dot/bar size and maximum symbol size for individual component camera types

Head cameras

For all head cameras, maximum symbol size is limited only by the size of the component body.

Camera type	Resolution in µm/Pixel	Minimum dot / bar size in µm
23	17.1	80
28	50	250
29	26.7	130
30	17.2	80
38	17.1	80
41	10.1	50
45	10	50
48	11.2	50
49	5.5	25

Stationary cameras

For stationary cameras, the barcode should always be contained in a single camera image, even for multiple measurements. Therefore, there are specific limits for maximum barcode size for stationary cameras as shown in the table. As with the head cameras, the barcode group should be also completely inside the component shape body.

Camera type	Resolution in µm/Pixel	Minimum dot / bar size in µm	Maximum symbol size X/Y in mm
25	16	80	9.0
33	42	200	23.5
36	79	400	18.5

Compatibility

Component shapes with barcode on bottom size can also be used with software versions < 712.0, but data for barcode on bottom side will not be sent to software versions < 712.0.

9.2.5 Using regular expressions

A regular expression consists of ordinary characters (for example, letters a through z) and special characters, known as meta characters. Meta characters use the ECMAScript syntax:

Meta character	Behavior	Example
{n,m}	Quantifier expression Matches the preceding character or sub expression n to m times.	ab{2,4} matches "abb","abbb", and "abbbb" but not "ab" or "abbbbb".
{n,}	Matches the preceding character or sub expression n or more times.	ab{2, } matches "abb","abbb", and "abbbbb" but not "ab"
{,n}	Equivalent to **{0,n}**	ab{,2 } matches "a" and "abb", but not "abbb".
*	Matches the preceding character or sub expression zero or more times. Equivalent to **{0,}**	**ab*** matches "a" and "abb".
+	Matches the preceding character or sub expression one or more times. Equivalent to **{1,}**	**ab+** matches "ab" and "abb", but not "a".
?	Matches the preceding character or sub expression zero or one time. Equivalent to **{0, 1}**	**ab?** matches "a" and "ab", but not "abb".
[abc]	Character set. Matches any one of the specified characters	**[123]** matches "1", "2" or "3".
[a-z]	Range(s) of characters. Matches any character in the specified range(s)	**[1-3]** matches "1", "2" or "3". **[1-3A-C]** matches "1", "2", "3","A", "B", or "C".
[^]	Negative range of characters. Matches any character that is not in the specified range.	**[^0-9]** matches any character that is not in the range "0" through "9".
.	Matches any single character.	.. matches "ab", "cd", but not "a" or "abc".
(abc)	Sub expression	(abc)* matches "abc" and "abcabc" but not "ababc".
|	Indicates a choice between two or more items.	(A|B)C matches "AC" and "BC", but not "DC".
\	Escape character Uses the next character as an ordinary character instead of a meta-character.	ab*c matches "ab*c" ab\\c matches "ab\c".

Examples

ABC[0-9]{6} matches all codes starting with "ABC" followed by exactly 6 digits.

(A|B).*C matches all codes starting with "A" or "B" and ending with "C".

([1-9]|[1-9][0-9])\..* matches all codes starting with a number between 1 and 99 followed by "."

9.2.6 Version overview

This section gives an overview of SIPLACE Vision versions / Station SW versions / SIPLACE Pro versions required for certain features:

Station soft-ware	Vision version	SIPLACE Pro	Feature
705	4.3.1	9.2	Read barcode with PCB camera using Halcon (Dongle license)
707.1	5.1.2	11.2	Halcon (Dongle license) not required any more Mirrored data matrix symbols possible (707.1 SP2)
709.1	5.3.2	13.1	Barcode type Interleaved 2/5
711.0	5.5.1	15.0	QR Barcodes Barcode conditions
712.0	5.6.1	16.0	Barcode on bottom side of component

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ASM Assembly Systems GmbH & Co. KG
Rupert-Mayer-Straße 44
81379 Munich
Germany

www.asm-smt.com